Adaptive Design of Clustered Surveys and Experiments

Carlos Gonzalez-Perez*

Click here for the most recent version

October 12, 2026

Abstract

When designing clustered surveys or experiments on a budget, researchers face a trade-off between the number of clusters and the number of units per cluster. Unfortunately, the variance-minimizing allocation depends on the intra-cluster correlation (ICC), a property of the data the researcher has not yet collected. So, in practice, designers guess. I show that guessing is first-order expensive: a design built on a misspecified ICC forfeits precision of the same order as the target variance itself. Instead, I propose two pilot-based algorithms—one two-staged, one fully sequential—that learn the ICC while the experiment runs. They attain negligible excess variance compared to a designer which knows the true ICC, improve an order of magnitude upon any misspecified static design—maximin included—and are rate optimal. Moreover, adaptivity need not invalidate conventional inference. Under a symmetry assumption on the errors, mean estimators remain asymptotically normal conditional on the realized design, with no data discarded and no correction. The theory on inference invariance extends beyond the cluster setting and validates standard inference for popular sampling procedures such as Neyman allocations with variances computed from pilot data. Revisiting Fink et al. (2020b), I find that, for their main specification, adaptive allocation would have reduced the variance by up to 41% at equal cost or the reduced the cost by up to 42% at equal variance. Finally, I demonstrate feasibility of the methods by fielding a live online survey whose design was chosen in real time by the algorithm. The results show that uncertainty about the ICC can be learned during data collection at only second-order precision cost, without sacrificing conventional inference.

Keywords: Adaptive experimental design, optimal experimental design, clustered sampling, intra-cluster correlation, adaptive inference.

1 Introduction

When designing a clustered survey or clustered experiment the researcher faces two costs: fixed costs of sampling additional clusters—travel, training additional staff, negotiating with local authorities—and variable costs of sampling additional units. As a result, when the researcher operates on a fixed budget, she must trade-off *width* (the number of clusters, K) against *depth* (the number of units per cluster, N). For instance, Fink et al. (2020b), henceforth FJM, designed an experiment where

*University of Oxford, Department of Economics. Email: carlos.gonzalezperez@economics.ox.ac.uk. I would like to thank Maximilian Kasy, Davide Viviano, Günther Fink, Anders Kock, Rahul Singh and numerous seminar participants for feedback on this manuscript.

they gave favorable loans to farming households in Zambia with treatment at the village level. Their design included $K = 175$ villages and $N = 18$ households per village on average. In these settings, the variance-minimizing allocation of units versus clusters depends on a single statistical parameter: the intra-cluster correlation (ICC), the share of total variation that is between rather than within clusters. When units are highly correlated within clusters, there is little informational value in sampling additional units from the same cluster, and so the researcher is better-off by implementing a wide design (many clusters, few units per cluster).[1] The solution, which has been known at least since Cochran (1977), is simple and intuitive.

She cannot use it. The ICC is a property of the data she has not yet collected, therefore unknown at the time of the design. So, in practice, designers guess this parameter. Some apply rules of thumb, some transport ICC estimates from prior studies, and some run non-optimally sized pilots.[2] The consequences of guessing are not innocuous. I show that any design built on a misspecified ICC incurs excess variance of the same order as the oracle variance itself. In other words, the researcher pays, in variance, an amount comparable to what her budget buys. For example, I show that the static design implemented by FJM might accrue up to 70% higher variance than the cost-optimal allocation.

The ICC has been reported to vary enormously across settings and outcomes. For instance, Knox and Chondros (2004) report variations in the ICC of one order of magnitude across covariates within the same population.[3] Guessing the ICC is therefore not only costly, but difficult. Learning this parameter from pilot data is also non-trivial. The ICC is a function of the within- and between-cluster variances, where the latter can only be estimated by sampling many clusters—the expenditure the design is meant to economize. The designer then faces a *circularity*: selecting the optimal design requires knowing the ICC, but learning the ICC requires sampling many clusters, which may or may not be design-optimal. Cluster-level costs are often sunk once incurred, so mistakes that take place during learning cannot be undone.

This manuscript makes a threefold contribution. Firstly, I establish formal optimality guarantees in the joint allocation problem for two algorithms that do not know the ICC. These algorithms do not only escape the circularity, but induce negligible excess variance compared to a designer who knows the ICC ex ante, are rate optimal, and improve an order of magnitude upon any guessing-like policy (Theorems 5.4 and 5.5). For instance, in the context of the FJM application, my algorithms achieve up to 41% lower variance compared to the authors' design at no additional cost. Second, under a symmetry assumption on the errors, I show that inference can proceed as if the design had been fixed in advance, i.e. estimators of population means remain asymptotically normal with the usual variance after conditioning on the realized design (Theorem 6.8). The result rests on a general principle: under symmetry, error signs and error magnitudes are independent. *Even* designs—coordinate-wise sign-invariant functionals—only depend on the magnitudes, hence signs remain unrestricted after conditioning on the design, and so a central limit theorem can run through them without compromising inference. This invariance principle extends beyond the cluster setting and includes many popular sampling methods such as Neyman allocations with pilot-estimated variances (Appendix D). This means

[1] In the extreme case where observations are perfectly correlated within clusters, the analyst extracts all of a cluster's information from a single unit, hence she would sample one unit from as many clusters as the budget allows.

[2] The design in Fink et al. (2020b) used an ICC-informed power calculation estimated through a small pilot, leading to a selection of 175 villages with up to 22 households each. Yet, they did not optimize the allocation of sampling effort across villages and households against the relative costs of those margins. Their register only documents: "to be powered to detect an increase in farm production value of 10%" (Fink et al., 2014).

[3] Knox and Chondros (2004) report variation of 0.055 to 0.451 on demographic covariates and from 0.005 to 0.059 on morbidity covariates within the same target population.

that the 41% lower variance achieved by the adaptive algorithms can enter the formula for regular clustered standard errors without modification or data exclusion. Third, I demonstrate the method end to end: through simulations, through a semi-synthetic reconstruction of FJM, and through an online survey which I field in real time using one of my algorithms.

I start this manuscript by characterizing the optimal design under known ICC as a function of budget in a minimal model with nested errors, independent clusters, linear costs and a hard budget, as in Cochran (1977). This allocation yields an oracle variance which decreases linearly with budget. I then propose two algorithms that implement the optimal design in the limit with no prior knowledge of the ICC. Both share the same architecture: run a carefully-sized pilot, learn the ICC, allocate the remaining budget through plug-in rules. The first of the two, Explore-then-Commit Cluster Selection (`EtC-CS`) implements a simple two-stage design by taking a worst-case approach in the pilot, Adaptive Cluster Selection (`ACS`) samples clusters sequentially, stops under a data-dependent rule, and commits to the final design. The expected excess variance of the algorithms is negligible compared to the oracle variance, and a full order of the budget smaller than the excess variance of any static design built on a misspecified ICC, including the maximin optimal design of van Breukelen and Candel (2015). I further show the adaptive algorithms are rate unimprovable by deriving matching lower bounds on the problem. Although both algorithms display bounds of the same rate, `ACS` outperforms `EtC-CS` in simulations at the expense of greater adaptivity requirements.

It is well-known that data-dependent designs can compromise conventional inference. Most results in this literature tackle this issue by excluding pilot data (Viviano, 2022), re-weighting the terminal estimator (Yin and Shen, 2005), imposing strong normality assumptions (Rinott, 1978), conditioning on designs only with degenerate limiting probabilities (Hahn et al., 2011), assessing distortions through simulation (Wittes et al., 1999), or some combination of the above. My algorithms also select the design adaptively, so they could be subject to the same limitations. They are not. Under a symmetry assumption on the errors, inference on sample averages is invariant to adaptive data collection whenever the design depends only on (asymptotically) even functions of the data, such as sample variances. Asymptotic normality is then preserved with no correction and without discarding pilot observations. Unlike Hahn et al. (2011), my theory allows for conditioning on design events with non-degenerate limiting probability, which is what makes conditional and unconditional inference meaningfully distinct. While the driving heuristic of the theory leverages exact evenness, feasible designs are never exactly even as sample variances depend on squared residuals rather than squared errors. Still, I show that the same conclusions extend to the feasible class of asymptotically even policies by coupling them with unfeasible even surrogates at the relevant rate and imposing an anti-concentration requirement. Simulations verify the inference results and delineate their limits: once symmetry or (asymptotic) evenness fails, conditional inference can be distorted.

The theory results on performance are also borne out in simulations and in a reconstruction of FJM. FJM ran a clustered experiment, randomly selecting 175 villages and up to 22 households per village, to identify the effects of relaxing liquidity constraints on rural Zambian households through subsidized loans. FJM approached the design with more care than is typical: they ran a small pilot survey, estimated the ICC from the sample, and implemented a design "sufficiently powered to detect an increase in farm production of 10%". Yet neither the pilot nor the main experiment explicitly optimized the width-depth allocation against its cost structure—and far from exceptional, this is the

norm.[4] As a consequence, under a plausible grid of fixed-to-variable cost ratios, FJM could have achieved substantially lower variance for the same budget, or spent a substantially smaller budget at fixed precision. In particular, I show that both algorithms attain nearly zero excess variance, what reduces the variance of their estimator by up to 41% and the standard errors of their main specification by up to 19% at equal costs. Conversely, my algorithms could have achieved the same precision as FJM with 42% lower budget. The application also illustrates how to test for error symmetry—the load-bearing assumption entering the inference theory. Finally, I demonstrate that the methods are implementable by using `ACS` to field an adaptive online survey in real time. The survey measures people's ability to distinguish AI-generated from human-written text across languages. Each language forms a cluster, as outcomes are likely correlated within languages, and carries a genuine fixed cost, as materials must be produced and checked by paid native speakers. On top of that, each additional respondent also entails a variable cost.

This paper connects several literatures. Survey theory has long studied cost-optimal clustered designs through an algorithmic approach (Bethel, 1989; Barcaroli et al., 2022), but has rarely integrated ICC estimation within design optimization. Biostatistics, by contrast, has a long tradition on learning nuisance parameters through pilots dating back to Stein (1945). Within the internal-pilot literature (Wittes and Brittain, 1990), the closest papers to mine are Lake et al. (2002) and van Schie and Moerbeek (2014), where the authors re-estimate the ICC within a cluster trial, but each adjusts a single margin—width in the former, depth in the latter—and neither embeds a cost structure nor derives an optimality theory. In econometrics, my paper is closest to the two-wave tradition of Hahn et al. (2011) and, in structure, to Viviano (2022), from which it differs in decision space—clustered sampling rather than network experiments—and in requiring no sample splitting for inference. The performance criterion is the cluster-sampling analog of Neyman regret (Dai et al., 2023), and the inference results extend an invariance tradition in sample-size re-estimation running from Stein (1945), where, for asymptotically even policies, I can replace normality by a mild error symmetry assumption, in a spirit reminiscent of randomization inference under group invariance (Canay et al., 2017).[5]

2 Literature Review

Clustered Designs in Survey Theory. Optimal multi-stage survey design under a budget constraint is a classical problem in survey theory, dating back to Tschuprow (1923), Hansen and Hurwitz (1942) and Kish (1965).[6] In its simplest form, a researcher conducting a multi-stage survey first partitions the population into non-overlapping clusters, then samples K clusters at random, and finally samples N units at random within each selected cluster. The objective of this literature is the one I adopt here: to minimize the variance of a population estimator through the allocation of sampling effort across clusters and units. My adaptive designs are benchmarked against the cost-optimal oracle allocation that knows the ICC as derived by Cochran (1977).[7] Since Bethel's algorithm (Bethel, 1989), this literature has extended Cochran's logic to multi-domain, multi-variable and multi-objective settings, with an emphasis on algorithmic performance and the central role of the intra-cluster correlation (Cicchitelli et al., 1992; Campbell et al., 2004; Ballin and Barcaroli, 2013; Barcaroli et al., 2022). Yet

[4] If anything, by pre-estimating the ICC via a pilot, FJM display additional care relative to the large body of designs built on guesstimates borrowed from the literature or previous studies.

[5] All proofs in this paper, including a standalone treatment of the general inference invariance theory, are collected in the Appendix.

[6] See Biemer and Lyberg (2003) for a modern textbook treatment.

[7] Raudenbush (1997) and Moerbeek et al. (2000) obtain analogous oracle allocations for cluster designs.

it almost never estimates the ICC *inside* the design problem. Barcaroli et al. (2022) assume it can be recovered from auxiliary data, while applied guidance alternates between recruiting as many clusters as logistics permit (Tam, 2005) and fixing the number of clusters ex ante (Sedgwick, 2015). Neither prescription is optimal in general.

Adaptive Designs and ICC Re-estimation in Biostatistics. The idea of using an initial sample to learn a nuisance parameter that governs the design has a long history in statistics and biostatistics. It begins with Stein (1945), who proposes a two-stage rule for Gaussian data with unknown variance and preserves the t distribution of the second-stage statistic by choosing the total sample size as a function of the first-stage variance estimate. Rinott (1978) and Dudewicz and Dalal (1975) extend this logic to best-group identification, and Yin and Shen (2005) embed ICC re-estimation in a fully sequential GEE trial.More recently, this idea was revived by the *internal pilot* literature. Wittes and Brittain (1990) and Wittes et al. (1999) argue that nuisance parameters can be estimated mid-trial and the design revised accordingly, with the final analysis conducted on all data as if the design had been fixed ex ante. Much of this literature assesses performance and inference through simulation under Gaussianity, reporting type I error inflation that varies with the timing and size of the pilot, without formal guarantees. Lake et al. (2002) and van Schie and Moerbeek (2014) are the closest antecedents of my paper as they specialize the internal pilots thesis to ICC estimation in the context of cluster-randomized medical trials. Two differences stand out: they optimize a single margin—width in Lake et al. (2002), depth in van Schie and Moerbeek (2014)—whereas I optimize (K, N) jointly. Also, they do not impose a cost structure and do not derive an optimality theory. A parallel literature acknowledges ICC uncertainty but holds the design fixed, either by placing Bayesian priors on the ICC (Spiegelhalter, 2001; Turner et al., 2005) or by adopting maximin designs (van Breukelen and Candel, 2015, 2018, 2021). These approaches optimize against ICC uncertainty within a static design class, but do not benchmark the resulting loss against the known-ICC oracle. I establish worst-case excess variance bounds of $O(1/B^2)$ that cannot be improved in rate, and I show that my algorithms asymptotically dominate every misspecified static policy over the ICC range—including the maximin design of van Breukelen and Candel (2015), whose worst-case excess variance is $\Theta(1/B)$ (Lemma 5.2). To my knowledge, this is the first paper to treat the clusters-versus-units allocation as a sequential learning problem, and the first to provide formal optimality guarantees for an adaptive cluster design.

The Econometrics of Experimental Design. My work relates thematically to the econometric literature on optimal but non-adaptive design through stratification and covariate adjustment (Cytrynbaum, 2021; Bai, 2022; Cytrynbaum, 2024; Bai et al., 2024). That literature shares my variance-minimization objective, and occasionally my attention to sampling costs, but neither address the cluster-versus-unit margin nor the learning problem. Closer in structure is the two-wave tradition initiated by Hahn et al. (2011), in which a pilot estimates the parameters that enter the optimal design: treatment probabilities in Hahn et al. (2011), and optimal stratification in Chambaz et al. (2015) and Tabord-Meehan (2023). Nearest to my setting is Viviano (2022), who designs two-wave experiments under network interference by minimizing estimator variance using pilot-estimated variance components, and derives regret guarantees, asymptotic normality, and a characterization of the regret-minimizing pilot size. I share the pilot-then-optimize architecture and the regret criterion, but my decision space is the clustered sampling design and my inference results require no data exclusion: Viviano (2022) must remove both the pilot units and their network neighbors from the main experiment to keep the design unconfounded, whereas I use the full sample. My work also connects to

adaptive experimentation and policy learning in economics (Kasy and Sautmann, 2021; Adusumilli, 2026; Cesa-Bianchi et al., 2025; Li et al., 2026).

Neyman Regret in Sequential Experiments. My performance criterion—expected excess variance relative to the variance-minimizing oracle design—is the clustered-sampling analogue of the *Neyman regret* introduced by Dai et al. (2023) for adaptive treatment allocation. Working in a design-based framework, they attain $\tilde{O}(\sqrt{T})$ Neyman regret through clipped online gradient descent and show that explore-then-commit designs can suffer linear Neyman regret when early units are unrepresentative.[8] Subsequent work attains logarithmic rates in stochastic environments (Neopane et al., 2024), and does so even under the stronger notion of Neyman regret in Neopane et al. (2025). Both rates are consistent with my guarantees.

Inference with Adaptive Designs. Adaptive data collection frequently invalidates conventional inference. In particular, adaptivity can bias sample means (Nie et al., 2018) and destroy asymptotic normality (Hadad et al., 2021).[9] The remedies fall into two traditions: an *adjustment* tradition, which reweights the observations entering the estimator to restore martingale central limit behavior (Hadad et al., 2021; Zhang et al., 2021; Shen and Fisher, 1999; Yin and Shen, 2005); and an *invariance* tradition, which shows that, under normality assumptions, designs depending on the sample variance are independent of the sample mean (Stein, 1945; Rinott, 1978; Dudewicz and Dalal, 1975). The same invariance is used implicitly in the internal pilot and blinded sample size re-estimation literatures (Gould and Shih, 1992; Kieser and Friede, 2003). Proposition 6.6 and Theorem 6.8 in this paper belong to the latter, but I generalize this route in three directions: from normality to symmetry of the errors, from the sample variance to any asymptotically even functional of the data, and from finite-stage adaptivity to arbitrary designs. The resulting class contains my algorithms, but also Neyman allocations computed from pilot variances, and the optimal second-stage propensity scores of Hahn et al. (2011).[10] In addition, the contrast with the class of conditioning events in Hahn et al. (2011) is instructive. While their validity rests on the second-stage design converging to a deterministic limit, my theory allows for conditioning on events with non-degenerate probability. The symmetry assumption plays a role reminiscent of group invariance in randomization inference (Canay et al., 2017). The boundary of my results, however, remains sharp: adaptation through *odd* functionals of the data, like estimated effects or futility rules, reintroduces the pathologies that motivate the adjustment tradition.

Other Connections. My work is more distantly related to the treatment of nested errors in panel data (Wooldridge, 2003), to standard error adjustment in clustered experiments (Abadie et al., 2023), and to the optimal-design tradition of Kiefer and Wolfowitz (1960), which underpins modern algorithms for linear bandits.[11] These connections are conceptual rather than methodological.

[8] Their linear-regret example is specific to the design-based environment they study. I work in the stochastic *iid* setting, in which explore-then-commit designs such as `EtC-CS` do not incur linear regret.

[9] See Wager (2024) for a textbook treatment.

[10] Propensity scores are ratios of pilot-estimated conditional standard deviations, therefore even. Conditional-mean heterogeneity enters the semiparametric efficiency bound additively and does not depend on the assignment probability, so it drops out of the optimization, and the score depends on the pilot only through second moments.

[11] See Lattimore and Szepesvári (2020) for a textbook treatment.

3 Cluster Design Problem

3.1 Model

Consider a population partitioned into equally sized clusters. The designer samples K clusters, indexed by $k = 1, \ldots, K$ and N^k units from sampled cluster k indexed by $i = 1, \ldots, N^k$.[12]

Assumption 3.1. *Linear Additive Model and Error Distributions. Let the outcome of interest y_{ki} follow an additive model with nested errors,*

$$y_{ki} = \mu + u_k + \varepsilon_{ki} \tag{1}$$

where μ is the population mean and estimand, u_k is the cluster-level error and ε_{ki} is the unit-level error. $(u_k)_k$ is an iid sequence with $\mathbb{E}[u_k] = 0$, $Var\,(u_k) = \tau^2 > 0$; $(\varepsilon_{ki})_{k,i}$ is an iid sequence with $\mathbb{E}[\varepsilon_{ki}] = 0$, $Var\,(\varepsilon_{ki}) = \sigma^2 > 0$; and $(u_k)_k \perp\!\!\!\perp (\varepsilon_{ki})_{k,i}$.

Define the variance ratio $\lambda_0 = \sigma^2/\tau^2$. λ_0 maps one-to-one to the intra-cluster correlation, $\text{ICC} = \tau^2/(\tau^2 + \sigma^2) = 1/(1 + \lambda_0)$, and is unknown to the designer.[13]

Assumption 3.2. *Known Compact Parameter Space. Let $\omega = (\tau^2, \lambda_0) \in \Omega$, where $\Omega = [\tau^2_m, \tau^2_M] \times [\lambda_m, \lambda_M]$ is known to the designer, and $\tau^2_m > 0, \lambda_m > 0$.*

3.2 Variance Minimization in Survey Design

For each cluster k, the designer decides how many units N^k to sample to minimize the variance of an estimator of the population mean μ. In the homogeneous-variance setting the natural estimator $\hat{\mu}$ is simply the average of averages

$$\hat{\mu} = \frac{1}{K}\sum_{k=1}^{K} \hat{\mu}^k_{N^k}, \qquad \hat{\mu}^k_{N^k} = \frac{1}{N^k}\sum_{i=1}^{N^k} y_{ki}, \quad k : N^k > 0$$

where $K = \sum \mathbb{1}(N^k > 0)$ is the number of sampled clusters and $\hat{\mu}^k_{N^k}$ is the sample average of cluster k.[14] Appendix A shows that, for any fixed number of sampled clusters, the variance-minimizing allocation is balanced, i.e. $N^k \in \{0, N\}$ for all k.[15] Under this simplification, let $W(K, N) = \text{Var}(\hat{\mu}_{K,N}) = \frac{\tau^2}{K}(1 + \lambda_0/N)$ be the variance of the balanced estimator. The goal of the designer is to minimize the variance of the estimator subject to some known linear costs and a hard budget. Let F be the fixed cost of sampling an additional cluster, V be the variable cost of sampling an additional unit, and B be the hard budget, then[16,17]

$$\min_{K \geq 2, N \geq 2} W(K, N) \qquad \text{s.t. } K(F + VN) \leq B \tag{2}$$

[12] The model implicitly assumes the existence of a super-population with an infinite amount of clusters and units.

[13] Section 9 introduces a more general model with heterogeneous cluster variances τ^2_k.

[14] When the design size is fixed ex-ante, and both clusters and units are selected through sequential randomization, then $\hat{\mu}$ is also unbiased for μ and variance minimization is equivalent to mean-squared error minimization.

[15] Balancedness is specific to the unweighted average of averages under homogeneous variances. It fails for the GLS-weighted average with heterogeneous cluster variances studied in Section 9.

[16] Assume $B \geq 2(F + 2V)$. The assumption is without loss as the manuscript focuses on the $B \to \infty$ regime.

[17] The bounds $K \geq 2$, $N \geq 2$ are imposed for technical reasons that affect the estimation of unknown parameters, see Section 4. In practice the restriction is with little loss, as nearly all clustered surveys and experiments consider larger samples anyway.

For the performance analysis, I work with the continuous relaxation of (2), treating (K, N) as real-valued.[18] I further rule out corner solutions through an additional assumption,

Assumption 3.3. *Interior Solution.* $\lambda_m F/V \geq 4$ *and* $B \geq 2\left(F + \sqrt{\lambda_M FV}\right)$.[19]

Under Assumptions 3.1, 3.2 and 3.3, the first-order conditions delivers Cochran's oracle allocation (Cochran, 1977),

$$N^*(\lambda_0) = \sqrt{\lambda_0\ F/V}, \qquad K^*(\lambda_0) = \frac{B}{F + VN^*(\lambda_0)}, \qquad W_0 = \frac{\tau^2\left(\sqrt{V\lambda_0} + \sqrt{F}\right)^2}{B} = \Theta(1/B) \tag{3}$$

where $W_0 = W(K^*(\lambda_0), N^*(\lambda_0))$ is the oracle variance, as it depends on the unknown variance ratio λ_0. More generally, for $\lambda \in [\lambda_m, \lambda_M]$ define the optimal map as $\lambda \mapsto (K^*(\lambda), N^*(\lambda))$.

Clustered Experimental Design. The survey design problem in (2) extends naturally to the two canonical clustered experimental designs: treatment assigned at the cluster level, where half the clusters are assigned to treatment and half to control; and clusters as blocks, where within each cluster half the units are assigned to treatment and half to control, usually assuming no interference. To do so, the model in (1) is extended to the potential outcomes framework via $y_{ki}(d_{ki}, \boldsymbol{d}_{k,-i})$, where $d_{ki} \in \{0, 1\}$ is unit i's own treatment and $\boldsymbol{d}_{k,-i}$ collects the assignments of the remaining units in cluster k. In both cases, the design variance takes the form $W(K, N) = \frac{c}{K}\left(\tau^2 + \sigma^2/N\right)$ for some constant c, under the substitutions in Table 1. Appendix A introduces both settings in greater detail.

Problem	**Estimand**	**Estimator**	c	τ^2	σ^2
Survey	$\mathbb{E}[y_{ki}]$	$\frac{1}{KN}\sum_k^K \sum_i^N y_{ki}$	1	τ^2	σ^2
Cluster-Level Treatment	$\mathbb{E}[y_{ki}(1, \mathbf{1}) - y_{ki}(0, \mathbf{0})]$	$\frac{2}{K}\left(\sum_{k:d_k=1} \hat{\mu}_N^k - \sum_{k:d_k=0} \hat{\mu}_N^k\right)$	2	$\tau_1^2 + \tau_0^2$	$\sigma_1^2 + \sigma_0^2$
Clusters as Blocks	$\mathbb{E}[y_{ki}(1, \cdot) - y_{ki}(0, \cdot)]$	$\frac{2}{KN}\sum_k^K \left(\sum_{i:d_{ki}=1} y_{ki} - \sum_{i:d_{ki}=0} y_{ki}\right)$	1	$\tau_1^2 + \tau_0^2 - 2\tau_{01}$	$2(\sigma_1^2 + \sigma_0^2)$

Table 1: Canonical clustered design problems

Notes: All three problems share the optimal-map functional form in (3). (τ_d^2, σ_d^2) are the cluster- and unit-level variances of potential outcome d, and $\tau_{01} = \mathrm{Cov}(u_k^1, u_k^0)$. Under cluster-level assignment $d_{ki} = d_k$ and the estimand is the TATE; under clusters as blocks, assuming no interference, the estimand is the ATE. The blocks design admits a degenerate corner: when $u_k^1 = u_k^0$, within-cluster differencing removes the cluster-level noise entirely ($\tau^2 = 0$), and the variance is minimized by sampling a single cluster. Assumption 3.2 excludes this case by requiring $\tau_m^2 > 0$.

[18] In simulations, all algorithms are mapped to integer allocations through $(\lfloor K^\pi \rfloor, \lfloor N^\pi \rfloor)$. This rounding criterion guarantees implementability, eases cross-algorithm comparison, and removes differences stemming from integer approximation. In practice, alternative terminal budget allocations may achieve superior performance.

[19] The assumption is stated in primitives and it is equivalent to requiring $N^*(\lambda) \geq 2$ and $K^*(\lambda) \geq 2$ for every $\lambda \in [\lambda_m, \lambda_M]$, with $K^*(\lambda), N^*(\lambda)$ defined in (3).

4 Adaptive Sampling under Unknown ICC

4.1 Estimating the Variance Ratio

When the variance ratio is unknown, every attempt to induce a near-optimal design will require estimating λ_0. To do so, fix $K \geq 2$ and $N \geq 2$, and define

$$S^2_{k,N} = \frac{1}{N-1}\sum_{i=1}^{N}(y_{ki} - \hat{\mu}^k_N)^2, \qquad \hat{w}^2_{K,N} = \frac{1}{K}\sum_{k=1}^{K} S^2_{k,N}, \qquad \hat{b}^2_{K,N} = \frac{1}{K-1}\sum_{k=1}^{K}(\hat{\mu}^k_N - \hat{\mu})^2,$$
$$\hat{\tau}^2_{K,N} = \Pi_{[\tau^2_m, \tau^2_M]}\left(\hat{b}^2_{K,N} - \frac{\hat{w}^2_{K,N}}{N}\right), \qquad \hat{\lambda}_{K,N} = g(\hat{w}^2_{K,N}, \hat{b}^2_{K,N}) = \Pi_{[\lambda_m, \lambda_M]}\left(\frac{\hat{w}^2_{K,N}}{\hat{\tau}^2_{K,N}}\right)$$

where $\hat{w}^2$ is the within-cluster sample variance, $\hat{b}^2$ is the between-cluster sample variance, $\Pi_{[a,b]}(x) = \min\{b, \max\{a, x\}\}$ is the clip operator and $\hat{\lambda}$ is the (clipped) sample estimator of λ. Lemma 4.1 shows consistency of these estimators under K and N asymptotics.

Lemma 4.1. *Consistency and Asymptotic Regimes. Let $\theta^2_N := \tau^2 + \frac{\sigma^2}{N}$ be the variance of a cluster sample mean, and let $\overset{x}{\to}$ denote convergence in x under $T \to \infty$ asymptotics. Under Assumptions 3.1 and 3.2: (i) $\hat{w}^2_{K,N} \overset{a.s.}{\underset{N}{\to}} \sigma^2$ (with $K \geq 1$ fixed) and $\hat{w}^2_{K,N} \overset{a.s.}{\underset{K}{\to}} \sigma^2$ (with $N \geq 2$ fixed); (ii) $\hat{b}^2_{K,N} \overset{a.s.}{\underset{K}{\to}} \theta^2_N$ (with $N \geq 2$ fixed); and (iii) $\hat{\lambda}_{K,N} \overset{a.s.}{\underset{K}{\to}} \lambda_0$ (with $N \geq 2$ fixed).*

Width is the binding margin for design and for learning alike. Both optimal design and learning require asymptotic sampling of clusters, an observation that shapes the adaptive algorithms and explains their success. *Design.* For any $\tau^2 > 0$, $\hat{\mu}$ is consistent for μ if and only if $K \to \infty$. Intuitively, increasing N averages away the individual-level noise ε_{ki}, but only additional clusters average away the cluster-level noise u_k. The optimal map encodes this asymmetry, while $N^*(\lambda)$ does not depend on B, $\lim_B K^*(\lambda) = \infty$ for any fixed λ. *Learning* presents a similar asymmetry. Lemma 4.1 shows that while the within-cluster variance can be estimated consistently under either N- or K-asymptotics, the between-cluster variance—and therefore λ_0—can be estimated consistently only under K-asymptotics. Three consequences follow. First, in the limit, any candidate to optimal design policy must set K to infinity and keep N bounded; second, the asymptotic alignment between design and learning makes the $O(1/B^2)$ excess-variance rate attainable in a way that I make explicit in Section 5; and third, despite the asymptotic alignment, learning and design compete at finite budgets, as sampling additional clusters improves estimation of λ_0 but risks committing resources to width beyond what the oracle allocation would prescribe. I expand on the latter consequence below.

4.2 Pilot-Based Policies and Implementability

For the remaining of this paper, I focus on a policy class that I refer to as pilot-based policies. A pilot design or simply a pilot is a dataset collected during an initial stage with size (K_{pB}, N_p), where only the width of the pilot is allowed to vary with B. By contrast, the terminal design (K_B, N_B) is the total sample collected by the designer, including the initial pilot data.

Definition 4.2. *Pilot-Based Policies. A policy π is pilot-based if collects a pilot of (possibly random) size (K^π_{pB}, N^π_p), estimates $\hat{\lambda}_{pB} = \hat{\lambda}_{K^\pi_{pB}, N^\pi_p}$, and implements a terminal design (K^π_B, N^π_B) such that*

$$K^\pi_B = K^*(\hat{\lambda}_{pB}), \qquad N^\pi_B = N^*(\hat{\lambda}_{pB})$$

In words, in a pilot-based policy, the designer collects an initial set of observations that she uses for λ_0 estimation. She then computes the terminal design by applying the optimal map to the pilot estimate $\hat{\lambda}_p$. Crucially, *not all pilot-based policies can be run.* Clusters and units cannot be unsampled, so pilot observations must be retained in the terminal sample. Definition 4.3 introduces the notion of *implementability* and Lemma 4.4 summarizes a key property of implementable pilot-based policies.

Definition 4.3. *Implementability. A pilot-based policy π is implementable if its pilot is almost surely nested within its terminal design,*

$$K_{pB} \leq K_B^\pi = K^*(\hat{\lambda}_{pB}), \qquad N_p \leq N_B^\pi = N^*(\hat{\lambda}_{pB}) \qquad a.s.$$

Lemma 4.4. *Implementable Pilot-Based Policies with Diverging K Converge to the Oracle. Define the scaled width $\nu_B^\pi = K_B^\pi/B$, and the oracle constants $\nu_0 = K^*(\lambda_0)/B = (F + \sqrt{FV\lambda_0})^{-1}$ and $N_0 = N^*(\lambda_0)$. Let π be an implementable pilot-based policy with pilot design (K_{pB}^π, N_p^π), and suppose $K_{pB}^\pi \xrightarrow[B]{p} \infty$. Then, under Assumptions 3.1, 3.2 and 3.3, $\hat{\lambda}_{pB} \xrightarrow[B]{p} \lambda_0$, and therefore*[20]

$$|\nu_B^\pi - \nu_0| \xrightarrow[B]{p} 0, \qquad |N_B^\pi - N_0| \xrightarrow[B]{p} 0$$

Two remarks. First, the takeaway of Lemma 4.4 is that implementable pilot-based policies which sample a diverging number of clusters within the pilot induce the optimal design in the limit. Convergence to the oracle design, however, had to be established at two different scales. The oracle depth N_0 is budget-free, so convergence can be established at the unit scale. On the other hand, the oracle width $K^*(\lambda_0)$ is itself proportional to B, so the design and its target diverge together and only the budget-normalized ratio settles. Second, Lemma 4.4 concretizes for pilot based policies the tension between learning and optimal design implementation discussed in the previous section. Optimal design is realized by convergence of $\hat{\lambda}_{pB} \to \lambda_0$, which benefits from an asymptotically large number of clusters entering the pilot. However, the implementability requirement entering the lemma caps the pilot size by the very design the pilot produces $K^*(\hat{\lambda}_{pB})$. $\hat{\lambda}_{pB}$ is unavailable until the pilot has been sampled, so a designer who explores aggressively improves the precision of the λ_0 estimator but risks committing resources that the terminal design cannot absorb. The two algorithms below resolve this circularity in different ways—the first by making the pilot conservative enough that implementability can never bind, the second by monitoring the online estimator and stopping the moment the pilot is about to oversample.

4.3 Explore-then-Commit Cluster Selection

For convenience define the extreme widths and depths,

$$K_M = \lfloor K^*(\lambda_M) \rfloor, \qquad K_m = \lfloor K^*(\lambda_m) \rfloor, \qquad N_m = \lfloor N^*(\lambda_m) \rfloor, \qquad N_M = \lfloor N^*(\lambda_M) \rfloor \tag{4}$$

Since $K^*(\cdot)$ is strictly decreasing and $N^*(\cdot)$ strictly increasing in λ, K_M is the narrowest and N_m the shallowest design the optimal map can ever prescribe on $[\lambda_m, \lambda_M]$, while K_m is the widest and N_M the deepest. The simplest resolution to the circularity above is to choose a pilot small enough in both margins so that implementability holds regardless of what the pilot finds, i.e. the narrow-and-shallow

[20] Write $N_0 = \lfloor N_0 \rfloor$. If the terminal depth is rounded to an integer and $N_0 \notin \mathbb{N}$, the second statement is equivalent to $\mathbb{P}\big(\lfloor N_B^\pi \rfloor = N_0\big) \to 1$.

pilot $(K_{pB}, N_p) = (K_M, N_m)$. Since the clipped estimator satisfies $\hat{\lambda}_{pB} \in [\lambda_m, \lambda_M]$ by construction, the usual assumptions return

$$K_{pB} = K_M \leq K^*(\lambda_M) \leq K^*(\hat{\lambda}_{pB}) = K_B^\pi, \qquad N_p = N_m \leq N^*(\lambda_m) \leq N^*(\hat{\lambda}_{pB}) = N_B^\pi$$

and implementability holds almost surely. I refer to this two-stage policy as Explore-then-Commit Cluster Selection (`EtC-CS`).

Algorithm 1 `EtC-CS`

Input B, F, V, Ω

Sample N_m units from each of K_M clusters

observe $(y_{ki})_{i \leq N_m,\, k \leq K_M}$

compute $\hat{\lambda}_p = \hat{\lambda}_{K_M, N_m}$

Set $\left(K_B^{\texttt{EtC-CS}}, N_B^{\texttt{EtC-CS}}\right) = \left(K^*(\hat{\lambda}_p),\, N^*(\hat{\lambda}_p)\right)$

While theorists emphasize the superior performance of fully adaptive designs, practitioners often face logistical obstacles that prevent their deployment in the field (Chow and Corey, 2011). In that sense, the simplicity of `EtC-CS` is an asset: it attains rate-optimal performance guarantees (Theorem 5.4) with a single design update. Nevertheless, the reader may wonder what could be achieved were the designer given access to further adaptivity.

4.4 Adaptive Cluster Selection

Example 4.5. *Consider a researcher choosing a sampling width in* $[20, 60]$*, i.e. for the given budget,* $K_M = 20$ *and* $K_m = 60$*.* ***EtC-CS*** *would then collect a pilot of* 20 *clusters, compute* $\hat{\lambda}_p = \hat{\lambda}_{20,N_m}$*, and implement* $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$*. Suppose the design prescribed by the pilot is* $K^*(\hat{\lambda}_p) = 50$*. Committing immediately would be wasteful. The researcher knows she will recruit thirty more clusters in any case, so she could use some of them to improve her estimate of* λ_0 *before fixing the terminal design, at negligible risk of oversampling.*

Building on this heuristic, I propose an algorithm that samples clusters sequentially and commits according to a data-dependent stopping rule. I refer to this algorithm as Adaptive Cluster Selection (`ACS`), whose pilot width is determined by the stopping time

$$s = \inf\left\{k \in \mathbb{N} : k \geq K_M \;\text{ and }\; k + 1 > K^*(\hat{\lambda}_{k,N_m})\right\} \tag{5}$$

In words, the algorithm samples $N_p = N_m$ units from one additional cluster at a time, until recruiting one more cluster would constitute oversampling according to the online estimate of the variance ratio. Two observations on the exploration path induced by `ACS`. First, $s \in [K_M, K_m]$ almost surely.[21] Second, instead of fixing the depth and exploring only along the width margin, the designer could try to maximize the cost-adjusted marginal return to exploration by considering richer exploration paths. In particular, at each step she could sample an additional unit from a previously visited cluster or from a fresh cluster. Nonetheless, such a path is not only computationally intractable but also impractical to administer in the field. In fact, concentrating exploration along the width margin is the right simplification. Depth and width are not symmetric inputs to learning: increasing N_p lowers the

[21] The lower bound is immediate from the index set in (5). For the upper bound, suppose $s > K_M$, so that cluster s was actually explored; but then, the stopping condition must have failed at $k = s - 1$, giving $s \leq K^*(\hat{\lambda}_{s-1,N_m}) \leq K^*(\lambda_m)$. Since s is an integer, $s \leq K_m$. When $s = K_M$ the upper bound holds trivially under Assumption 3.3.

asymptotic variance of $\hat{\lambda}$ only through a constant, whereas increasing K_p improves the rate at which $\hat{\lambda}$ concentrates ($K_p^{-1/2}$ for any fixed N_p).[22] Therefore, holding N_p fixed and exploring along the width margin keeps the rule simple while preserving the convergence guarantees of Lemma 4.4.

Algorithm 2 `ACS`

Input B, F, V, Ω
Sample N_m units from each of K_M clusters
  observe $(y_{ki})_{i \le N_m,\, k \le K_M}$
  set $k = K_M$ and **compute** $\hat{\lambda}_k = \hat{\lambda}_{k,N_m}$
while $k+1 \le K^*(\hat{\lambda}_k)$
  sample N_m units from cluster $k+1$ and **observe** $(y_{k+1,i})_{i \le N_m}$
  update $k = k+1$ and **recompute** $\hat{\lambda}_k = \hat{\lambda}_{k,N_m}$
Set $s = k$ and **recover** $\hat{\lambda}_{\texttt{ACS}} = \hat{\lambda}_{\max\{s-1,\, K_M\},\, N_m}$
Set $\left(K_B^{\texttt{ACS}}, N_B^{\texttt{ACS}}\right) = \left(K^*(\hat{\lambda}_{\texttt{ACS}}),\ N^*(\hat{\lambda}_{\texttt{ACS}})\right)$

`ACS` opens with the same worst-case pilot as `EtC-CS`, then recruits clusters one at a time and stops just before oversampling according to the online estimate $\hat{\lambda}_k$. It therefore uses weakly more information than `EtC-CS` without compromising implementability. The main difference between the two algorithms is that while `EtC-CS` may recruit a large number of clusters after committing, `ACS` concentrates nearly all cluster exploration within the pilot. Since λ_0 estimation depends critically on the width margin—which is also the expensive dimension in practice—moving clusters into the pilot improves estimation of the optimal design at no additional sampling cost. The gain does not change the order of the excess-variance bounds derived in Section 5, but it is visible at small budgets in the simulations of Section 7. The improvement in performance is, however, not free: `ACS` requires the designer to revise the sample size after every cluster, so the applied designer must weigh the logistic inconvenience against the expected gain when choosing between the two.

Remark 4.6. *Committing One Cluster Early.* ***ACS*** *estimates λ_0 from $\max\{s-1, K_M\}$ clusters rather than all s. By definition of the stopping time, $s+1 > K^*(\hat{\lambda}_{s,N_p})$, and if cluster s was explored at all then $s \le K^*(\hat{\lambda}_{s-1,N_p})$. Together these give $K^*(\hat{\lambda}_{s,N_p}) - 1 < s \le K^*(\hat{\lambda}_{s-1,N_p})$, from which the implementability requirement $s \le K^*(\hat{\lambda}_{s,N_p})$ cannot be inferred—except when $s = K_M$, where it holds under Assumption 3.3. Committing at $\max\{s-1, K_M\}$ instead delivers an almost surely implementable design at the cost of one cluster's worth of information. For large pilots $\hat{\lambda}_{s,N_p} \approx \hat{\lambda}_{s-1,N_p}$, so the implications for the terminal design are immaterial by continuity of the optimal map, and at most one cluster is explored outside the pilot.*

4.5 Convergence to the Oracle Design

Both algorithms satisfy the premises of Lemma 4.4, hence, inducing the optimal design in the limit.

Corollary 4.7. ***EtC-CS*** *and* ***ACS*** *Converge to the Oracle Design. Both algorithms are pilot-based with fixed pilot depth $N_p = N_m \ge 2$ and pilot widths $K_{pB} = K_M$ for* ***EtC-CS*** *and $K_{pB} = \max\{s-1, K_M\}$ for* ***ACS****. Both are implementable—by construction for* ***EtC-CS****, and by committing one cluster early for* ***ACS*** *(Remark 4.6)—and both satisfy $K_{pB} \ge K_M \to \infty$ almost surely, since $K^*(\lambda)$ is monotonic in B*

[22] See Lemma E.15, where the asymptotic variance of $\sqrt{K}(\hat{\lambda}_{K,N_p} - \lambda_0)$ is finite and depends on N_p, but the rate does not.

for every fixed λ. The conditions of Lemma 4.4 therefore hold, and

$$\left(\nu_B^\pi, N_B^\pi\right) \xrightarrow[B]{p} (\nu_0, N_0), \qquad \pi \in \{\texttt{EtC-CS}, \texttt{ACS}\}$$

5 Excess Variance and Optimality

This section evaluates the sampling policies of Section 4 against the oracle. The performance criterion is the design excess variance of a policy Δ_π, sometimes referred to in the literature as *Neyman regret* (Dai et al., 2023): the difference between the variance of the estimator induced by the policy's design (K_B^π, N_B^π) and the oracle variance W_0,

$$\Delta_\pi = W\left(K_B^\pi, N_B^\pi\right) - W_0 \;\geq\; 0 \tag{6}$$

This criterion isolates the efficiency loss associated to the selection of a terminal design, while holding fixed the sampling variance associated with any given design. For adaptive policies, however, the design remains data-dependent, and so Δ_π is a random variable. As is natural in the learning literature, I evaluate policies by its expectation $\mathbb{E}[\Delta_\pi]$.[23] For static policies the design is non-random and the expectation is simply redundant. Every policy considered in this paper, adaptive or not, selects its terminal design by feeding a value of λ into the optimal map (3), whose excess variance can be written in closed form as

Lemma 5.1. *Excess Variance for Optimal Maps. Under Assumptions 3.1, 3.2, 3.3, any policy π which induces a terminal design $(K_B^\pi, N_B^\pi) = (K^*(\lambda), N^*(\lambda))$ for some input λ has an excess variance of the form*

$$\Delta_\pi = W\left(K^*(\lambda), N^*(\lambda)\right) - W_0 \;=\; \frac{\tau^2\sqrt{FV}}{B} \cdot \frac{\left(\sqrt{\lambda} - \sqrt{\lambda_0}\right)^2}{\sqrt{\lambda}} \qquad \forall \lambda \in [\lambda_m, \lambda_M] \tag{7}$$

Two features of (7) are worth discussing. First, the cost of misspecification enters through the square of a deviation because the oracle satisfies the first-order condition at λ_0. Misspecification errors therefore affect variance only to second order on λ. Second, the entire expression carries a factor $1/B$, so any policy whose input is a fixed distance from λ_0 inherits the order of the oracle variance itself. It immediately follows that any policy which imports a misspecified guess from the literature, including the minimax optimal, will incur excess variance of order $\Theta(1/B)$. By contrast, by inputting a converging estimator $\hat{\lambda}_p$ into the optimal map, the learning algorithms `EtC-CS` and `ACS` improve substantially compared to the static benchmark. In particular, I derive upper bounds on their expected excess variance of order $1/B^2$—a magnitude of B below the variance of the oracle itself—, hence asymptotically negligible. Finally, I prove that no policy can improve on the rate of the learning algorithms by deriving matching lower bounds of $\Omega(1/B^2)$ on a generalized problem, so the algorithms are also rate optimal. Table 2 summarizes these rates.

[23] The expectation is taken over the filtration generated by the interaction between the policy and the environment. See Lattimore and Szepesvári (2020) for a textbook treatment.

Policy	Metric	Rate	Result
Oracle	Variance	$\Theta(1/B)$	Equation (3)
Misspecified static policies including minimax	Excess variance	$\Theta(1/B)$	Lemma 5.2
`EtC-CS` and `ACS`	Excess variance	$O(1/B^2)$	Theorem 5.4
Minimax lower bound	Excess variance	$\Omega(1/B^2)$	Theorem 5.5

Table 2: Performance rates

Notes: Results hold under the usual Assumptions 3.1, 3.2 and 3.3, and the additional subgaussian Assumption 5.3.

5.1 Static Policies and the Cost of Guessing

It is common practice to insert an ICC guess from the literature or a previous field survey into the optimal map. Figure 1 illustrates the risks of blind imputation by plotting the optimal map $(K^*(\lambda), N^*(\lambda))$ and the excess variance as a function of λ for different F/V ratios. The excess variance presents a characteristic U-shape with a minimum at $\lambda = \lambda_0$. Lemma 5.2 quantifies the risk by locating the excess variance of these policies at a rate of $\Theta(1/B)$. A second approach, developed by van Breukelen and Candel,[24] replaces the guess with a minimax compromise over the parameter space which also incurs excess variance of order $\Theta(1/B)$. The takeaway is that ICC misspecification is a first-order concern which is as costly as the sampling noise the design sets out to minimize.

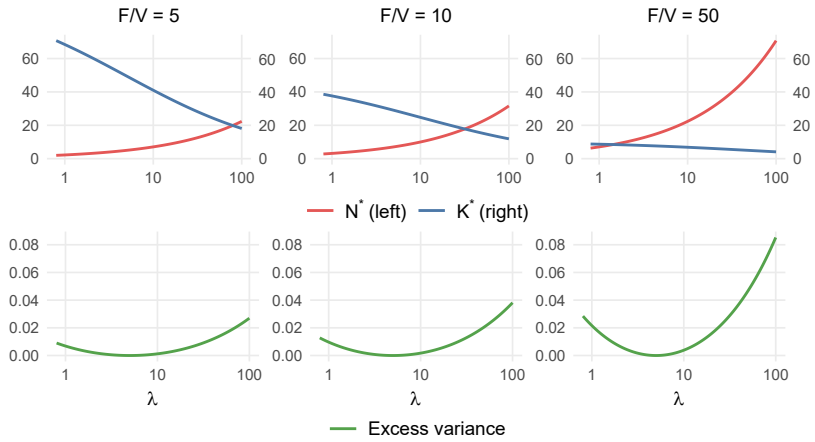


Figure 1: Design and excess variance under λ misspecification

Notes: Log-scale for λ. $\mu = 0$, $\omega = (1, 5)$, $B = 1000, V = 2$.

Lemma 5.2. *Misspecified and Minimax-Optimal Static Policies. Let $\lambda_0, \lambda_b \in [\lambda_m, \lambda_M]$ with $\lambda_b = \lambda_0 + b$. Under Assumptions 3.1, 3.2 and 3.3,*

(i) Any static input. *Define the static policy $\pi_b : (K_B^{\pi_b}, N_B^{\pi_b}) = (K^*(\lambda_b), N^*(\lambda_b))$. Then, the excess variance of π_b satisfies*

$$\frac{\tau^2\sqrt{FV}}{4\lambda_M^{3/2}} \cdot \frac{b^2}{B} \leq \Delta_{\pi_b} \leq \frac{\tau^2\sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{b^2}{B} \qquad \Delta_{\pi_b} = \Theta(1/B) \;\; \forall b \neq 0$$

(ii) The minimax-optimal input. *Let*

$$\lambda_{mM} = \underset{\lambda \in [\lambda_m, \lambda_M]}{\arg\min} \max_{\omega = (\tau^2, \lambda_0) \in \Omega} \Big[W\big(K^*(\lambda), N^*(\lambda); \omega\big) - W\big(K^*(\lambda_0), N^*(\lambda_0); \omega\big) \Big]$$

[24] See van Breukelen and Candel (2015, 2018, 2021).

Define the minimax-optimal policy $\pi_{mM} : (K^*(\lambda_{mM}), N^*(\lambda_{mM}))$. Then, the minimax input and its worst-case excess variance are given by

$$\lambda_{mM} = \left(\frac{\sqrt{\lambda_m} + \sqrt{\lambda_M}}{2}\right)^2, \qquad \max_{\omega\in\Omega} \Delta_{\pi_{mM}} = \frac{\tau_M^2\sqrt{FV}}{2B} \cdot \frac{\left(\sqrt{\lambda_M} - \sqrt{\lambda_m}\right)^2}{\sqrt{\lambda_m} + \sqrt{\lambda_M}} \;=\; \Theta(1/B)$$

Part (ii) is worth emphasizing. Optimizing over the static class does not change the order of the penalty, it only pins down the constant. This constant is governed by width of the designer's prior uncertainty ($\sqrt{\lambda_M} - \sqrt{\lambda_m}$), such that more uncertain designers pay a higher price for their lack of adaptivity. $\Theta(1/B)$ is the benchmark rate to improve upon.

5.2 Upper Bounds for the Learning Algorithms

I now show that learning improves on the static benchmark by a full order of B. Under some basic algebra, evaluating (7) at the pilot estimate and using $\omega \in \Omega$,

$$\Delta_\pi \;=\; \frac{\tau^2\sqrt{FV}}{B} \cdot \frac{\left(\sqrt{\hat{\lambda}_{pB}} - \sqrt{\lambda_0}\right)^2}{\sqrt{\hat{\lambda}_{pB}}} \leq \frac{\tau_M^2\sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{\left(\hat{\lambda}_{pB} - \lambda_0\right)^2}{B} \tag{8}$$

so the problem immediately reduces to bounding $\mathbb{E}\left[(\hat{\lambda}_{pB} - \lambda_0)^2\right]$. To that end, I impose a standard subgaussianity condition, which delivers sufficient concentration of the input estimators.[25]

Assumption 5.3. *Errors Subgaussianity. u_k is κ_u-subgaussian and ε_{ki} is κ_ε-subgaussian.*

Theorem 5.4. *Expected Excess Variance of `EtC-CS` and `ACS`. Under Assumptions 3.1, 3.2, 3.3 and 5.3, there exist finite constants $C_{\texttt{EtC-CS}}$ and $C_{\texttt{ACS}}$, independent of B, such that*

$$\mathbb{E}\left[\Delta_{\texttt{EtC-CS}}\right] \leq \frac{C_{\texttt{EtC-CS}}}{B^2}, \qquad \mathbb{E}\left[\Delta_{\texttt{ACS}}\right] \leq \frac{C_{\texttt{ACS}}}{B^2}$$

The Theorem implies that the ICC can be learned only at second-order cost as both `EtC-CS` and `ACS` incur a penalty that vanishes relative to the oracle variance itself. As anticipated in Section 4, `ACS` delivers superior average performance in simulations, since it exploits a weakly larger set of clusters in the estimation of λ_0. However, this does not translate into an improved rate because the two pilot widths K_{pB} are of the same order. They both lie in $[K_M, K_m]$, an interval whose endpoints are each $\Theta(B)$, so $K_{pB}^{\texttt{ACS}}/K_{pB}^{\texttt{EtC-CS}} = O(1)$ almost surely.[26]

Why $1/B^2$? The result rests on a $1/B \times 1/B$ mechanism. The first term follows directly from the order of the oracle stationarity itself, discussed in (7). The second term is nonetheless fortuitous and relates directly to the partial alignment argument between learning and optimal design implementation presented earlier. Mathematically, Lemma C.5 shows that both algorithms satisfy $\mathbb{E}\left[(\hat{\lambda}_{pB} - \lambda_0)^2\right] \leq C/K_M$ for some constant C. Since $K_M = \Theta(B)$, the result follows, but this observation requires further attention. While setting $K_{pB} = \Theta(B)$ is unequivocally good for estimation, the compatibility

[25] Analogous results hold under weaker conditions such as subexponential tails of the errors.

[26] Recovering the rate is technically more demanding under sequential sampling, since the deviations must be bounded for a pilot of random width, rather than deterministic. Appendix C circumvents this technicality by noting that $\hat{\lambda}_k - \lambda_0$ can be decomposed asymptotically as a function of martingales—which are amenable to a maximal version of Bernstein's inequality—and that K_m/K_M is bounded uniformly in B.

between estimation and optimal design remains surprising. Were the optimal design to require only $\Theta(1)$ clusters, a pilot of width $\Theta(B)$ would be infeasible, consistent learning of λ_0 could fail, and the B^{-2} rate would no longer follow.

5.3 Lower Bound

Finally, I show that the two algorithms are essentially unimprovable by deriving a matching lower bound of order $\Omega(1/B^2)$. In particular, for every sampling policy π there exists an environment $\omega \in \Omega$ in which it incurs excess variance of that order. `EtC-CS` and `ACS` are therefore minimax rate optimal in the joint (K, N) design problem, and can be improved by no more than a constant factor. The bound is derived for the generalized problem of Appendix A, in which the learner may sample a different number of units N^k from each cluster and revisit clusters at will.

Theorem 5.5. *Lower Bound on Expected Excess Variance. There exist constants $C > 0$ and $B_0 < \infty$, not depending on B or the policy, such that for every $\pi \in \Pi$ and every $B \geq B_0$,[27] there exists a data generating process indexed by $\omega \in \Omega$ such that*

$$\mathbb{E}_\pi[\Delta_\pi] \;\geq\; C/B^2$$

The proof proceeds in three steps. *First*, I work with an easier version of the problem in which the cluster error u_k is revealed to the learner the first time she visits cluster k. In the cluster setting, observations within a cluster are correlated, so once a cluster has been visited its subsequent observations are not independent conditional on the history. Revealing u_k on the first visit avoids this nuisance and restores the conditional independence on which standard divergence-decomposition arguments rely.[28] This "augmented experiment" gives the learner weakly more information than the benchmark experiment, so any lower bound established there carries over to the original problem. *Second*, within the augmented environment I consider two data generating processes y_- and y_+ of the form (1), indexed by

$$\omega_- = (\tau_-^2, \lambda_-), \qquad \lambda_- = \sigma^2/\tau_-^2, \qquad \omega_+ = (\tau_+^2, \lambda_+), \qquad \lambda_+ = \sigma^2/\tau_+^2$$

The two environments share the same mean μ and the same unit-level variance σ^2, and so they only differ in the cluster-level variance τ^2. For a small $\delta_B > 0$, the cluster variances are chosen so that $\lambda_- = \lambda - \delta_B$ and $\lambda_+ = \lambda + \delta_B$. Recall from Lemma 5.2 that optimizing against the wrong instance is costly; for instance, if implementing $(K^*(\lambda_+), N^*(\lambda_+))$ when the truth is ω_-, the designer would have implemented a design too narrow and too deep compared to the oracle. See Figure 2.

[27] The policy class Π is defined in Appendix C. The restriction excludes only policies that fail to terminate in finite time or violate the budget constraint with positive probability. Policies in Π may sample a different number of units from different clusters, revisit clusters freely, and sample a different number of units on each visit.

[28] See Chapter 15 in Lattimore and Szepesvári (2020).

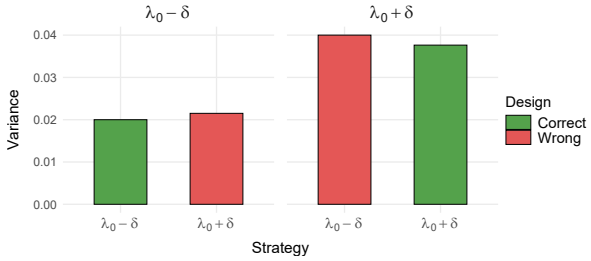


Figure 2: Optimal policies across environments (y_+, y_-).

Notes: $W(K^*(\cdot), N^*(\cdot))$ under ω_- (columns 1, 2) and ω_+ (columns 3, 4) for policy $(K^*(\lambda_-), N^*(\lambda_-))$ (columns 1, 3) and policy $(K^*(\lambda_+), N^*(\lambda_+))$ (columns 2, 4). $\mu = 0$, $\omega = (1 \pm \delta, 5/(1 \pm \delta))$, $\delta = 0.5$, $B = 2000$, $F = 20$, $V = 2$.

Third, δ_B is chosen to balance the trade-off between identification and punishment. A large δ_B makes the two environments easy to tell apart, so the learner optimizes correctly most of the times, but is penalized heavily when she does not. Conversely, a small δ_B makes them hard to distinguish, but shrinks the cost of choosing the wrong design. The two forces are balanced at $\delta_B \asymp B^{-1/2}$, as is customary in learning problems, at which point the resulting bound is maximized at order $1/B^2$.

6 Inference

Adaptive data collection generally invalidates conventional inference. Hence, the reduction in variance could end up being of limited use if it cannot be paired with suitable inference. This section shows that, under a symmetry restriction on the errors, there is no trade-off between adaptivity—thus, variance reduction—and inference. Inference on population means remains invariant after conditioning on designs induced by *even* policies, and remains asymptotically invariant after conditioning on designs induced by *asymptotically even* policies. Informally, a policy is even if its terminal designs depend on the data only through even functionals of the errors, such as squared errors; and asymptotically even if its terminal designs are close to the ones generated by even policies, as happens with policies driven by sample variances—which are close to policies driven by population variances. When the design is of either kind, group averages remain asymptotically normal under symmetry, with no correction and no exclusion of pilot data. This principle covers both `EtC-CS` and `ACS` as they are functions of sequences of (sample) variance ratios.

Why design-conditional inference? I study inference conditional on the realized design, rather than averaged over counterfactual designs for four reasons. First, it is the notion of validity that standard practice already implicitly adopts. A fixed (K, N) design is typically the outcome of a deliberation over alternatives, yet it is treated as non-random. The conditioning is innocuous when the selection draws only on information independent of the sample, but it does when the design depends on the data, as it is the case with adaptation. Design-conditional validity therefore places adaptive and static designs on the same inferential footing. Second, design-conditional inference is the honest guarantee. A reader who sees an interval reported alongside the realized design cares whether that interval covers the target for the experiment actually run, and not over experiments that never took place. An interval with correct unconditional coverage can systematically under-cover some realized designs and over-cover others. Third, the designs induced by `EtC-CS` and `ACS` remain genuinely random at the relevant scale, so the conditioning events are non-degenerate and the conditional statement has content (Section 6.3). Finally, design-conditional validity is the stronger result: averaging over designs

recovers unconditional validity, and so the analyst is covered whichever notion of coverage they adopt. The price of this strengthening is a symmetry restriction on the errors, which is mild compared to the Gaussianity assumption invoked in the classical two-stage and internal-pilot literatures.

The section proceeds in three steps. Section 6.1 first discusses the Gaussian benchmark, where the result is exact and finite-sample. Section 6.2 weakens Gaussianity into symmetry and establishes an asymptotic result for exactly even designs at every design scale. Section 6.3 extends the results to the feasible class, and identifies the unique scale at which conditioning events are both well defined and non-degenerate. The exposition is mostly confined to the cluster design problem, but connections to the general theory in Appendix D, which applies well beyond clustered sampling, are made throughout.

6.1 Exact Design Independence under Gaussian Errors

It is well known that, under exact Gaussianity, sample averages and the vector of sample residuals are independent. As such, this result has been exploited repeatedly in the sample size re-estimation and the internal pilot literatures.[29] It follows that any design which depends on the data only through the residuals is independent of the sample average, and so design-conditional inference reduces to unconditional inference at every finite sample size.[30] Policies depending on sample variances would just be a subset of this policy class, thus, this result covers both `EtC-CS` and `ACS`, which are functions of sequences of sample variance ratios. Mathematically, the relevant policy class can be characterized as the set of policies which are measurable with respect to the stopped σ-algebra, where the stopping time depends only on past residuals.[31]

Proposition 6.1. *Exact Conditional Normality under Gaussian Errors. Under Assumptions 3.1, 3.2 and 3.3, let $u_k \sim N(0,\tau^2)$ and $\varepsilon_{ki} \sim N(0,\sigma^2)$. Then, for $\pi \in \{\texttt{EtC-CS}, \texttt{ACS}\}$, every budget B, and every design event $A_B \in \mathscr{P}^\pi$ with $\mathbb{P}(A_B) > 0$,*

$$\mathcal{L}\left(\frac{\sqrt{K_B^\pi}\big(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu\big)}{\theta_{N_B^\pi}} \;\middle|\; A_B \right) = N(0,1) \quad \textit{exactly}$$

Three notes on Proposition 6.1. First, exactness holds only for the infeasible standardization $\theta_{N_B^\pi}$. When replaced by the sample standard deviation $\hat{b}_{K_B^\pi, N_B^\pi}$, the theory requires an asymptotic argument because the numerator and denominator are computed from the same cluster means. Second, the argument relies on the pilot being nested within the terminal design—$K_{pB} \leq K_B^\pi$ and $N_p \leq N_B^\pi$—, guaranteed under implementability. Nesting makes the terminal average orthogonal to every residual generating the stopped algebra, so that pilot observations need not be discarded. Third, the proposition delineates what Gaussianity can and cannot buy. Under Gaussianity the invariant class is large. The design may depend arbitrarily on the residuals—not just through the sample variances but also through any *odd* functional of the residuals—and still deliver exact conditional inference at every budget. When Gaussianity is weakened to symmetry, the invariance policy class also shrinks. The conditioning σ-algebra contracts from the residuals to their magnitudes, and so the policy class contracts from any residual-driven policy to even policies. What no assumption rescues are policies that condition on the

[29] See Stein (1945); Dudewicz and Dalal (1975); Rinott (1978) for the former and Wittes and Brittain (1990); Wittes et al. (1999); Lake et al. (2002); van Schie and Moerbeek (2014) for the latter.

[30] Equivalently, the sample average is sufficient for μ while the residual vector is ancillary, so independence also follows from Basu's theorem. Joint Gaussianity delivers it directly through zero covariance.

[31] Appendices D.2 and E.1 collect the formal definitions of the residual σ-algebra R_j and the stopped σ-algebras $\mathscr{P}^{\texttt{EtC-CS}}$ and $\mathscr{P}^{\texttt{ACS}}$.

level of the data rather than on demeaned outcomes. A rule that reads the pilot mean—such as the naive policy of Section 7—conditions on the direction removed through demeaning; so its design events lie outside the invariance class, and conditional normality fails even under Gaussianity (Figure 8).

6.2 Inference for Even Designs under Symmetry

Exact Gaussianity is a strong restriction. By reducing the invariance policy class from functionals of the residuals to (asymptotically) even functionals of the data, I can significantly weaken the distributional assumption on the errors: from Gaussianity to symmetry. Moreover, my results are robust to studentization.

Assumption 6.2. *Error Symmetry. Let* $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$. *Then* $C_k \overset{d}{=} -C_k$.

Because of independence of u_k and ε_{ki}, a simple sufficient alternative for Assumption 3.1 is that u_k and ε_{ki} are each symmetric about zero. The symmetry assumption is familiar and comparatively benign, as it underlies most of sign and permutation tests, as well as symmetry-based randomization inference (Canay et al., 2017). Additionally, it is far weaker than the parametric assumptions practitioners routinely impose in power calculations, and it is partially testable.[32] The restriction however excludes important error distribution families such as binary and count outcomes, as well as censored and one-sided distributions.

As hinted in the introduction, Assumption 6.2 permits a decomposition of the errors into independent signs and magnitudes. Conditioning on any event that depends on the magnitudes alone—such as events generated by even functionals of the data—leaves the signs unrestricted, so that a central limit theorem can run through them independently. Formally,

Lemma 6.3. *Cluster Sign Factorization. Fix* N *and let* $(\zeta_k)_k$ *be an iid Rademacher sequence independent of* $\big((C_k)_k, (U_k)_k\big)$. *Define*

$$M_{k,N} = u_k + \frac{1}{N}\sum_{i=1}^{N} \varepsilon_{k,i}, \qquad \chi_{k,N} = \mathbb{1}(M_{k,N} \neq 0)\,\mathrm{sign}(M_{k,N}) + \mathbb{1}(M_{k,N} = 0)\,\zeta_k, \qquad R_{k,N} = |M_{k,N}|$$

so that $M_{k,N} = \chi_{k,N} R_{k,N}$, *where* $M_{k,N}$ *is the average error of cluster* k. *Under Assumptions 3.1 and 6.2,* $(\chi_{k,N})_k$ *is iid Rademacher and* $(\chi_{k,N})_k \perp\!\!\!\perp (R_{k,N})_k$.

Corollary 6.4. *Signs are Independent of the Even* σ*-algebra. For a fixed* N_p *and an iid sequence* $(U_k)_k \sim \mathrm{Unif}[0,1]$ *independent of* $\big((C_k)_k, (\zeta_k)_k\big)$, *define the even* σ*-algebra*

$$\mathcal{G}_\infty := \sigma\Big(\big(S^2_{k,N_p}\big)_{k\geq 1},\ \big(M^2_{k,N}\big)_{k\geq 1,\, N\geq 1},\ (U_k)_{k\geq 1}\Big)$$

Then, under Assumptions 3.1 and 6.2, for every fixed $N \geq 1$ *the sign sequence* $(\chi_{k,N})_{k\geq 1}$ *is iid Rademacher and independent of* $\mathcal{G}_\infty$, *hence independent of every event* $A \in \mathcal{G}_\infty$.

Next, I introduce the class of policies which are measurable with respect to the even σ-algebra.

Definition 6.5. *A policy* π *is*

(i) ***Even*** *if the terminal design* (K_B^π, N_B^π) *is* $\mathcal{G}_\infty$*-measurable;*

[32] Section 8 implements a symmetry test on the average cluster errors $M_{k,N} = u_k + \frac{1}{N}\sum_{i=1}^{N} \varepsilon_{ki}$ using the data of Fink et al. (2020b).

(ii) ***Non-starving*** *if* $\mathbb{P}(\nu_B^\pi \geq c) \to 1$ *for some constant* $c > 0$*;*

(iii) N***-consistent*** *if* $N_B^\pi \xrightarrow[B]{p} N_{\underline{0}} = \lfloor N^*(\lambda_0) \rfloor \in \mathbb{N}$.

Evenness is the substantive requirement: an even policy induces design events in $\mathcal{G}_\infty$, so conditioning on the even σ-algebra them freezes the design and the magnitudes but leaves the signs unrestricted. The other two conditions are technical but do real work. Any asymptotic argument will require a divergent sequence, which is guaranteed by non-starvation. As such, it is also present in the general theory of Appendix D. The linear form of the divergence is immaterial for exactly even policies, but becomes necessary when transferring results to the feasible class. N-consistency is specific to the cluster problem: it stabilizes the depth margin so that the analysis can proceed along K-asymptotics alone. By Corollary 4.7, `EtC-CS` and `ACS` are both non-starving and N-consistent under flooring—but neither is exactly even.

Proposition 6.6. *Uniform Mixing for Even Designs. Let* $\big((K_B^\pi, N_B^\pi), (A_B)_B\big)$ *be a design and a sequence of events induced by an even, non-starving,* N*-consistent policy. Define the scaled sum*

$$Z_B^\pi = \sqrt{K_B^\pi}\,\big(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu\big) = \frac{1}{\sqrt{K_B^\pi}} \sum_{k=1}^{K_B^\pi} M_{k, N_B^\pi}$$

and the conditional characteristic function as $\phi_B(t) = \mathbb{E}\big[\exp(itZ_B^\pi) \mid \mathcal{G}_\infty\big]$*. Write* $\phi(t) = \exp(-\theta_{N_{\underline{0}}}^2 t^2/2)$ *for the Gaussian characteristic function with oracle depth* $N_{\underline{0}}$*. Then, under Assumptions 3.1, 3.2, 3.3 and 6.2,*

$$\Big|\mathbb{E}\big[\phi_B(t)\,\mathbb{1}_{A_B}\big] - \phi(t)\,\mathbb{P}(A_B)\Big| \to 0 \qquad \forall t \in \mathbb{R},$$

uniformly over $A_B \in \mathcal{G}_\infty$*. Moreover, for any such sequence with* $\liminf_B \mathbb{P}(A_B) > 0$*,*[33]

$$\mathcal{L}\big(Z_B^\pi \mid A_B\big) \implies N\big(0, \theta_{N_{\underline{0}}}^2\big), \qquad \mathcal{L}\left(\frac{Z_B^\pi}{\hat{b}_{K_B^\pi, N_B^\pi}} \,\middle|\, A_B\right) \implies N(0,1)$$

Proposition 6.6 displays a very strong mode of convergence: the scaled sums converge $\mathcal{G}_\infty$-uniformly mixing. The implication is that asymptotic normality survives conditioning on any $\mathcal{G}_\infty$-measurable event, at any scale, terminal or not. Nonetheless, this result is of limited use on its own as the policy class—policies which have access to the true squared errors—is very thin. At best, policies may sample according to sequences of nearly even statistics such as sample variances. Recall that sample variances demean the observations by the sample average, so a policy depending on a sequence of sample variances could induce designs depending on the relative signs, hence not in $\mathcal{G}_\infty$.[34] The next section shows that the results extend to the feasible class, once the set of conditioning events is restricted to the appropriate scale.

[33] For exactly even designs, symmetry buys more than asymptotic normality. Conditional on $\mathcal{G}_\infty$ the signs $(\chi_{k,N})_k$ are *iid* Rademacher (Corollary 6.4), so sign-flip randomization on the centered cluster means yields an exact finite-sample test of $H_0 : \mu = \mu_0$, in the spirit of symmetry-based randomization inference (Canay et al., 2017). This finite-sample property however does not extend to feasible policies. `EtC-CS` and `ACS` are only asymptotically even, so for them the sign-flip test is only asymptotically valid, just like the normal approximation pursued here.

[34] Concretely, for a simple model $X_s = \mu + \varepsilon_s$, the sample variance can be written as $\hat{\sigma}_m^2 = \frac{m-1}{m^2}\sum_{s\leq m}\varepsilon_s^2 - \frac{2}{m^2}U_m$, where $U_m = \sum_{1\leq i<j\leq m}\varepsilon_i\varepsilon_j$. For $m = 2$, $U_2 = |\varepsilon_1\varepsilon_2|\text{sign}(\varepsilon_1)\text{sign}(\varepsilon_2)$ which is clearly not even. Therefore, a policy depending on $\hat{\sigma}_2^2$ may induce designs outside the even σ-algebra. The leakage is nonetheless asymptotically negligible relative to the even component, which is what licenses the notion of asymptotic evenness below. See Appendix D for the general statement.

6.3 Inference for Asymptotically Even Designs under Symmetry

Definition 6.7. *Asymptotically Even Policies. Let* $K_{0B} = K^*(\lambda_0)$ *and define* $\Xi_B^\pi = (K_B^\pi - K_{0B})/\sqrt{B}$. *A policy* π *is asymptotically even if there exists a* $\mathcal{G}_\infty$*-measurable pair* $(\tilde{K}_B, \tilde{N}_B)$*, with* $\tilde{\Xi}_B = (\tilde{K}_B - K_{0B})/\sqrt{B}$*, such that*

$$\left|\Xi_B^\pi - \tilde{\Xi}_B\right| \xrightarrow[B]{p} 0, \qquad \left|N_B^\pi - \tilde{N}_B\right| \xrightarrow[B]{p} 0$$

In essence, a policy is asymptotically even if its design is close in probability to one induced by an even policy, with closeness measured on the $\sqrt{B}$ scale for width and on the unit scale for depth. Observe that the definition places no restriction on the surrogate. The even design $(\tilde{K}_B, \tilde{N}_B)$ need be neither feasible nor related to π in any substantive way. However, in practice, the analyst may construct the even surrogate by replacing the squared residuals entering π with the true squared errors $M_{k,N}^2$, and then establishing convergence of the two designs at the relevant scale. The transition from exact to asymptotic evenness restricts the class of admissible conditioning events to the set of events that agree asymptotically under the two policies—in the sense that $\mathbb{P}(A_B \triangle \tilde{A}_B) \to 0$. When events are expressed on the same scale as the convergence of the designs, survival is warranted under a mild margin condition. The natural conditioning class is then given by

$$(A_B)_B : A_B = \left\{(\Xi_B^\pi, N_B^\pi) \in C\right\}, \qquad C \subseteq \mathbb{R} \times \mathbb{N} \text{ Borel} \tag{9}$$

Which scale should the analyst condition on? Three scales are *a priori* available in the width dimension: unit, $1/\sqrt{B}$ and $1/B$. For exactly even policies the choice is immaterial. By Proposition 6.6, any $\mathcal{G}_\infty$ event at any scale leaves the limiting law of Z_B^π unchanged. For asymptotically even policies the choice is decisive. K_B^π is proportional to B and Lemma E.11 shows that it fluctuates over a window of width $O_p(\sqrt{B})$ around the oracle, therefore the feasible and surrogate designs cannot couple at the unit scale, $\mathbb{P}(A_B \triangle \tilde{A}_B) \not\to 0$ and inference results do not transfer. At the scale of $1/B$, conditioning is possible but asymptotically vacuous. By Lemma 4.4, $\nu_B^\pi \xrightarrow[B]{p} \nu_0$, so $\mathbb{P}(\nu_B^\pi \in C) \to \mathbb{1}\{\nu_0 \in C\} \in \{0, 1\}$ for every C whose boundary ν_0 avoids, and the conditional law degenerates to the unconditional one. At that scale, a random-index central limit theorem in the spirit of Anscombe (1952) and Chow and Robbins (1965) suffices without ever invoking symmetry or policy symmetry. At the intermediate $1/\sqrt{B}$ scale, the designer can obtain both: the coupling between feasible and surrogate designs survives and the conditioning events remain non-degenerate. $1/\sqrt{B}$-scale events are particularly relevant as they include events involving quantiles of the realized width. These events lie outside the reach of Anscombe (1952), and so require the special treatment discussed in this manuscript. Table 3 summarizes the scale discussion.

Scale	Does Surrogate Coupling Survive?	Non-degenerate Events?	Tools
Unit	No	–	–
$1/B$	Yes	No	Anscombe
$1/\sqrt{B}$	Yes	Yes	Symmetry and Evenness

Table 3: Scale of conditioning events

Theorem 6.8. *Inference for Asymptotically Even Designs. Let* π *be an asymptotically even, non-starving,* N*-consistent policy, and let* $(A_B)_B$ *be a sequence of events as in* (9) *satisfying the margin*

condition

$$\lim_{\eta \downarrow 0} \limsup_{B} \mathbb{P}\Big(\operatorname{dist}\big((\Xi_B^\pi, N_B^\pi), \partial C\big) \le \eta\Big) = 0 \tag{10}$$

Then, under Assumptions 3.1, 3.2, 3.3 and 6.2,[35]

$$\Big|\mathbb{E}[\exp(itZ_B^\pi)\mathbb{1}_{A_B}] - \phi(t)\mathbb{P}(A_B)\Big| \to 0 \quad \forall t \in \mathbb{R}$$

If in addition $\liminf_B \mathbb{P}(A_B) > 0$*, then*

$$\mathcal{L}\left(Z_B^\pi \mid A_B\right) \implies N(0, \theta^2_{N_0}), \qquad \mathcal{L}\left(\frac{Z_B^\pi}{\hat{b}_{K_B^\pi, N_B^\pi}} \,\middle|\, A_B\right) \implies N(0,1)$$

Condition (10) is an anti-concentration requirement on the choice of the set C. The scaled design must not pile up on the boundary of the conditioning set, since it is there that the feasible and surrogate designs can disagree. In practice this condition is mild, and follows from a central limit theorem—requiring finite fourth moments of the errors—for the statistic governing the design, $\hat{\lambda}$ in the context of the cluster design problem.

It remains to verify that the two algorithms are asymptotically even. Propositions E.12 and E.13 do so constructively, building surrogates `EtC̃-CS` and `AC̃S` that replace the nearly even statistic $\hat{b}^2_{K,N}$ with its exactly even counterpart $K^{-1}\sum_{k\le K} M^2_{k,N}$.[36] For `EtC-CS` the coupling is immediate: the design depends on the single statistic $\hat{\lambda}_p$ and its two versions differ by $O_p(1/B)$, so the induced widths differ by $O_p(1)$—an order below the required $o_p(\sqrt{B})$. `ACS` is more delicate. Replacing residuals by errors would still leave the analyst with the difficult task of coupling two λ_0 estimates evaluated at two different random stopping times. I avoid the difficulty by giving the surrogate a deterministic stopping rule, $\tilde{s} = K_{0B}$, localizing the `ACS` stopping time in an $O_p(\sqrt{B})$ window around K_{0B}, and applying a maximal inequality.

Theorem 6.9. *Design-Robust Inference for `EtC-CS` and `ACS`. Let* (K_B^π, N_B^π) *be the design induced by* $\pi \in \{$`EtC-CS`, `ACS`$\}$*, and let* $A_B = \{(\Xi_B^\pi, N_B^\pi) \in C\}$ *for any* C *satisfying* (10) *and* $\liminf_B \mathbb{P}(A_B) > 0$*. Then, under Assumptions 3.1, 3.2, 3.3 and 6.2,*

$$\mathcal{L}\left(\frac{Z_B^\pi}{\hat{b}_{K_B^\pi, N_B^\pi}} \,\middle|\, A_B\right) \implies N(0,1)$$

The takeaway from Theorem 6.9 is that *nothing needs to change.* The estimator is the ordinary average of cluster averages computed on the full sample—pilot observations included—the standard error is the usual clustered one, and the normal approximation is valid conditional on the realized design. No reweighting, no sample splitting, and no discarding of exploration data is required. A full array of simulations verify this prescription.

The boundary of the result. Policy evenness and error symmetry are not merely sufficient. The simulation evidence shows that relaxing either can destroy the invariance. A policy that adapts through

[35] Clearly, the uniform mixing statement cannot be established for Z_B^π, as this scaled sum is not even $\mathcal{G}_\infty$ measurable. Yet, a qualitatively similar result holds pointwise along the given sequence of events. In particular, the two conditional limit laws, which are what an applied analyst uses, carry over unchanged. See Appendix E for a complete treatment.

[36] The within-cluster variance $\hat{w}^2_{K,N}$ is both even and feasible, so it requires no adjustment.

odd functionals of the data—estimated treatment effects, interim averages, futility rules—leaks information about the signs and breaks the sign–magnitude factorization of Lemma 6.3. For such policies the design-conditional mean of Z_B^π need not vanish, and the pathologies documented by Nie et al. (2018) and Hadad et al. (2021) reappear, along with the necessity to introduce the reweighting corrections developed in those papers. Similarly, error asymmetry allows exactly even policies to pick up information about the error signs, breaking, once again, the sign–magnitude factorization. Section 7 makes the boundary concrete by assessing Theorem 6.8 in a grid which interacts policy evenness and error symmetry.

6.4 Beyond Clustered Sampling

Nothing in the preceding argument is specific to clusters. Stripped to its essentials, the mechanism only requires units to be sampled adaptively from a collection of populations, errors to be symmetric, and the design to read even functionals of the data. In fact, these principles can be captured through a much more general theory of inference invariance that is presented as a standalone technical note in Appendix D. The leading application of this theory is the oldest design rule in the book: Neyman allocations with plug-in variances. At a high level, a designer would like to estimate a set of populations means, where each population has a different variance. The designer could then estimate the variance of each arm through a uniform pilot, and then allocate the remaining observations proportionally to the estimated standard deviations. The second-stage of the design is driven by sample variances, so the policy is asymptotically even and the general theory applies off the shelf.

Corollary 6.10. *Two-Stage Neyman Allocations (Informal). Imagine a population with K arms, each containing symmetric iid errors, a strictly positive variance, and finite fourth moments. Consider the two-stage implementable design that samples all arms uniformly over a fixed share of the horizon and then sets the terminal counts proportional to the pilot standard deviations. Conditional on any sufficiently coarse event—quantile cells of the realized counts included—the studentized arm means are jointly asymptotically normal: computed on the full sample, with the usual standard errors, and with no correction. Proposition D.35 and Corollary D.37 contain the formal statements.*

The same reasoning covers other rules driven by even moments. For instance, the two-stage propensity score allocation of Hahn et al. (2011) is a function of the ratio of pilot-estimated conditional standard deviations, hence asymptotically even. The boundary, however, travels with the principle: adaptation through odd functionals of the data—interim means, estimated treatment effects, futility rules—or asymmetry are not covered by the general theory.

7 Simulations

This Section confronts the theoretical prescriptions above with simulation evidence. Section 7.1 does so for the excess-variance results in Section 5, and Section 7.2 for the conditional normality results in Section 6. Data are generated from the survey model in (1) with $\mu = 0$. Every reported statistic is computed over 10^5 independent replications of the complete design-and-estimation cycle, so that the randomness of the pilot, the induced design, and the terminal estimator, are all accounted for. The cost structure is $(F, V) = (20, 2)$ throughout and the designer's parameter space is $\Omega = [\tau_m^2, \tau_M^2] \times [\lambda_m, \lambda_M] = [0.25, 4] \times [1, 100]$.[37] The baseline environment is $\omega = (\tau^2, \lambda_0) = (1, 5)$ at a budget

[37] These bounds enter the algorithms directly, and not only through Assumption 3.2. The pilot width is $K_p = K_M$ and the pilot depth is $N_p = N_m$.

of $B = 2{,}000$, which places the oracle design at $\big(K^*(\lambda_0), N^*(\lambda_0)\big) = (58.6, 7.1)$ and the pilot at $(K_M, N_m) = (24, 3)$. Assumption 3.3 holds at every point of the grid reported below. Errors are Gaussian in the performance exercises, and take a range of symmetric and asymmetric shapes in the inference exercises.

7.1 Performance

For each algorithm, Figure 3 reports the distribution of the terminal design (K_B^π, N_B^π) and the pilot estimate $\hat{\lambda}_{pB}$ that enters the optimal map. Both algorithms concentrate around the oracle—as prescribed by Corollary 4.7—with `ACS` inducing tighter designs, as it commits later and therefore estimates λ_0 from a weakly larger set of clusters. The distribution of $\hat{\lambda}_{pB}$ in the right-hand panel is sharp around the truth but visibly right-skewed at finite samples. The reason is that $\hat{\tau}^2$ is recovered as a difference of two random objects $\hat{b}^2 - \hat{w}^2/N_p$, so when τ^2 is small relative to σ^2, $\hat{\tau}^2$ is estimated as a noisy residual between two large and separately noisy quantities. Because $\hat{\tau}^2$ enters the ratio through the convex map $\hat{\tau}^2 \mapsto 1/\hat{\tau}^2$, any symmetric noise in the denominator becomes a heavy right tail in $\hat{\lambda}_{pB}$. The $\hat{\lambda}_{pB}$ skew is however partially absorbed by the design, provided $\hat{\lambda}_{pB}$ enters the optimal map through a square root which compresses the upper tail and the excess variance is flat to the right of λ_0 and steep to its left. Figure 13 in Appendix F shows that both algorithms track the oracle design relative well with minor depth oversampling and width undersampling in finite samples.

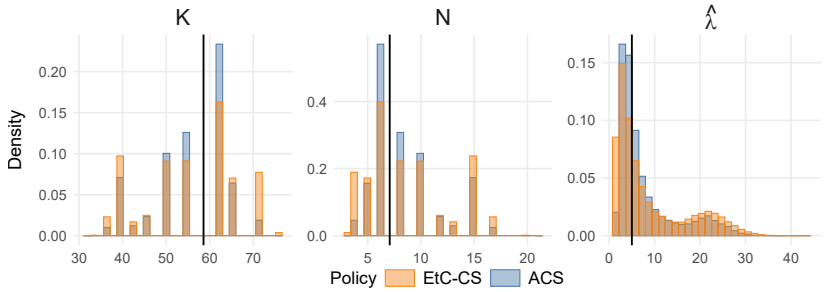


Figure 3: Design (K_B^π, N_B^π) and pilot estimate $\hat{\lambda}_{pB}$ for `EtC-CS` and `ACS`

Notes: Solid black lines mark the oracle values $K_{\underline{0}}^*$, $N_{\underline{0}}^*$ and λ_0. 10^5 replications. $\omega = (1, 5)$, $B = 2{,}000$, $F = 20$, $V = 2$.

Figure 4 plots excess variance against λ_0 for the two adaptive algorithms and a set of five static policies, each indexed by their imputed variance ratio. In this setting, the minimax policy is given by $\lambda_{mM} = 30.25$, so the static policy with $\lambda = 31.6$ is nearly identical to the maximin design in van Breukelen and Candel (2015). Static policies do not incur sampling risk, so their excess variance is deterministic. Moreover, their excess variance vanishes where the imputed value is close to the true value and rises on either side, whereas the excess variance of the adaptive algorithms stays uniformly close to the oracle. The median performance of the algorithms (right) is even better than its mean performance (left) as the adaptive designs occasionally draw an unlucky pilot and pay for it. Performance is also best at small λ_0, where τ^2 is large relative to σ^2 and the pilot estimate is well-conditioned. The red markers give the worst excess of each static policy in the grid, which fall at the endpoints by convexity. The comparison becomes clear. A static policy can beat the algorithms in a neighbourhood of its own imputed value, but the worst case of either adaptive algorithm improves on the worst case of every static competitor, minimax design included. Adaptation buys robustness against misspecification of the intra-cluster correlation.

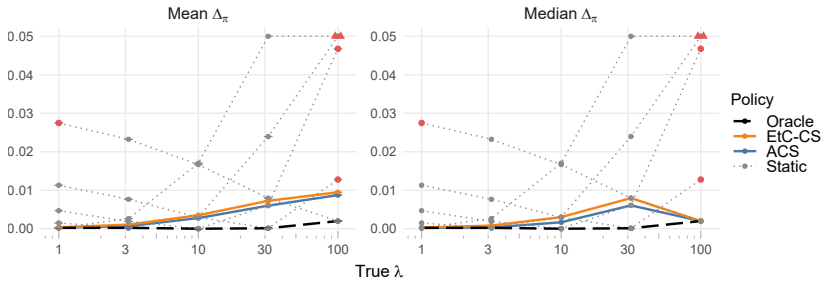


Figure 4: Excess variance across λ_0

Notes: Static policies imputed at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $B = 2{,}000$, $F = 20$, $V = 2$. Red triangles signal that worst case excess was capped at $\Delta_\pi = .05$ for display purposes.

Finally, Figure 5 fixes the environment and moves along the budget line in absolute terms (left) as in terms of rate (right). The left panel shows that for this parameter combination the excess variance of `EtC-CS` and `ACS` is already small at modest budgets and falls sharply until converging with oracle variance. The right panel shows slopes approximately of -1 for the static policies and of -2 for the adaptive algorithms, what is consistent with the excess variance results in Lemma 5.2 and Theorem 5.4, i.e. $\Theta(1/B)$ and $O(1/B^2)$, respectively. `ACS` dominates `EtC-CS` uniformly, with the gap being widest at small budgets. Recall that both pilot widths are of the same order, so the late-commitment advantage manifests only through a constant and not a rate, and so it is more salient at small budgets.

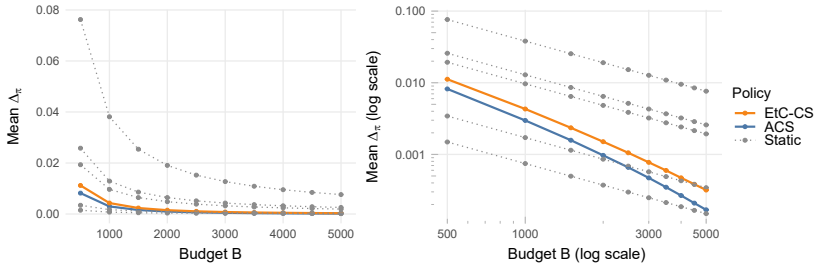


Figure 5: Mean excess variance across budget

Notes: Static policies imputed at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $F = 20$, $V = 2$.

7.2 Conditional Inference

I now turn to inference with an emphasis on `ACS`.[38] Figure 6 reports, for a grid of symmetric error distributions—including bimodal distributions and Laplace (heavy-tailed) laws—the modal-design conditional density and the unconditional density of $Z_B^\pi/\theta_{N_B^\pi}$ alongside the standard normal density.[39] As per Theorem 6.8, conditioning on the realized design leaves the limiting law unchanged, hence the two coincide throughout. Figure 7 displays more demanding evidence by reporting cell-by-cell the share of replications of the interval $\hat{\mu}_{K^\pi,N^\pi} \pm 1.96\,\hat{b}_{K^\pi,N^\pi}/\sqrt{K^\pi}$ that cover μ. As prescribed by Theorem 6.9, the coverage holds on average over designs and is close to nominal in every cell with a mild tilt as a

[38] `ACS` interacts with the data far more than `EtC-CS`. It conducts nearly all cluster exploration inside the pilot, so the design not only contains more data, but depends on the whole sequence $(\hat{\lambda}_{k,N_p})_{k\le s}$ rather than on a single statistic $\hat{\lambda}_p$. It is therefore the more demanding test of the results in Section 6.

[39] Figure 14 in the Appendix displays the full grid of combinations.

byproduct of studentization.[40]

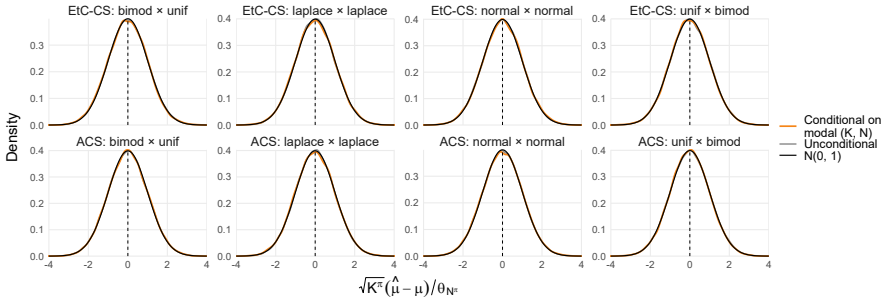


Figure 6: Empirical density of $Z_B^\pi/\theta_{N_B^\pi}$ at the modal design and unconditionally under symmetry

Notes: Distribution of u_k (left) and ε_{ki} (right). Distributions considered: Bimodal-Uniform, Laplace-Laplace, Normal-Normal, Uniform-Bimodal. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 2{,}000$, $F = 20$, $V = 2$.

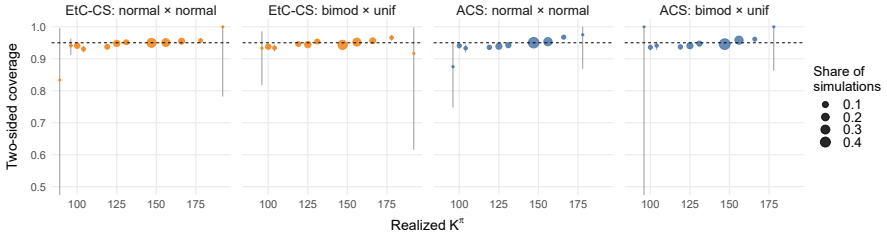


Figure 7: Design-conditional coverage of $\hat{\mu}_{K_B^\pi, N_B^\pi} \pm 1.96\, \hat{b}_{K_B^\pi, N_B^\pi}/\sqrt{K_B^\pi}$

Notes: Normal–normal and bimodal–uniform errors. Grey lines are Clopper–Pearson 95% intervals at each realized K_B^π. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 5{,}000$, $F = 20$, $V = 2$.

I next document the role of evenness within the inference framework. Theorem 6.8 implies that $\mathbb{E}[Z_B^\pi \mid A_B] \to \mathbb{E}[Z_B^\pi] = 0$ for any asymptotically even policy. I show that the symmetry requirement bites by comparing three asymptotically even policies—a naive-even policiy, `EtC-CS` and `ACS`—against a naive-odd policy that reads levels (and signs). The naive-odd and naive-even policies follow the same structure. They first run a pilot with K_M clusters at N_m units, just like `EtC-CS`. In a second stage they set

$$\text{(Odd)}\quad K^o = \begin{cases} K_M & \text{if } \bar{M}_{K_p,N_p} \geq 0 \\ K_m & \text{if } \bar{M}_{K_p,N_p} < 0 \end{cases} \qquad \text{(Even)}\quad K^e = \begin{cases} K_M & \text{if } K_p^{-1}\sum_k |M_{k,N_p}| \geq \mathbb{E}|M_{k,N_p}| \\ K_m & \text{if } K_p^{-1}\sum_k |M_{k,N_p}| < \mathbb{E}|M_{k,N_p}| \end{cases}$$

In both cases, the terminal depth is chosen to exhaust the budget.[41] In this example, the odd policy reads the sign of the pilot mean, so conditioning on it carries information about the signs $(\chi_{k,N})_k$ themselves, and the factorization of Lemma 6.3 fails. Conversely, the even policy depends only on magnitudes, and the inference principle applies. Figure 8 partitions replications into quintiles

[40] The tilt is driven by a selection effect. The feasible variance $\hat{b}^2_{K^\pi,N^\pi}$ depends on the error signs, and therefore is correlated with the conditioning event in finite samples. The bias disappears approximately at a rate of $B^{-1/2}$ as $\hat{b}^2_{K^\pi,N^\pi} \to \theta^2_{N_0}$, exactly as described in the Appendix. Still, the tilt balances across cells, leaving unconditional coverage unaffected according to the Anscombe route outlined in the text, which requires no symmetry. Observe that the tilt is also absent in Figure 6, as it standardizes by $\theta_{N_B^\pi}$.

[41] $\mathbb{E}|M_{k,N}| = \theta_{N_p}\sqrt{2/\pi}$ is the exact mean of the half-normal under Gaussian errors. Hence, by a CLT argument, the designer should expect roughly 1/2 of the designs below the cutoff and 1/2 above.

$(C_j)_j$ of the realized width and plots $\mathbb{E}[Z_B^\pi \mid C_j]$ for each. The conditional expectations of `EtC-CS` and `ACS` are indistinguishable from zero across cells, just like the ones of the naive-even, while those of the odd policy live visibly away from zero and change sign as predicted.[42]

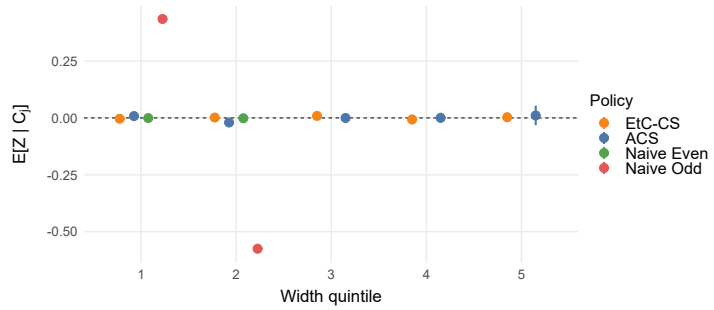


Figure 8: Design-conditional expectation $\mathbb{E}[Z_B^\pi \mid C_j]$ over width quintiles

Notes: $(C_j)_j$ are quintiles of the realized width. Gaussian errors. The naive-even and naive-odd policies are two-valued, so only two quintiles carry positive mass. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 2{,}000$, $F = 20$, $V = 2$.

Figure 9 plots the maximum cell-frequency discrepancy $D_B = \sup_{t\in\mathcal{T}} \max_j \left|\mathbb{E}\left[e^{itZ_B^\pi}\mathbb{1}_{C_j}\right] - \phi(t)\,\mathbb{P}(C_j)\right|$ against B. Theorem 6.8 claims that D_B should vanish for asymptotically even policies under symmetry. Four combinations are shown: the naive-odd policy under Gaussian errors, `ACS` under asymmetric errors—$u_k \sim \Gamma(\alpha, \beta)$ with $\alpha = \beta = 1$ so that $\text{Skew}(u_k) = 2/\sqrt{\alpha} = 2$—, `ACS` under Laplace errors, and `ACS` under Gaussian errors.

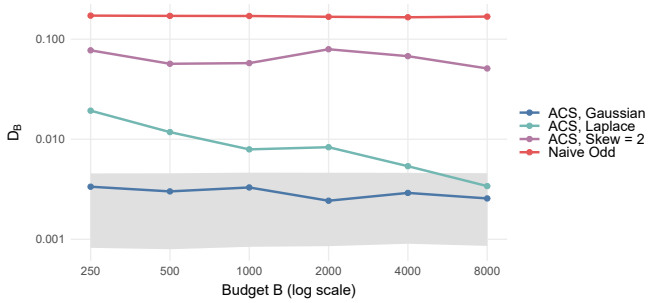


Figure 9: Maximal discrepancy between the conditional and normal characteristic functions

Notes: $(C_j)_j$ are quintiles of the realized width. Grey band represents worst-case 95% range of D_B over B under exact conditional normality via simulation. Symmetric errors are $N(0, 1)$, asymmetric errors are centered $\Gamma(1, 1)$. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $F = 20$, $V = 2$.

The figure separates the two ingredients entering the inference theory cleanly. Either dropping evenness or symmetry breaks the result: Maximal discrepancies are uniformly bounded away from zero for both the naive-odd policy with Gaussian errors and the `ACS` policy with asymmetric errors. As a result, the simulation evidence suggests that, in general, neither condition can be dispensed with. By contrast, where both hold, the theory cannot be rejected. D_B for `ACS` under Laplace errors declines steadily in B, exactly as predicted by Theorem 6.9; and for Gaussian errors sits uniformly at the simulation noise floor as predicted by Proposition 6.1.[43]

[42] Observe that the odd policy does not only read the signs but also the levels. As a result, although the policy deals with Gaussian errors, it is not measurable with respect to the filtration of residuals $(\mathcal{R}_j)_j$ and so it is not covered by Proposition 6.1.

[43] The grey band gives the 95% range of D_B under an exact null, obtained by drawing designs from the empirical

Finally, Figure 10 pins down the monotonicity within the symmetry assumption by plotting the cell-conditional expectations $\mathbb{E}[Z_B^\pi \mid C_j]$ against $\text{Skew}(u_k)$. At exact symmetry all cells are centered at zero, but as skewness rises the conditional expectations fan out. The result is expected. Under asymmetry the design cell becomes informative about the signs themselves, and the more extreme the cell, the more it reveals—which is why the departures are largest in cells 1 and 5. `ACS` departs more than `EtC-CS` at every level of skewness, consistent with its design depending on a longer sequence of statistics, therefore leaking weakly more information into the terminal average when conditioning.

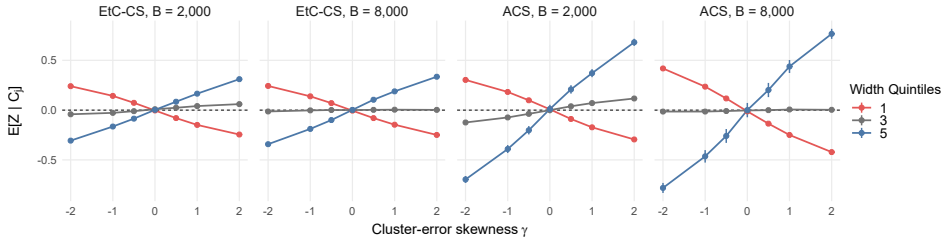


Figure 10: Design-conditional expectation $\mathbb{E}[Z_B^\pi \mid C_j]$ as a function of $\text{Skew}(u_k)$

Notes: $(C_j)_j$ are quintiles of the realized width. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $F = 20$, $V = 2$.

8 Empirical Application: Fink et al. (2020b)

8.1 Setting and Design Problem

FJM study whether relaxing liquidity constraints during the agricultural lean season changes harvest outcomes and labor allocation among smallholder farmers in Chipata District, Zambia. Farmers were offered favorable loans, in cash or in maize, repayable after the harvest. The experiment was implemented through a cluster design with $K = 175$ villages and a target of $N = 22$ households per village. The treatment was assigned at the village level in three equal arms (cash loan, maize loan and control) after a baseline survey.[44] The experiment is a natural setting for implementing the adaptive sampling procedure described in this paper: large number of clusters, ex-ante fixed budget, and a prominent role of fixed costs.

Importantly, unlike many other studies, FJM are not inattentive to the ICC. The authors run a small pilot and estimate the ICC, which they later use to power the study appropriately but not in a cost-aware fashion.[45] Conventional power calculations fix a detectable effect and solve for a sample size, but leave aside the cost structure.[46] Instead, the realized design appears to be driven by logistical concerns as the 22 household target was simply the size of the smallest village in their sample.[47] These practices are not unique to FJM, but prevalent in field experiments and medical trials. In fact, this is

design distribution of `ACS` and generating the corresponding statistics as independent standard normals, so that D_B is zero by construction and the band measures Monte Carlo error alone.

[44] Realized depth effectively ranged from 5 to 22 households, averaging 17.9, leading to a baseline sample of 3,139 households, of which 3,028 appear in the year-one survey. Appendix G.1 describes the sampling frame, the outcome variables, and the residualization process entering my analysis in detail.

[45] The experiment register indicates that the study is powered to detect a ten percent increase in farm production value, see Fink et al. (2024). Further design details can be found in Fink et al. (2020a) and the pre-analysis plan Fink et al. (2014).

[46] There is also no evidence that the size of the pilot was selected according to a statistical or cost-relevant criterion.

[47] Other features of the design point in the same direction. The initial design considered sampling a round five villages per administrative block, and villages were excluded from the initial list of 250 for reasons such as the impossibility of reaching them by vehicle during the rainy season. Fixed costs and logistics are then plausibly first-order, as they are in most field experiments. Yet, none of these concerns imply that the realized design ended up being cost-suboptimal.

the *status quo* that the manuscript seeks to replace.

There are two caveats in the counterfactual exercise below. First, I do not observe the cost parameters (F, V), so every result below is stated across a grid of ratios $F/V \in [1, 100]$.[48] Hence, none of my claims should suggest that the realized design was suboptimal at the ratio FJM actually faced, even if the design was not intentionally selected in a cost-aware fashion. Second, the exercise uses baseline data alone to identify the design. Post-treatment outcomes would in principle be available (Kasy and Sautmann, 2021), but the one-year horizon of this experiment leaves no room for adaptation within it. I therefore assume the variance ratio is stable across survey rounds t for every arm d, so that $\tau^2_{td} = \tau^2_d$ and $\lambda_{td} = \lambda_{0d}$. Time homogeneity together with random assignment at baseline further implies that $\tau^2_d = \tau^2$ and $\lambda_{0d} = \lambda_0$. Appendix G.2 tests the set of restrictions and cannot reject any.

The designer's problem. With three arms of equal size, the intention-to-treat (ITT) estimator for arm d specializes the cluster-level treatment estimator of Section 3 and Appendix A.2.1 to

$$\hat{\mathrm{ITT}}^{(d)}_{K,N} \;=\; \hat{\mu}_d - \hat{\mu}_0 \;=\; \frac{3}{K}\left(\sum_{k\,:\,D_k=d}\frac{1}{N}\sum_{i=1}^{N} y_{ki} \;-\; \sum_{k\,:\,D_k=0}\frac{1}{N}\sum_{i=1}^{N} y_{ki}\right)$$

Under the homogeneity assumptions the designer's minimax problem across arms collapses to a single objective,[49]

$$\min_{K,N}\ \max_{d\in\{1,2\}}\ \mathrm{Var}\big(\hat{\mathrm{ITT}}^{(d)}_{K,N}\big) \;=\; \min_{K,N}\ \frac{6\tau^2}{K}\left(1+\frac{\lambda}{N}\right) \qquad \text{s.t.} \quad K(F+NV)\le B$$

This is the objective of Section 3 scaled by a constant, and so the optimal map and oracle allocation remain unchanged. Both arms cost approximately the same per household—about 60 ZMW excluding administration—so V should be read as the average per-household cost across arms.

Design inputs. I estimate λ_0 from the full baseline sample and calibrate the designer's interval $[\lambda_m, \lambda_M]$ through a cluster bootstrap of $\hat{\lambda}$, taking λ_m just below the 2.5th percentile and λ_M just above the 97.5th.[50] Table 4 reports the results for the two outcomes I study: log(1 + harvest value), the register's primary outcome, and log(1 + ganyu mandays), the informal-labor measure the published paper emphasizes. Both are residualized according to the covariates included in the paper's specification.

Outcome	λ_0 $(=\hat{\lambda}_{K,N})$	ICC	$Q_{0.025}(\hat{\lambda}^{(b)}_{K,N})$	$Q_{0.975}(\hat{\lambda}^{(b)}_{K,N})$	λ_m	λ_M
log(harvest value + 1)	9.658	0.094	7.072	14.295	7	15
log(ganyu mandays + 1)	27.683	0.035	18.337	53.187	18	54

Table 4: Design inputs

Notes: The design inputs are estimated from the full baseline sample. Percentiles from a 2,000-replication cluster bootstrap holding village sizes fixed. Covariate-residualized data.

[48] In practice, the distinction between fixed and variable costs may be blurry. In conversations with the authors and practitioners in the area, the realized F/V ratio was probably between 1 and 2. Yet, I take a (very) conservative approach to the discussion and limit my claims to the grid of ratios $[1, 100]$.

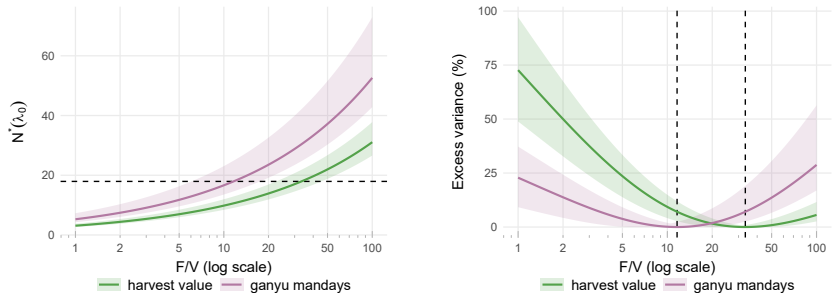
[49] The current manuscript does not take any stance on how to trade across different outcome variables.

[50] This λ box is possibly too wide in a way that hurts the implementation of the adaptive algorithms. In practice, the authors probably had access to prior information that could narrow the interval. The intervals for both variables satisfy Assumption 3.3 over the full F/V grid.

Is inference valid here? The results of Section 6 rest on Assumption 6.2, which requires the whole error block $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ to satisfy symmetry. That restriction is not directly testable, since the analyst only observes cluster averages and not the block itself. The test on symmetry is therefore limited to the cluster average residuals $M_{k,N}$, which is implied at every depth by the symmetry of C_k. I use the symmetry test of Miao et al. (2006) to test this implication on the centered residuals $\hat{M}_{k,N}$ at $N \in \{4, 8, 12, 16, 20\}$. The results are calibrated against a Rademacher sign-flip null and aggregated across subsampling draws of size N by the dependence-robust rule of Meinshausen et al. (2009). Symmetry is not rejected for either outcome at any depth, at any conventional level. While this evidence cannot be considered final towards testing Assumption 6.2, the results are consistent with block symmetry. Appendix G.3 develops the testing methodology in greater detail and reports the full set of results. Under the symmetry null, standard errors and significance levels can be computed in the ordinary way: on the full sample, with no correction, and using Gaussian asymptotics.

8.2 Was the Realized Design Cost-Optimal?

I first inspect if the realized design can be rationalized by some cost structure, and how much is lost across the grid. Normalizing $V = 1$ and varying F, each ratio implies a budget $B = 175 \times (F + 17.9V)$ and an oracle design $(K_B^*(\lambda_0), N^*(\lambda_0))$. Panel (a) of Figure 11 plots the oracle depth against the realized $N = 17.9$, and Panel (b) plots the excess variance of the realized design relative to the oracle.



(a) Depth (b) Excess variance

Figure 11: Realized FJM design against the oracle across F/V cost ratios

Notes: Excess variance relative to the variance of the oracle design (%). V set to 1. Realized average depth $N = 17.9$ (dashed, left), rationalizing ratios (dashed, right). $B = 175 \times (F + 17.9V)$. Ribbons are 95% cluster-bootstrap intervals, obtained by plugging-in the bootstrap percentiles of $\hat{\lambda}$ from Table 4. Covariate-residualized data.

The realized design is rationalized at a cost ratio of 33.2 for harvest value and 11.6 for ganyu mandays. Away from the rationalized costs ratios the excess variance grows quickly, with oracle depths ranging from 3 to 52 units per village. At the low end of the cost grid, the realized design carries an excess variance of about 70% for harvest value, and, at the high end, of about 30% for ganyu mandays. Removing an excess variance of 70% corresponds to a variance reduction of roughly 41% percent. I emphasize once again that the current analysis remains silent on whether the realized FJM design is or is not cost optimal, yet, the broader point prevails: without proper accounting of the ICC and the cost structure, no design is safe from landing on the steep part of the excess variance curve.[51]

[51] Cluster availability is not the binding constraint at any point on the grid. The village list from which FJM sampled contains 733 villages with ten or more smallholder households, 853 with five or more, and 973 with three or more. Additional districts around Chipata District within Eastern Province were also available had they been needed.

8.3 What an Adaptive Design Would Have Delivered

Counterfactual designs cannot be run on the realized data, since any design at the same budget requires either more villages or more households per village than those surveyed by the authors. I therefore build a semi-synthetic population under the additive model $y_{ki} - \boldsymbol{x}_{ki}\boldsymbol{\beta} = \mu + u_k + \varepsilon_{ki}$, resampling the rescaled empirical cluster- and household-level residuals to match the first and second moments $(\hat{\mu}, \hat{\tau}^2, \hat{\sigma}^2)$ exactly. Appendix G.4 documents the construction and validates it against the realized outcome distribution. Figure 12 reports the excess variance of the realized design, `EtC-CS` and `ACS` on the semi-synthetic population across the cost grid.

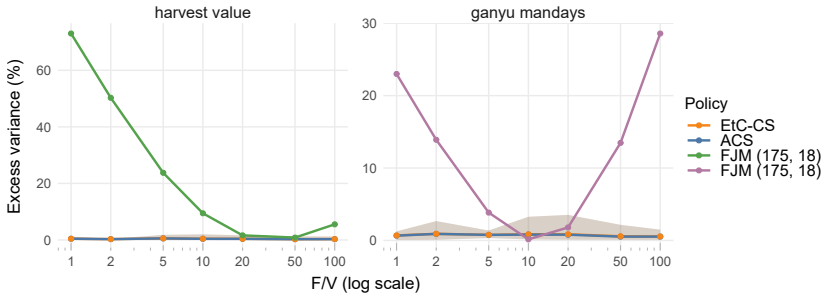


Figure 12: Excess variance across cost ratios

Notes: Excess variance relative to the variance of the oracle design (%). Means over 10,000 simulations, bands are 95% intervals across simulations.

Three results stand out from the figure. First, both algorithms are essentially optimal at every cost ratio, with mean excess variance close to zero and narrow confidence bands, especially for the harvest value outcome. Second, both algorithms behave in a similar way. In this application, the budget supports several hundred villages, so even the worst case conservative pilot (K_M, N_m) entering `EtC-CS` identifies λ_0 with high precision, lowering the informational value of the extra clusters in `ACS`. Third, both algorithms dominate the realized design across the grid except at the rationalized ratio—which is unknown to the designer, hence the rationale for the exercise.

Consequences for the reported estimates. Finally, I ask what the design would have meant for the results in the paper. Applying the same construction to year-one outcomes, I re-estimate the treatment-effect specification of Fink et al. (2020b), Table 4 Panel B, on data generated under each counterfactual design. Tables 5 and 6 report the results. The synthetic replication of the realized design (column 3) tracks the real data (column 2) closely. The gap in standard errors comes from the synthetic design being balanced while the realized design is not, a feature that is pronounced for ganyu mandays.[52] Point estimates are stable across designs—as they should be because the design affects precision, not identification—while standard errors fall by up to 17 percent for harvest value and 19 percent for ganyu mandays relative to the balanced realized design (column 3). The reduction is even more salient when compared to the implemented unbalanced design (column 2).

[52] My estimates differ slightly from those in Fink et al. (2020a) (column 1) because the published table pools year one and year two, whereas I use year one alone. Generating year-two data would have required modeling treatment rotation, differential attrition, and post-default exclusions, none of which the additive model accommodates cleanly.

	Paper	FJM	FJM	F/V						
	(Real)	(Real)	(Synthetic)	1	2	5	10	20	50	100
K/N	175/17.9	175/17.9	175/18	774/3	543/4	332/7	245/10	194/14	165/22	157/32
Cash	0.105*	0.112**	0.112***	0.109***	0.111***	0.114***	0.113***	0.112***	0.111***	0.111***
	(0.041)	(0.046)	(0.042)	(0.035)	(0.038)	(0.039)	(0.041)	(0.042)	(0.042)	(0.040)
Maize	0.056	0.037	0.036	0.034	0.037	0.037	0.036	0.037	0.035	0.038
	(0.054)	(0.047)	(0.042)	(0.035)	(0.038)	(0.039)	(0.041)	(0.042)	(0.041)	(0.040)

Table 5: Treatment-effect estimates for log(harvest value + 1), year-one outcomes.

Notes: Column 1 is reproduced from Fink et al. (2020a), Table 4 Panel B (years one and two pooled). Counterfactual designs are generated by `ACS`; coefficients and standard errors are means over 1,000 simulations, and K/N reports the mean realized design. CR1 cluster standard errors, as in the original. Stars from the normal approximation: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	FJM	FJM	F/V						
	(Real)	(Synthetic)	1	2	5	10	20	50	100
K/N	175/17.9	175/18	521/6	357/8	236/12	181/17	151/24	135/38	135/53
Cash	-0.182*	-0.180**	-0.186***	-0.180**	-0.184**	-0.186**	-0.182**	-0.182**	-0.178**
	(0.096)	(0.080)	(0.065)	(0.071)	(0.076)	(0.079)	(0.080)	(0.078)	(0.074)
Maize	-0.210**	-0.209***	-0.211***	-0.209***	-0.213***	-0.212***	-0.208***	-0.209***	-0.209***
	(0.101)	(0.080)	(0.065)	(0.071)	(0.077)	(0.079)	(0.080)	(0.078)	(0.075)

Table 6: Treatment-effect estimates for log(ganyu mandays + 1), year-one outcomes.

Notes: This specification is not reported in Fink et al. (2020b). Counterfactual designs are generated by `ACS`; coefficients and standard errors are means over 1,000 simulations, and K/N reports the mean realized design. CR1 cluster standard errors. Stars from the normal approximation: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9 Extension: Heterogeneous Cluster Variances

The baseline model explicitly imposes variance homogeneity across clusters. This section relaxes the restriction by allowing clusters to carry a set of observable covariates that shift the dispersion of their cluster-level errors,

$$y_{ki} = \mu + u_k + \varepsilon_{ki}, \qquad \mathbb{E}\big[u_k^2 \mid X_k\big] = \tau_k^2 = \ell(X_k, \gamma), \qquad (X_k)_k \; iid \in [x_m, x_M], \tag{11}$$

where the link ℓ is known up to a finite-dimensional parameter $\gamma \in \Gamma$, and the remaining error structure is unchanged: cross-cluster independence, homogeneous unit-level variance σ^2 with $\varepsilon_{ki} \perp\!\!\!\perp X_k$, and cross-error terms independence $u_k \perp\!\!\!\perp \varepsilon_{ki}$.[53] I further introduce two new population objects, namely the arithmetic and harmonic cluster variance means, and their corresponding variance ratios. Expectations are taken with respect to the distribution of the covariates, which need not be known to the designer beyond its support $[x_m, x_M]$.

$$\bar{\tau}^2 := \mathbb{E}\big[\tau_k^2\big], \qquad \tau_{\mathrm{h}}^2 := \frac{1}{\mathbb{E}\big[1/\tau_k^2\big]}, \qquad \bar{\lambda}_0 := \frac{\sigma^2}{\bar{\tau}^2}, \qquad \lambda_{0\mathrm{h}} := \frac{\sigma^2}{\tau_{\mathrm{h}}^2}$$

[53] Appendix H collects the formal assumptions (H1)–(H4): the error structure and variance link, compactness—which bounds $\tau_k^2 \in [\tau_m^2, \tau_M^2]$ away from zero and degeneracy—interiority, and subgaussianity.

Timeline. The designer first commits to an estimator class by selecting a set on non-negative weights entering a weighted average of cluster means $\hat{\mu}_\omega = \sum_k \omega_k \hat{\mu}^k_{N^k} / \sum_k \omega_k$. When the designer recruits a new cluster, she first pays the fixed cost F, then she observes the covariates X_k, and finally she selects the number of units N^k, possibly as a function of the cluster covariates. By contrast, cluster recruitment cannot depend on the cluster's own covariate vector, as X_k is not realized at the time of recruitment. I further assume that recruited clusters cannot be rejected *ex post*. The designer's problem can be written as

$$\min_{K,\{N^k\}} \mathbb{E}\big[\mathrm{Var}\big(\hat{\mu}_\omega \mid X^K\big)\big] = \min_{K,\{N^k\}} \mathbb{E}\left[\frac{\sum_{k=1}^K \omega_k^2 \left(\tau_k^2 + \frac{\sigma^2}{N^k}\right)}{\left(\sum_{k=1}^K \omega_k\right)^2}\right] \quad \text{s.t.} \quad FK + V\sum_{k=1}^K N^k \leq B \ \text{ a.s.} \tag{12}$$

where $X^K = (X_1, \ldots, X_K)$. I study (12) for the two leading estimators: the unweighted average with weights $\omega_k = 1/K$, and the GLS average with weights $\omega_k = 1/(\tau_k^2 + \sigma^2/N^k)$.[54]

Heterogeneity is irrelevant for the unweighted average. Replacing $\omega_k = 1/K$ in (12) shows that the cluster variances $(\tau_k^2)_k$ do not interact with depth. Intuitively, once cluster k is recruited, τ_k^2 enters the conditional variance additively and sunk, so no choice of N^k touches it. Depth only buys a reduction of σ^2/N^k, whose marginal value does not depend on X_k. The variance-minimizing solution then yields a balanced covariate-free allocation, and the problem reduces to the original formulation. Lemma 9.1 formalizes this result.

Lemma 9.1. *Design Irrelevance under Equal Weights. Under (H1)–(H3), random recruitment and no cluster rejection, the design problem for the unweighted average collapses to the homogeneous problem of Section 3 evaluated at $\bar{\tau}^2$. The oracle allocation equals $\big(K^*(\bar{\lambda}_0), N^*(\bar{\lambda}_0)\big)$, and so it is constant across X^K realizations. The oracle variance is given by $W_u^* = \bar{\tau}^2\big(\sqrt{V\bar{\lambda}_0} + \sqrt{F}\big)^2/B$.*

Corollary 9.2. *`EtC-CS` is Robust to Heterogeneity. Under (H1)–(H4), the sequence of cluster errors is unconditionally iid, so `EtC-CS` retains its $O(1/B^2)$ excess-variance guarantee against the oracle W_u^*.*

The bottom line is that heterogeneity does not introduce a motive for unbalanced sampling on its own. The real motivation for unbalanced sampling will only emerge from the interaction between heterogeneity and weighted estimation.

Clairvoyant Designer and GLS weights. By construction, GLS weights deliver $\mathrm{Var}(\hat{\mu}_\omega \mid X^K) = 1/\sum_k \omega_k$ with $\omega_k = (\tau_k^2 + \sigma^2/N^k)^{-1}$, so depth choices now interact with the heterogeneous variances τ_k^2. Feasible solutions are adapted, as N^k can only depend on the covariates of observed clusters but not future ones, thus, no closed-form solution exists for (12). Nonetheless, the solution to the problem can be approximated efficiently. In particular, consider a *clairvoyant* designer who observes the full covariate vector X^K before sampling. Her optimal allocation could then equalize $\tau_k^2 N^k$ across clusters making depth inversely proportional to the cluster variance.The reason is that, from the perspective of depth, τ_k^2 operates as an irreducible noise floor. Noisy clusters will be downweighted anyway through the GLS factors, so depth efforts can be focused on lowering the individual variance σ^2/N^k of quiet clusters. Substituting the optimal profile back into the variance expression returns

$$\mathrm{Var}\big(\hat{\mu}_\omega \mid K, X^K\big) = \frac{1}{K}\left(\tau^2_{\mathrm{h},K} + \frac{\sigma^2}{\bar{N}}\right), \qquad \tau^2_{\mathrm{h},K} := \frac{K}{\sum_k 1/\tau_k^2} \tag{13}$$

[54] GLS weights are given by the inverse of the conditional variance of the cluster mean given the realized X_k.

The clairvoyant variance is then analogous to the benchmark equation with the realized harmonic mean $\tau^2_{h,K}$ in place of τ^2 and the average depth $\bar{N}$ in place of N. Observe that the only clairvoyant object in (13) is $\tau^2_{\mathrm{h},K}$, which can be lower bounded efficiently. For a fixed K, replacing $\tau^2_{\mathrm{h},K}$ by the population counterpart τ^2_{h} delivers a valid lower bound via Jensen which converges at the parametric rate provided $\tau^2_{\mathrm{h},K} \to \tau^2_{\mathrm{h}}$. The substitution is nearly costless and the clairvoyant variance becomes nearly attainable to a standard oracle designer.

Proposition 9.3. *The Weighted Oracle. Under (H1)–(H3), every feasible policy with deterministic width satisfies*

$$\mathbb{E}\big[\mathrm{Var}\big(\hat{\mu}_\omega \mid X^K\big)\big] \;\geq\; W^*_\omega := \frac{\big(\sqrt{F\tau^2_{\mathrm{h}}} + \sqrt{V\sigma^2}\big)^2}{B}$$

which is attained up to $O(1/B^2)$ by the oracle maps

$$K^*_\omega = \frac{B}{F + \sqrt{\lambda_{0\mathrm{h}} FV}}, \qquad N^{k*}_\omega = \frac{\tau^2_{\mathrm{h}}}{\tau^2_k}\sqrt{\lambda_{0\mathrm{h}}\frac{F}{V}} \tag{14}$$

The optimal design resembles once again the solution to the benchmark problem evaluated at $(\tau^2_{\mathrm{h}}, \lambda_{0\mathrm{h}})$, with an additional depth scaling term of $\tau^2_{\mathrm{h}}/\tau^2_k$. All the machinery developed for the homogeneous setting will therefore transport nearly verbatim to the GLS-heterogeneity setting.

Proposition 9.4. *Ignoring Heterogeneity Incurs a First-Order Loss. Under (H1)–(H2),*

$$W^*_u - W^*_\omega = \frac{F\big(\bar{\tau}^2 - \tau^2_{\mathrm{h}}\big) + 2\sqrt{VF\sigma^2}\,\big(\bar{\tau} - \tau_{\mathrm{h}}\big)}{B} \;\geq\; 0 \tag{15}$$

with equality if and only if τ^2_k is degenerate. The gap is otherwise $\Theta(1/B)$ and increasing in the dispersion of τ^2_k.

The excess variance from ignoring heterogeneity is of the same order as λ misspecification in the homogeneous setting, and so it becomes a first-order concern to the designer.

Estimation and algorithm. Input identification is severely more challenging than in the homogeneous setting, especially regarding τ^2_k. Each cluster contributes a single draw of u_k, so τ^2_k is only identified by pooling across clusters through the link. To do so, I first construct conditionally unbiased pseudo-cluster errors $q_{k,K}$; then, I estimate the γ vector via nonlinear least squares of $q_{k,K}$ on $\ell(X_k, \cdot)$. These estimates can then be used to commit to a depth profile, which will guide a sequential exploration process until the budget is exhausted. Observe that even after estimation the learner cannot commit to a fixed design, as the costs depend on the number of units N^k, which in turn depend on the unrealized covariates $\{X_k\}_{k=K_p}^K$. The weighted analogue of `EtC-CS`, denoted `wEtC-CS`, concretizes this heuristic by running a conservative pilot in the first stage.[55]

Conjecture 9.5. *Performance of `wEtC-CS`. Under (H1)–(H4) and the NLS conditions of Appendix H,* $\mathbb{E}\big[\mathrm{Var}\big(\hat{\mu}_{\hat{\omega}} \mid X^K\big)\big] - W^*_\omega = O(1/B^2)$.

Formalizing the conjecture on performance requires localizing the random terminal width around $K^*_\omega(\lambda_{0\mathrm{h}})$, which I expect to follow the same structure as the optimality proof for `EtC-CS` in the homogeneous setting. In that sense, the open challenge does not rely on a single step but on

[55] Further details can be found in Appendix H.

their combination. First, the algorithm does not chase the clairvoyant allocation, but the oracle maps in (14). As per Proposition 9.3, the additional excess variance coming from this substitution is also $O(1/B^2)$, and so it is costless in rate. Second, to achieve the desired rate $O(1/B^2)$ the intermediate inputs $(\hat{w}_K^2, \hat{b}_K^2, \hat{\gamma}, \hat{\tau}_{\mathrm{h}}^2, \hat{\lambda}_{\mathrm{h}})$ must be estimated at the parametric rate.The main difficulty lies in the estimation of $\hat{\gamma}$, which is conducted via nonlinear least squares on cross-sectionally dependent pseudo-outcomes. The maximal Bernstein arguments used in Appendix C must then be strengthened so they hold uniformly in the parameter space $\Omega \times \Gamma$. Third, the structure of the algorithm differs in one important point compared to the benchmark case: it does not plug-in first stage estimates into the optimal map. Under heterogeneity, sampling costs $F + VN^k(X_k)$ are random, hence the learner cannot commit to a terminal width right after the pilot. Instead, she commits to a depth profile that wastes at most $O(1)$ in budget.[56] This observation can be leveraged to localize $K_B^{\texttt{wEtC-CS}}$ within $O_p(\sqrt{B})$ of the oracle $K_\omega^*(\lambda_{0\mathrm{h}})$, possibly through a law of large numbers on the stopped random walk with increments governed by the concentration properties of the input estimators. A further technicality on the application of Jensen's inequality to random widths, especially relevant for lower bounds, must also be accounted for (Remark H.2).

Finally, each of the deviations above must be quantified in terms of excess variance. The collapsed objective (13) is smooth and first-order stationary along the budget line at the oracle, so scale deviations of order $O_p(\sqrt{B})$ should cost $O(1/B^2)$. No individual step seems to induce deviations incompatible with the target rate of $O(1/B^2)$ attained in the homogeneous setting, yet these deviations operate jointly, and therefore must be dealt with jointly. I leave this to future work.

10 Conclusion

This is the first paper to provide optimality guarantees for the joint width-versus-depth allocation problem without any knowledge on the ICC or any constraints on the policy class. Optimality is achieved by two algorithms—a two-stage rule and a fully sequential rule—that learn the intra-cluster correlation within the experiment and implement the oracle design in the limit. They achieve expected excess variance of order $1/B^2$, a full order of the budget below the $\Theta(1/B)$ cost of misspecified static designs, maximin designs included. By deriving a matching lower bound, I show that their rate cannot be improved upon.

Variance gains do not come at the expense of inference. Under a symmetry restriction on the errors, estimators remain asymptotically normal after conditioning on asymptotically even designs. No reweighting, no sample splitting, no discarded pilot data. The invariance theory also allows for conditioning on design events with non-degenerate limiting probability, where earlier two-stage arguments do not apply. This principle extends beyond clustered designs and validates standard inference for popular sampling procedures like Neyman allocations computed from pilot variances. Its boundary, however, remains equally sharp: simulations show that asymmetric errors or odd policies can restore the distortions identified by the reweighting literature.

The empirical results indicate that the stakes are economically meaningful. In the reconstruction of Fink et al. (2020b), the adaptive allocation delivered up to 41% lower variance for the same money compared to the realized design. Moreover, the methods are implementable as confirmed by the live survey fielded in real time by one of the algorithms. Some of the modeling decisions shed light

[56] i.e. the learner effectively optimizes with respect to a budget $B' = B - O(1)$

on the next steps. For instance, symmetry still excludes important distributions such as binary and censored outcomes; costs are assumed to be known, linear and irreversible; and the current framework only accommodates for a single outcome variable. Despite the outlined limitations, the results provide a foundation for cost-aware data collection in multistage surveys and clustered randomized trials.

References

Abadie, A., Athey, S., Imbens, G. W., and Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *The Quarterly Journal of Economics*, 138(1):1–35.

Adusumilli, K. (2026). How to sample and when to stop sampling: The generalised wald problem and minimax policies. *Review of Economic Studies*, 93(1):1–34.

Anscombe, F. J. (1952). Large-sample theory of sequential estimation. In *Mathematical Proceedings of the Cambridge Philosophical Society*, volume 48, pages 600–607. Cambridge University Press.

Bai, Y. (2022). Optimality of matched-pair designs in randomized controlled trials. *American Economic Review*, 112(12):3911–3940.

Bai, Y., Jiang, L., Romano, J. P., Shaikh, A. M., and Zhang, Y. (2024). Covariate adjustment in experiments with matched pairs. *Journal of Econometrics*, 241(1):105740.

Ballin, M. and Barcaroli, G. (2013). Joint determination of optimal stratification and sample allocation using genetic algorithm. *Survey Methodology*, 39(2):369–393.

Barcaroli, G., Fasulo, A., Guandalini, A., and Terribili, M. D. (2022). Two-stage sampling design and sample selection with the r package r2beat. *arXiv preprint arXiv:2209.06532*.

Bethel, J. (1989). Sample allocation in multivariate surveys. *Survey methodology*, 15(1):47–57.

Biemer, P. P. and Lyberg, L. E. (2003). *Introduction to survey quality*. John Wiley & Sons.

Bruhn, M. and McKenzie, D. (2009). In pursuit of balance: Randomization in practice in development field experiments. *American economic journal: applied economics*, 1(4):200–232.

Campbell, M. K., Grimshaw, J. M., and Elbourne, D. R. (2004). Intracluster correlation coefficients in cluster randomized trials: empirical insights into how should they be reported. *BMC Medical Research Methodology*, 4(1):9.

Canay, I. A., Romano, J. P., and Shaikh, A. M. (2017). Randomization tests under an approximate symmetry assumption. *Econometrica*, 85(3):1013–1030.

Cesa-Bianchi, N., Colomboni, R., and Kasy, M. (2025). Adaptive maximization of social welfare. *Econometrica*, 93(3):1073–1104.

Chambaz, A., van Der Laan, M. J., and Zheng, W. (2015). Targeted covariate-adjusted response-adaptive lasso-based randomized controlled trials. *Modern adaptive randomized clinical trials: statistical, operational, and regulatory aspects*, pages 345–368.

Chow, S.-C. and Corey, R. (2011). Benefits, challenges and obstacles of adaptive clinical trial designs. *Orphanet journal of rare diseases*, 6(1):79.

Chow, Y. S. and Robbins, H. (1965). On the asymptotic theory of fixed-width sequential confidence intervals for the mean. *The Annals of Mathematical Statistics*, 36(2):457–462.

Cicchitelli, G., Herzel, A., and Montanari, G. E. (1992). *Il campionamento statistico*. il Mulino.

Cochran, W. G. (1977). *Sampling techniques*, volume 3. wiley New York.

Cytrynbaum, M. (2021). Optimal stratification of survey experiments. *arXiv preprint arXiv:2111.08157*.

Cytrynbaum, M. (2024). Covariate adjustment in stratified experiments. *Quantitative Economics*, 15(4):971–998.

Dai, J., Gradu, P., and Harshaw, C. (2023). Clip-OGD: An experimental design for adaptive Neyman allocation in sequential experiments. In *Advances in Neural Information Processing Systems*, volume 36, pages 32235–32269.

Dudewicz, E. J. and Dalal, S. R. (1975). Allocation of observations in ranking and selection with unequal variances. *Sankhyā: The Indian Journal of Statistics, Series B*, pages 28–78.

Fink, G., Jack, B. K., and Masiye, F. (2014). Food constraints, yield uncertainty and "ganyu" labour: Pre-analysis plan. Pre-analysis plan. AEA RCT Registry ID: AEARCTR-0000130.

Fink, G., Jack, B. K., and Masiye, F. (2020a). Online appendix to seasonal liquidity, rural labor markets, and agricultural production. Online appendix. Appendix to American Economic Review, 110(11): 3351–3392.

Fink, G., Jack, B. K., and Masiye, F. (2020b). Seasonal liquidity, rural labor markets, and agricultural production. *American Economic Review*, 110(11):3351–3392.

Fink, G., Jack, B. K., and Masiye, F. (2024). Food constraints, yield uncertainty and "ganyu" labour. *AEA RCT Registry*.

Gould, A. L. and Shih, W. J. (1992). Sample size re-estimation without unblinding for normally distributed outcomes with unknown variance. *Communications in Statistics – Theory and Methods*, 21(10):2833–2853.

Hadad, V., Hirshberg, D. A., Zhan, R., Wager, S., and Athey, S. (2021). Confidence intervals for policy evaluation in adaptive experiments. *Proceedings of the national academy of sciences*, 118(15):e2014602118.

Hahn, J., Hirano, K., and Karlan, D. (2011). Adaptive experimental design using the propensity score. *Journal of Business & Economic Statistics*, 29(1):96–108.

Hansen, M. H. and Hurwitz, W. N. (1942). Relative efficiencies of various sampling units in population inquiries. *Journal of the American Statistical Association*, 37(217):89–94.

Kasy, M. and Sautmann, A. (2021). Adaptive treatment assignment in experiments for policy choice. *Econometrica*, 89(1):113–132.

Kiefer, J. and Wolfowitz, J. (1960). The equivalence of two extremum problems. *Canadian Journal of Mathematics*, 12:363–366.

Kieser, M. and Friede, T. (2003). Simple procedures for blinded sample size adjustment that do not affect the type I error rate. *Statistics in Medicine*, 22(23):3571–3581.

Kish, L. (1965). Survey sampling. new york: John wesley & sons. *Am Polit Sci Rev*, 59(4):1025.

Knox, S. A. and Chondros, P. (2004). Observed intra-cluster correlation coefficients in a cluster survey sample of patient encounters in general practice in australia. *BMC Medical Research Methodology*, 4(1):30.

Lake, S., Kammann, E., Klar, N., and Betensky, R. (2002). Sample size re-estimation in cluster randomization trials. *Statistics in Medicine*, 21(10):1337–1350.

Lattimore, T. and Szepesvári, C. (2020). *Bandit algorithms*. Cambridge University Press.

Li, D., Raymond, L., and Bergman, P. (2026). Hiring as exploration. *Review of Economic Studies*, 93(2):1200–1240.

Lu, J. (2025). Big data analysis. *Springer Books*.

Meinshausen, N., Meier, L., and Bühlmann, P. (2009). P-values for high-dimensional regression. *Journal of the American Statistical Association*, 104(488):1671–1681.

Miao, W., Gel, Y. R., and Gastwirth, J. L. (2006). A new test of symmetry about an unknown median. In *Random walk, sequential analysis and related topics: A festschrift in honor of Yuan-Shih Chow*, pages 199–214. World Scientific.

Moerbeek, M., van Breukelen, G. J. P., and Berger, M. P. F. (2000). Design issues for experiments in multilevel populations. *Journal of Educational and Behavioral Statistics*, 25(3):271–284.

Neopane, O., Ramdas, A., and Singh, A. (2024). Logarithmic neyman regret for adaptive estimation of the average treatment effect. *arXiv preprint arXiv:2411.14341*.

Neopane, O., Ramdas, A., and Singh, A. (2025). Optimistic algorithms for adaptive estimation of the average treatment effect. *arXiv preprint arXiv:2502.04673*.

Nie, X., Tian, X., Taylor, J., and Zou, J. (2018). Why adaptively collected data have negative bias and how to correct for it. In *International Conference on Artificial Intelligence and Statistics*, pages 1261–1269. PMLR.

Raudenbush, S. W. (1997). Statistical analysis and optimal design for cluster randomized trials. *Psychological Methods*, 2(2):173–185.

Rinott, Y. (1978). On two-stage selection procedures and related probability-inequalities. *Communications in Statistics-Theory and methods*, 7(8):799–811.

Rudelson, M. and Vershynin, R. (2013). Hanson-wright inequality and sub-gaussian concentration.

Sedgwick, P. (2015). Multistage sampling. *Bmj*, 351.

Shen, Y. and Fisher, L. (1999). Statistical inference for self-designing clinical trials with a one-sided hypothesis. *Biometrics*, 55(1):190–197.

Spiegelhalter, D. J. (2001). Bayesian methods for cluster randomized trials with continuous responses. *Statistics in Medicine*, 20(3):435–452.

Stein, C. (1945). A two-sample test for a linear hypothesis whose power is independent of the variance. *Annals of Mathematical Statistics*, 16(3):243–258.

Tabord-Meehan, M. (2023). Stratification trees for adaptive randomisation in randomised controlled trials. *Review of Economic Studies*, 90(5):2646–2673.

Tam, B. (2005). Household sample surveys in developing and transition countries.

Tschuprow, A. (1923). On the two different aspects of the representative method: the method of stratified son the mathematical expectation of the moments of frequency distributions in the case of correlated observation sampling and the method of purposive selection. *Metron*, 2(646-683):192.

Turner, R. M., Prevost, A. T., and Thompson, S. G. (2005). Prior distributions for the intracluster correlation coefficient, based on multiple previous estimates, and their application in cluster randomized trials. *Clinical Trials*, 2(2):108–118.

van Breukelen, G. J. P. and Candel, M. J. J. M. (2015). Efficient design of cluster randomized and multicentre trials with unknown intraclass correlation. *Statistical Methods in Medical Research*, 24(5):540–556.

van Breukelen, G. J. P. and Candel, M. J. J. M. (2018). Efficient design of cluster randomized trials with treatment-dependent costs and treatment-dependent unknown variances. *Statistics in Medicine*, 37:3027–3046.

van Breukelen, G. J. P. and Candel, M. J. J. M. (2021). Maximin design of cluster randomized trials with heterogeneous costs and variances. *Biometrical Journal*, 63(7):1444–1463.

van Schie, S. and Moerbeek, M. (2014). Re-estimating sample size in cluster randomised trials with active recruitment within clusters. *Statistics in Medicine*, 33(19):3253–3268.

Viviano, D. (2022). Experimental design under network interference. *arXiv preprint arXiv:2003.08421*.

Wager, S. (2024). Causal inference: A statistical learning approach. *URL https://web. stanford. edu/swager/causal_inf_book. pdf*.

Wittes, J. and Brittain, E. (1990). The role of internal pilot studies in increasing the efficiency of clinical trials. *Statistics in Medicine*, 9(1-2):65–72.

Wittes, J., Schabenberger, O., Zucker, D. M., Brittain, E., and Proschan, M. (1999). Internal pilot studies I: Type I error rate of the naive t-test. *Statistics in Medicine*, 18(24):3481–3491.

Wooldridge, J. M. (2003). Cluster-sample methods in applied econometrics. *American Economic Review*, 93(2):133–138.

Yin, G. and Shen, Y. (2005). Adaptive design and estimation in randomized clinical trials with correlated observations. *Biometrics*, 61(2):362–369.

Zhang, K., Janson, L., and Murphy, S. (2021). Statistical inference with m-estimators on adaptively collected data. *Advances in neural information processing systems*, 34:7460–7471.

A Proofs and Appendix to Section 3

A.1 Generalized Problem

Let Assumptions 3.1, 3.2 and 3.3 hold throughout. Consider the general problem in which the designer samples an arbitrary number of units $N^k \geq 0$ from each cluster k subject to the budget constraint. Let $\boldsymbol{N} = (N^k)_k$, and define $\mathcal{K} = \{k : N^k > 0\}$ with $K = |\mathcal{K}|$. The variance of the average-of-averages estimator is given by

$$W(\boldsymbol{N}) = \text{Var}(\hat{\mu}_{\boldsymbol{N}}) = \text{Var}\left(\frac{1}{K}\sum_{k\in\mathcal{K}}\frac{1}{N^k}\sum_{i=1}^{N^k} y_{ki}\right) = \text{Var}\left(\mu + \frac{1}{K}\sum_{k\in\mathcal{K}} u_k + \frac{1}{K}\sum_{k\in\mathcal{K}}\frac{1}{N^k}\sum_{i=1}^{N^k}\varepsilon_{ki}\right)$$
$$= \frac{K}{K^2}\tau^2 + \frac{1}{K^2}\sum_{k\in\mathcal{K}}\frac{\sigma^2}{N^k} = \frac{\tau^2}{K}\left(1 + \frac{\lambda_0}{K}\sum_{k\in\mathcal{K}}\frac{1}{N^k}\right) = \frac{\tau^2}{K}\left(1 + \frac{\lambda_0}{N^H(\boldsymbol{N})}\right)$$

where the fourth equality uses cross-cluster independence and $u_k \perp\!\!\!\perp \varepsilon_{ki}$ (Assumption 3.1), and

$$N^H(\boldsymbol{N}) = \left(\frac{1}{K}\sum_{k\in\mathcal{K}}\frac{1}{N^k}\right)^{-1}$$

is the harmonic mean of the sampled depths. As expected, when $N^k = N$ for all $k \in \mathcal{K}$, the expression reduces to (2). The generalized problem can be written as

$$\min_{\boldsymbol{N}} W(\boldsymbol{N}) \qquad \text{s.t.} \quad KF + V\sum_{k\in\mathcal{K}} N^k \leq B. \tag{16}$$

Conditional on $\mathcal{K}$, $W(\boldsymbol{N})$ is strictly decreasing in each N^k and so the budget constraint binds. Moreover, by conditioning on K and solving from the budget constraint, the total number of units $S = \sum_{k\in\mathcal{K}} N^k$ remains fixed at the optimum, i.e. $S = (B - KF)/V$. The variance minimization problem then reduces to minimizing $\sum_{k\in\mathcal{K}} 1/N^k$ subject to a constant total number of units S. Since $x \mapsto 1/x$ is strictly convex on $(0, \infty)$, Jensen's inequality gives

$$\frac{1}{K}\sum_{k\in\mathcal{K}}\frac{1}{N^k} \;\geq\; \frac{1}{\frac{1}{K}\sum_{k\in\mathcal{K}} N^k} \;=\; \frac{K}{S}$$

Equivalently $N^H(\boldsymbol{N}) \leq S/K$, with equality if and only if $N^k = S/K = N$ for all $k \in \mathcal{K}$. The variance minimizing allocation is therefore balanced, conditional on K. Next, impose a balanced allocation $N^k = N$ and solve for K from the budget constraint such that $K(N) = B/(F + VN)$. Finally,

$$W(K(N), N) = \frac{\tau^2(F + VN)}{B}\left(1 + \frac{\lambda_0}{N}\right) = \frac{\tau^2}{B}\left[F + V\lambda_0 + VN + \frac{F\lambda_0}{N}\right]$$

The expanded form makes both optimality conditions (FOC and SOC) immediate,

$$W_N = \frac{\tau^2}{B}\left[V - \frac{F\lambda_0}{N^2}\right] = 0 \implies N^*(\lambda_0) = \sqrt{\lambda_0 F/V}, \qquad W_{NN} = \frac{2F\lambda_0\tau^2}{BN^3} > 0 \;\text{ for } N > 0$$

The stationary point is a global minimum and interior in the feasible region due to strict convexity on $(0, \infty)$ and Assumption 3.3, respectively. Evaluating the variance object at $N^*(\lambda_0)$ returns the oracle

variance

$$W_0 = \frac{\tau^2}{B}\left[F + V\lambda_0 + 2\sqrt{FV\lambda_0}\right] \;=\; \frac{\tau^2\left(\sqrt{F}+\sqrt{V\lambda_0}\right)^2}{B}$$

The oracle allocation, optimal map and oracle variance of the generalized problem all coincide with those of the balanced problem in (3).

A.2 Experimental Design

Assumption A.1. *Linear Additive Model and Error Distributions under Potential Outcomes. For $d \in \{0,1\}$, let the potential outcomes of the outcome of interest y_{ki}^d follow an additive model with nested errors*

$$y_{ki}^d = \mu_d + u_k^d + \varepsilon_{ki}^d$$

where μ_d is the expected potential outcome under treatment $D_{ki} = d$, u_k^d is the cluster-level error for potential outcome d and ε_{ki}^d is the unit-level error for potential outcome d. $\{(u_k^0, u_k^1)\}_k$ is an iid sequence with $\mathbb{E}[u_k^d] = 0$, $\mathbb{E}[(u_k^d)^2] = \tau_d^2 > 0$, and $Cov(u_k^0, u_k^1) = \tau_{01}$; $\{(\varepsilon_{ki}^0, \varepsilon_{ki}^1)\}_{k,i}$ is an iid sequence with $\mathbb{E}[\varepsilon_{ki}^d] = 0$ and $\mathbb{E}[(\varepsilon_{ki}^d)^2] = \sigma_d^2 > 0$; $(u_k^d)_{k,d} \perp\!\!\!\perp (\varepsilon_{ki}^d)_{k,i,d}$. Moreover, treatment assignment is independent of the full collection of potential outcomes.

A.2.1 Clusters as Treatment Level

The designer samples $K \ll \overline{K}$ clusters and $N \ll \overline{N}$ units per cluster, and randomly allocates $K/2$ clusters to treatment and $K/2$ to control. Let $D_{ki} \in \{0,1\}$ denote the treatment status of unit i in cluster k. Let $y_{ki}(d, \boldsymbol{d})$ denote the potential outcome of unit i in cluster k when $D_{ki} = d$ and $(D_{kj})_{j\neq i} = \boldsymbol{d}$. Under cluster-level assignment $\boldsymbol{d} = (d, \ldots, d)$, so the potential outcome admits a simplified representation $y_{ki}(d, \boldsymbol{d}) = y_{ki}(d)$.

Assumption A.2. *Linear Additive Model and Error Distributions under Potential Outcomes. For $d \in \{0,1\}$, let the potential outcomes of interest $y_{ki}(d) = y_{ki}^d$ follow an additive model with nested errors*

$$y_{ki}^d = \mu_d + u_k^d + \varepsilon_{ki}^d$$

where μ_d is the expected potential outcome under treatment $D_{ki} = d$, u_k^d is the cluster-level error for potential outcome d and ε_{ki}^d is the unit-level error for potential outcome d. $\{(u_k^0, u_k^1)\}_k$ is an iid sequence with $\mathbb{E}[u_k^d] = 0$, $\mathbb{E}[(u_k^d)^2] = \tau_d^2 > 0$, and $Cov(u_k^0, u_k^1) = \tau_{01}$; $\{(\varepsilon_{ki}^0, \varepsilon_{ki}^1)\}_{k,i}$ is an iid sequence with $\mathbb{E}[\varepsilon_{ki}^d] = 0$ and $\mathbb{E}[(\varepsilon_{ki}^d)^2] = \sigma_d^2 > 0$; $(u_k^d)_{k,d} \perp\!\!\!\perp (\varepsilon_{ki}^d)_{k,i,d}$. Moreover, treatment assignment is independent of the full collection of potential outcomes.

The estimand is the total average treatment effect $\text{TATE} = \mathbb{E}[y_{ki}(1,\mathbf{1}) - y_{ki}(0,\mathbf{0})] = \mathbb{E}[y_{ki}(1) - y_{ki}(0)] = \mu_1 - \mu_0$, and the natural estimator is the difference in cluster means

$$\hat{\text{TATE}}_{K,N} = \frac{2}{K}\left(\sum_{k:D_k=1} \hat{\mu}_N^k - \sum_{k:D_k=0} \hat{\mu}_N^k\right)$$

where $D_k \in \{0,1\}$ is the treatment status of cluster k. Under Assumption A.2, $\hat{\text{TATE}}_{K,N}$ is unbiased for the TATE. Because a cluster contributes $u_k^{D_k}$ for exactly one d, the arms draw on disjoint sets of

clusters and no covariance term $\mathrm{Cov}(u_k^1, u_k^0)$ arises,

$$\mathrm{Var}\big(\hat{\mathrm{TATE}}_{K,N}\big) = \frac{4}{K^2}\cdot\frac{K}{2}\left(\tau_1^2 + \frac{\sigma_1^2}{N}\right) + \frac{4}{K^2}\cdot\frac{K}{2}\left(\tau_0^2 + \frac{\sigma_0^2}{N}\right) = \frac{2}{K}\left(\tau_1^2 + \tau_0^2 + \frac{\sigma_1^2+\sigma_0^2}{N}\right)$$

so that, in the notation of Table 1,

$$W(K,N) = \frac{2\tau^2}{K}\left(1+\frac{\lambda}{N}\right), \qquad \tau^2 = \tau_1^2+\tau_0^2, \qquad \sigma^2 = \sigma_0^2+\sigma_1^2, \qquad \lambda = \sigma^2/\tau^2$$

A.2.2 Clusters as Blocks under No Interference

The designer samples $K \ll \overline{K}$ clusters and $N \ll \overline{N}$ units per cluster, and randomly allocates treatment to $N/2$ units within each cluster. Assuming no interference $y_{ki}(d, \boldsymbol{d}) = y_{ki}(d)$, which is further assumed to follow the additive model with nested errors in Assumption A.2. The relevant estimand is the average treatment effect $\mathrm{ATE} = \mathbb{E}[y_{ki}(1) - y_{ki}(0)] = \mu_1 - \mu_0$, and the natural estimator is the average of within-cluster differences in means,

$$\hat{\mathrm{ATE}}_{K,N} = \frac{2}{KN}\sum_{k=1}^{K}\left(\sum_{i:D_{ki}=1} y_{ki} - \sum_{i:D_{ki}=0} y_{ki}\right)$$

Under Assumption A.2, $\hat{\mathrm{ATE}}_{K,N}$ is unbiased for the ATE. The term τ_{01} now appears in the variance of the estimator as both potential cluster errors (u_k^0, u_k^1) are active within the same cluster. In particular, $\mathrm{Var}(\hat{\mathrm{ATE}}_{K,N}) = \frac{(\tau_1^2+\tau_0^2-2\tau_{01})}{K} + \frac{2(\sigma_1^2+\sigma_0^2)}{KN}$. Re-writing into the notation of Table 1,

$$W(K,N) = \frac{\tau^2}{K}\left(1+\frac{\lambda}{N}\right), \qquad \tau^2 = \tau_1^2+\tau_0^2-2\tau_{01}, \qquad \sigma^2 = 2(\sigma_0^2+\sigma_1^2), \qquad \lambda = \sigma^2/\tau^2$$

Remark A.3. *Degenerate Corner. When $u_k^1 = u_k^0$ almost surely, $\tau_{01} = \tau_1^2 = \tau_0^2$ and hence $\tau^2 = 0$. In other words, within-cluster differencing removes the cluster-level noise entirely, so the variance is minimized by sampling a single cluster as deeply as the budget allows. Assumption 3.2 excludes this possibility by imposing $\tau_m^2 > 0$.*

B Proofs of Section 4

Proof of Lemma 4.1. Recall the definition $M_{k,N} = u_k + \bar{\varepsilon}_{k,N} = \hat{\mu}_N^k - \mu$ where $\bar{\varepsilon}_{k,N} = \frac{1}{N}\sum_{i=1}^N \varepsilon_{ki}$. Let $\bar{M}_{K,N} = \frac{1}{K}\sum_{k=1}^K M_{k,N}$. Under Assumption 3.1, $(M_{k,N})_k$ is *iid*, mean zero, with $\mathbb{E}[M_{k,N}^2] = \tau^2 + \sigma^2/N = \theta_N^2 < \infty$.

Unbiasedness. Since each $y_{ki} - \hat{\mu}_N^k = \varepsilon_{ki} - \bar{\varepsilon}_{k,N}$ has mean zero and variance $\frac{N-1}{N}\sigma^2$,

$$\mathbb{E}\big[S_{k,N}^2\big] = \frac{1}{N-1}\sum_{i=1}^{N}\mathrm{Var}\big(\varepsilon_{ki} - \bar{\varepsilon}_{k,N}\big) = \frac{1}{N-1}\cdot N\cdot\frac{N-1}{N}\sigma^2 = \sigma^2 \implies \mathbb{E}\big[\hat{w}_{K,N}^2\big] = \sigma^2$$

Moreover, by the sample-variance identity applied to the *iid* sequence $(\hat{\mu}_N^k)_k$,

$$\mathbb{E}\big[\hat{b}_{K,N}^2\big] = \frac{1}{K-1}\,\mathbb{E}\left[\sum_{k=1}^{K}\big(M_{k,N} - \bar{M}_{K,N}\big)^2\right] = \mathrm{Var}\big(M_{k,N}\big) = \tau^2 + \frac{\sigma^2}{N} = \theta_N^2$$

(i) Fix $N \geq 2$. Observe that $\hat{w}^2_{K,N} = \frac{1}{K}\sum_{k=1}^{K} S^2_{k,N}$ is an average of *iid* summands and $\mathbb{E}|S^2_{k,N}| = \sigma^2 < \infty$, with (cross-cluster) independence following from Assumption 3.1. The SLLN along the K dimension then yields $\hat{w}^2_{K,N} \xrightarrow[K]{\text{a.s.}} \sigma^2$.

Fix $K \geq 1$. Exchanging the order of summation and re-writing yields

$$\hat{w}^2_{K,N} = \frac{1}{K}\sum_{k=1}^{K} \frac{1}{N-1}\sum_{i=1}^{N} \left(\varepsilon_{ki} - \bar{\varepsilon}_{k,N}\right)^2 = \frac{1}{K}\sum_{k=1}^{K} \frac{N}{N-1}\left[\frac{1}{N}\sum_{i=1}^{N}\varepsilon^2_{ki} - \bar{\varepsilon}^2_{k,N}\right]$$

The SLLN gives $\frac{1}{N}\sum_i \varepsilon^2_{ki} \xrightarrow[N]{\text{a.s.}} \sigma^2$ and $\bar{\varepsilon}_{k,N} \xrightarrow[N]{\text{a.s.}} 0$, for each cluster k, and so the term in brackets converges a.s. to σ^2 under the CMT. Since $N/(N-1) \to 1$ and the outer average is taken over finitely many k, $\hat{w}^2_{K,N} \xrightarrow[N]{\text{a.s.}} \sigma^2$.

(ii) Fix $N \geq 2$. Using the same decomposition and re-writing,

$$\hat{b}^2_{K,N} = \frac{1}{K-1}\sum_{k=1}^{K} \left(M_{k,N} - \bar{M}_{K,N}\right)^2 = \frac{K}{K-1}\left[\frac{1}{K}\sum_{k=1}^{K} M^2_{k,N} - \bar{M}^2_{K,N}\right]$$

The SLLN applied to the *iid* sequences $(M^2_{k,N})_k$ and $(M_{k,N})_k$ gives $\frac{1}{K}\sum_k M^2_{k,N} \xrightarrow[K]{\text{a.s.}} \tau^2 + \sigma^2/N$ and $\bar{M}_{K,N} \xrightarrow[K]{\text{a.s.}} 0$, respectively. Since $K/(K-1) \to 1$, $\hat{b}^2_{K,N} \xrightarrow[K]{\text{a.s.}} \tau^2 + \sigma^2/N$ under the CMT.

(iii) Combining parts (i) and (ii) under the CMT, the unclipped cluster-variance estimator $\tilde{\tau}^2_{K,N}$ satisfies

$$\tilde{\tau}^2_{K,N} = \hat{b}^2_{K,N} - \frac{1}{N}\hat{w}^2_{K,N} \xrightarrow[K]{\text{a.s.}} \left(\tau^2 + \frac{\sigma^2}{N}\right) - \frac{\sigma^2}{N} = \tau^2$$

The projection $\Pi_{[a,b]}$ onto a closed interval is 1-Lipschitz, hence continuous, so $\hat{\tau}^2_{K,N} = \Pi_{[\tau^2_m,\tau^2_M]}(\tilde{\tau}^2_{K,N}) \xrightarrow[K]{\text{a.s.}} \Pi_{[\tau^2_m,\tau^2_M]}(\tau^2) = \tau^2$, where the final equality uses that $\tau^2 \in [\tau^2_m, \tau^2_M]$ by Assumption 3.2. Because $\tau^2_m > 0$, the denominator is bounded away from zero and $\hat{w}^2_{K,N}/\hat{\tau}^2_{K,N} \xrightarrow[K]{\text{a.s.}} \sigma^2/\tau^2 = \lambda_0$ by the CMT. Invoking the projection argument and using $\lambda_0 \in [\lambda_m, \lambda_M]$,

$$\hat{\lambda}_{K,N} = \Pi_{[\lambda_m,\lambda_M]}\left(\frac{\hat{w}^2_{K,N}}{\hat{\tau}^2_{K,N}}\right) \xrightarrow[K]{\text{a.s.}} \Pi_{[\lambda_m,\lambda_M]}(\lambda_0) = \lambda_0 \qquad \square$$

Lemma B.1. *Random-Index Composition. Let $(X_M)_{M\geq 1}$ be a sequence of random variables with $X_M \xrightarrow[M]{\text{a.s.}} X$, and let $(M_B)_B$ be a sequence of $\mathbb{N}$-valued random variables where $\mathbb{P}(M_B \geq m) \to 1$ for every fixed $m \in \mathbb{N}$. Then $X_{M_B} \xrightarrow[B]{p} X$.*

Proof of Lemma B.1. Fix $\epsilon, \delta > 0$ and set $T_M = \sup_{j\geq M}|X_j - X|$, which is measurable and non-increasing in M. Almost sure convergence of X_M implies $T_M \downarrow 0$ a.s., and so there exists a finite M_0 such that $\mathbb{P}(T_{M_0} > \epsilon) < \delta/2$. Since $M_B \xrightarrow[B]{p} \infty$, there exists a finite (budget) B_0 such that $\mathbb{P}(M_B < M_0) < \delta/2$ for all $B \geq B_0$. Observe that on the event $\{M_B \geq M_0\}$, $|X_{M_B} - X| \leq T_{M_0}$, hence for every $B \geq B_0$,

$$\mathbb{P}\left(|X_{M_B} - X| > \epsilon\right) \leq \mathbb{P}(M_B < M_0) + \mathbb{P}(T_{M_0} > \epsilon) < \delta \qquad \square$$

Remark B.2. *Lemma B.1 is invoked repeatedly to jump between a.s. convergence over a fixed diver-*

gent sequence and convergence in probability over a random divergent sequence, both in performance and inference results. Furthermore, observe that the lemma operates without any restriction on the dependence between the random index $(M_B)_B$ and the sequence $(X_M)_M$.

Proof of Lemma 4.4. By Lemma 4.1(iii), $\hat{\lambda}_{K,N_p} \xrightarrow[K]{\text{a.s.}} \lambda_0$ along any non-random diverging sequence of widths with $N_p \geq 2$ fixed. Since $K_{pB} \xrightarrow[p]{} \infty$, Lemma B.1 applies with $X_K = \hat{\lambda}_{K,N_p}$ and $M_B = K_{pB}$, therefore $\hat{\lambda}_{pB} \xrightarrow[K]{p} \lambda_0$. Define the map

$$h(\lambda) = (F + \sqrt{FV\lambda})^{-1}$$

which I will invoke repeatedly in the derivations below. The map $h(\cdot)$ is strictly decreasing, continuously differentiable and L_h-Lipschitz on $[\lambda_m, \lambda_M]$, provided $\lambda_m > 0$. L_h can be defined implicitly as $L_h = \sup_{\lambda \in [\lambda_m, \lambda_M]} |h'(\lambda)| < \infty$. Applying the CMT for any $\lambda \in [\lambda_m, \lambda_M]$ returns $\nu_B^\pi = h(\hat{\lambda}_{pB}) \xrightarrow[p]{} h(\lambda_0) = \nu_0$ and $N_B^\pi = N^*(\hat{\lambda}_{pB}) \xrightarrow[p]{} N_0$.

For the integer statement in the footnote, the flooring operator $\lfloor \cdot \rfloor$ is continuous at every non-integer point, so when $N_0 \notin \mathbb{N}$, $\lfloor N_B^\pi \rfloor \xrightarrow[p]{} N_{\underline{0}}$ by the CMT. Trivially, convergence in probability of an integer-valued sequence to an integer constant is equivalent to $\mathbb{P}(\lfloor N_B^\pi \rfloor = N_{\underline{0}}) \to 1$. □

C Proofs of Section 5

Proof of Equation (7). Let Assumptions 3.1, 3.2, 3.3 hold throughout. Starting from the results in Appendix A,

$$W(K(N), N) = \frac{\tau^2}{B}\left[F + V\lambda_0 + VN + \frac{F\lambda_0}{N}\right]$$

Recall $N^*(\lambda) = \sqrt{\lambda F/V}$, hence $VN^*(\lambda) = \sqrt{FV\lambda}$ and $F\lambda_0/N^*(\lambda) = \sqrt{FV}\ \lambda_0/\sqrt{\lambda}$. Therefore

$$W(K^*(\lambda), N^*(\lambda); (\tau^2, \lambda_0)) = \frac{\tau^2}{B}\left[F + V\lambda_0 + \sqrt{FV}\left(\sqrt{\lambda} + \frac{\lambda_0}{\sqrt{\lambda}}\right)\right]$$

Extracting $\sqrt{FV}$ as a common factor and substracting W_0 returns,

$$\Delta = \frac{\tau^2\sqrt{FV}}{B}\left(\sqrt{\lambda} - 2\sqrt{\lambda_0} + \frac{\lambda_0}{\sqrt{\lambda}}\right) = \frac{\tau^2\sqrt{FV}}{B} \cdot \frac{\lambda - 2\sqrt{\lambda\lambda_0} + \lambda_0}{\sqrt{\lambda}} = \frac{\tau^2\sqrt{FV}}{B} \cdot \frac{\left(\sqrt{\lambda} - \sqrt{\lambda_0}\right)^2}{\sqrt{\lambda}}$$

□

C.1 Static Policies

Proof of Lemma 5.2. For part *(i)*, observe that

$$\frac{\left(\sqrt{\lambda_b} - \sqrt{\lambda_0}\right)^2}{\sqrt{\lambda_b}} = \frac{(\lambda_b - \lambda_0)^2}{\left(\sqrt{\lambda_b} + \sqrt{\lambda_0}\right)^2\sqrt{\lambda_b}} \in \left[\frac{(\lambda_b - \lambda_0)^2}{4\lambda_M^{3/2}}, \frac{(\lambda_b - \lambda_0)^2}{4\lambda_m^{3/2}}\right]$$

where I used $\lambda_b, \lambda_0 \in [\lambda_m, \lambda_M]$. Bring this result into (7) and substitute $(\lambda_b - \lambda_0)^2 = b^2$. The resulting bounds are of order $\Theta(1/B)$ for $b \neq 0$ as desired. For part *(ii)*, note that $\frac{\tau^2\sqrt{FV}}{B} \cdot \frac{\left(\sqrt{\lambda} - \sqrt{\lambda_0}\right)^2}{\sqrt{\lambda}}$ is monotonically increasing in $\tau^2 > 0$, and so the maximum is achieved at $\tau^2 = \tau_M^2$. Additionally, for

some fixed λ, the map $\sqrt{\lambda_0} \mapsto (\sqrt{\lambda} - \sqrt{\lambda_0})^2$ is the square of an affine function, hence convex, and attains its maximum at the end points, i.e. the excess variance maximizer λ is in $\{\lambda_m, \lambda_M\}$, whichever yields a higher loss. The minimax policy is then given by

$$\min_{\lambda} \max \left\{ \frac{\left(\sqrt{\lambda} - \sqrt{\lambda_M}\right)^2}{\sqrt{\lambda}}, \frac{\left(\sqrt{\lambda} - \sqrt{\lambda_m}\right)^2}{\sqrt{\lambda}} \right\}$$

Solving recovers $\lambda_{mM} = \left(\frac{\sqrt{\lambda_m}+\sqrt{\lambda_M}}{2}\right)^2$. Finally, evaluate the worst case excess variance at $\lambda = \lambda_{mM}$,[57]

$$\max_{\omega \in \Omega} \Delta_{\pi_{mM}} = \frac{\tau_M^2 \sqrt{FV}}{B} \cdot \frac{\left(\frac{\sqrt{\lambda_m}+\sqrt{\lambda_M}}{2} - \sqrt{\lambda_M}\right)^2}{\frac{\sqrt{\lambda_m}+\sqrt{\lambda_M}}{2}} = \frac{\tau_M^2 \sqrt{FV}}{2B} \cdot \frac{\left(\sqrt{\lambda_m} - \sqrt{\lambda_M}\right)^2}{\sqrt{\lambda_m} + \sqrt{\lambda_M}} \quad \square$$

C.2 Upper Bounds

This section is organized as follows. I first introduce the notion of centered squared errors, that will be invoked repeatedly in performance and inference proofs. I then discuss some of their concentration properties under subgaussianity. Finally, I leverage these results in the derivation of an intermediate lemma that upper bounds the expected square deviation $\mathbb{E}[(\hat{\lambda}_p - \lambda_0)^2]$, as hinted in the main text. The proof of Theorem 5.4 will then follow as a simple corollary.

Definition C.1. *Centered Squared Errors. Let $\varepsilon_k = (\varepsilon_{k1}, \varepsilon_{k2}, \dots)$. Recall the definitions of $\bar{\varepsilon}_{k,N}$, $S^2_{k,N}$, $\hat{w}^2_{K,N}$, $M_{k,N}$ and $\bar{M}_{K,N}$ from Section 3 and Appendix B. Define*

$$\alpha_{k,N} = S^2_{k,N} - \sigma^2, \qquad \beta_{k,N} = M^2_{k,N} - \theta^2_N$$

where $\mathbb{E}[\alpha_{k,N}] = \mathbb{E}[\beta_{k,N}] = 0$.

Remark C.2. *Between-Variance Decomposition. The proofs below will exploit consistently the fact that the within- and between-cluster variances are near-martingales. The claim is exact for $\hat{w}^2_{K,N}$ as it is a sum of iid zero-mean errors,*

$$\hat{w}^2_{K,N} - \sigma^2 = \frac{1}{K} \sum_{k=1}^{K} \alpha_{k,N}$$

$\hat{b}^2_{K,N}$, however, cannot be written as the sum of iid zero-mean errors because individual terms are correlated through the de-meaning object $\bar{M}_{K,N}$ (the grand error mean). An approximate decomposition

[57] I am inputting $\lambda_0 = \lambda_M$. This is without loss as at $\lambda = \lambda_m$, the excess variance is equalized across endpoints $\{\lambda_m, \lambda_M\}$ and so any of the two endpoints would deliver the same excess variance.

remains possible

$$\begin{aligned}\hat{b}^2_{K,N} - \theta^2_N &= \frac{1}{K-1}\sum_{k=1}^{K}\left(M_{k,N} - \bar{M}_{K,N}\right)^2 - \theta^2_N \\ &= \frac{1}{K-1}\left(\sum_{k=1}^{K} M^2_{k,N} - K\bar{M}^2_{K,N}\right) - \theta^2_N \\ &= \frac{1}{K-1}\left(K\theta^2_N + \sum_{k=1}^{K}(M^2_{k,N} - \theta^2_N) - K\bar{M}^2_{K,N}\right) - \theta^2_N \\ &= \frac{1}{K-1}\sum_{k=1}^{K}\beta_{k,N} + \frac{\theta^2_N}{K-1} - \frac{K}{K-1}\bar{M}^2_{K,N} \approx \frac{1}{K-1}\sum_{k=1}^{K}\beta_{k,N}\end{aligned}$$

Lemma C.3. *Concentration of Centered Square Errors. Under Assumptions 3.1, 3.2, 3.3 and 5.3, $M_{k,N}, \alpha_{k,N}, \beta_{k,N}$ are subexponential with suitable parameters (κ, b).*

Proof of Lemma C.3. Consider some set of strictly positive constants $\{c_j\}_{j\leq 4}$. Define $\kappa_M = \sqrt{\kappa_u^2 + \kappa_\varepsilon^2/N}$. By independence of $u_k \perp\!\!\!\perp \bar{\varepsilon}_{k,N}$, $M_{k,N}$ is κ_M-subgaussian. Hence, it is subexponential with parameters $\kappa = \kappa_M^2, b_M = 0$. $\beta_{k,N}$ is the centered square of $M_{k,N}$—κ_M-subgaussian variable—thus it is subexponential with parameters $\kappa_B = c_1\kappa_M^4$, $b_B = c_2\kappa_M^2$. Next, consider the centering matrices

$$\boldsymbol{P}_N = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}', \qquad \tilde{\boldsymbol{P}}_N = \frac{1}{N-1}\boldsymbol{P}_N$$

then

$$\alpha_{k,N} = \frac{1}{N-1}\left(\varepsilon_k'\boldsymbol{P}_N\varepsilon_k - (N-1)\sigma^2\right) = \varepsilon_k'\tilde{\boldsymbol{P}}_N\varepsilon_k - \sigma^2$$

such that $\mathbb{E}[\varepsilon_k'\boldsymbol{P}_N\varepsilon_k] = \sigma^2\mathrm{tr}(\boldsymbol{P}_N) = \sigma^2(N-1)$ and $\alpha_{k,N}$ is centered. By Hanson-Wright inequality (Rudelson and Vershynin, 2013), $\alpha_{k,N}$ is subexponential with parameters $\kappa_A = c_3\kappa_\varepsilon^4\|\tilde{\boldsymbol{P}}_N\|_F^2$ and $b_A = c_4\kappa_\varepsilon^2\|\tilde{\boldsymbol{P}}_N\|$, where $\|\cdot\|_F$ is the Frobenius norm and $\|\cdot\|$ is the operator norm. Because $\boldsymbol{P}_N$ is an orthogonal projection of rank $N-1$, $\|\boldsymbol{P}_N\|_F^2 = (N-1)$ and $\|\boldsymbol{P}_N\| = 1$. After scaling, $\alpha_{k,N}$ is subexponential with parameters $\kappa_A = c_3\frac{\kappa_\varepsilon^4}{N-1}$ and $b_A = c_4\frac{\kappa_\varepsilon^2}{N-1}$. □

Lemma C.4. *Maximal Bernstein Inequality. Let $\{\xi_k\}$ be a sequence of iid, mean-zero, (κ, b)-subexponential random variables. Then, for every $K_+ \geq 1$ and $t > 0$,*

$$\mathbb{P}\left(\max_{1\leq K\leq K_+}\left|\sum_{k=1}^{K}\xi_k\right| \geq t\right) \leq 2\exp\left(-\frac{1}{2}\min\left(\frac{t^2}{\kappa\cdot K_+}, \frac{t}{b}\right)\right)$$

Proof of Lemma C.4. The proof is omitted from the present manuscript.[58] □

Lemma C.5. $\hat{\lambda}$ *Concentration for K-asymptotic Pilot Policies. Let π be an implementable pilot policy with pilot size (K_{pB}^π, N_p^π) where (i) $N_p^\pi = N_p$ fixed, and $K_{pB}^\pi \in \mathcal{I}_B = [K_{B-}, K_{B+}] \cap \mathbb{N}$ a.s.; (ii) $K_{B+}/K_{B-} \leq C_0$ for some finite positive constant C_0 not depending on B; and (iii) $K_{B-} \underset{B}{\to} \infty$. Under Assumptions 3.1, 3.2, 3.3 and 5.3, there exist two positive finite constants C_1 and B_0 such that*

[58] See Chapter 6 in Lu (2025) for a complete treatment.

for all $B \geq B_0$ and every $\varepsilon > 0$,

$$\mathbb{P}\left(\mid \hat{\lambda}_{K^{\pi}_{pB},N_p} - \lambda_0 \mid \geq \varepsilon\right) \leq 6\exp\left(-C_1 K_{B-}\varepsilon^2\right)$$

Consequently, for all $B \geq B_0$

$$\mathbb{E}\left[\left(\hat{\lambda}_{K^{\pi}_{pB},N_p} - \lambda_0\right)^2\right] \leq \frac{6}{C_1 K_{B-}}$$

Proof of Lemma C.5. Let $D_w := \sup_{K\in\mathcal{J}_B} \mid \hat{w}^2_{K,N_p} - \sigma^2 \mid$. For some $u > 0$

$$\begin{aligned}\mathbb{P}(D_w \geq u) &\leq \mathbb{P}\left(\frac{1}{K_{B-}} \max_{K_{B-}\leq K\leq K_{B+}} \left|\sum_{k=1}^{K} \alpha_{k,N_p}\right| \geq u\right) \leq 2\exp\left(-\frac{1}{2}\min\left(\frac{u^2K^2_{B-}}{K_{B+}\kappa_A}, \frac{uK_{B-}}{b_A}\right)\right)\\ &\leq 2\exp\left(-\frac{K_{B-}}{2}\min\left(\frac{u^2}{C_0\kappa_A}, \frac{u}{b_A}\right)\right)\end{aligned}$$

where the second inequality uses that α_{k,N_p} is sub-exponential with constants (κ_A, b_A) by Lemma C.3 and applies Lemma C.4 over α_{k,N_p} and $t = uK_{B-}$. Similarly, define $D_b := \sup_{K\in\mathcal{J}_B} \mid \hat{b}^2_{K,N_p} - \theta^2_{N_p} \mid$, and recall the decomposition in Remark C.2

$$\hat{b}^2_{K,N_p} - \theta^2_{N_p} = \frac{K}{K-1}\cdot\frac{1}{K}\sum_{k=1}^{K}\beta_{k,N_p} + \frac{\theta^2_{N_p}}{K-1} - \frac{K}{K-1}\left(\frac{1}{K}\sum_{k=1}^{K} M_{k,N_p}\right)^2$$

Define $D_B := \sup_{K\in\mathcal{J}_B} \mid \frac{1}{K}\sum_{k\leq K}\beta_{k,N_p} \mid$, $D_M := \sup_{K\in\mathcal{J}_B} \mid \frac{1}{K}\sum_{k\leq K} M_{k,N_p} \mid$, and $r_B := \frac{\tau^2_M(1+\lambda_M)}{K_{B-}-1} < \infty$ (by Assumption 3.2). Then, $D_b \leq 2D_B + 2D^2_M + r_B$, where I used that $K/(K-1) \leq 2$ for $K \geq 2$. Next, for every $u \geq 3r_B$,

$$\begin{aligned}\mathbb{P}(D_b \geq u) &\leq \mathbb{P}\left(D_B \geq \frac{u}{6}\right) + \mathbb{P}\left(D_M \geq \sqrt{\frac{u}{6}}\right) + \mathbb{P}\left(r_B \geq \frac{u}{3}\right)\\ &\leq 2\exp\left(-\frac{K_{B-}}{2}\min\left(\frac{u^2}{36C_0\kappa_B}, \frac{u}{6b_B}\right)\right) + 2\exp\left(-\frac{K_{B-}}{2}\frac{u}{6C_0\kappa_M}\right)\end{aligned}$$

where the third term is zero by selection of u and the main inequality follows from Lemma C.4 over $(\beta_{k,N_p}, M_{k,N_p})$. Recall that M_{k,N_p} is not only subexponential but subgaussian and so its bound does not include the subexponential refinement. Now I translate the bounds over $\hat{w}^2_{K,N_p}$ and $\hat{b}^2_{K,N_p}$ into bounds over $\hat{\tau}^2_{K,N_p}$ and $\hat{\lambda}_{K,N_p}$, by a double application of the triangle inequality. Specifically, $\tilde{\tau}^2_{K,N_p} - \tau^2 = (\hat{b}^2_{K,N_p} - \theta^2_{N_p}) - \frac{1}{N_p}(\hat{w}^2_{K,N_p} - \sigma^2)$, thus $\sup_{K\in\mathcal{J}_B} \mid \tilde{\tau}^2_{K,N_p} - \tau^2 \mid \leq D_b + D_w/N_p$. The same bound is mechanically preserved for the contraction $\hat{\tau}^2_{K,N_p} = \Pi_{[\tau^2_m,\tau^2_M]}(\tilde{\tau}^2_{K,N_p})$. Similarly,

$$\sup_{K\in\mathcal{J}_B} \mid \tilde{\lambda}_{K,N_p} - \lambda_0 \mid = \sup_{K\in\mathcal{J}_B}\left|\frac{\tau^2(\hat{w}^2_{K,N_p} - \sigma^2) - \sigma^2(\hat{\tau}^2_{K,N_p} - \tau^2)}{\tau^2\ \hat{\tau}^2_{K,N_p}}\right| \leq \frac{1}{\tau^2_m}\left[\left(1 + \frac{\lambda_M}{N_p}\right)D_w + \lambda_M D_b\right] \leq L(D_w + D_b)$$

with $L = (1+\lambda_M)/\tau^2_m$. The first inequality follows from $\hat{\tau}^2 \geq \tau^2_m$ by clipping, and $\sigma^2/\tau^2 = \lambda_0 \leq \lambda_M$ by compactness. The second inequality follows by noting that $N_p \geq 1$. The bound obtained for $\tilde{\lambda}$ is preserved after clipping. Therefore,

$$\mathbb{P}\left(\mid \hat{\lambda}_{K^{\pi}_{pB},N_p} - \lambda_0 \mid \geq \varepsilon\right) \leq \mathbb{P}\left(\sup_{K\in\mathcal{J}_B} \mid \hat{\lambda}_{K,N_p} - \lambda_0 \mid \geq \varepsilon\right) \leq \mathbb{P}\left(D_w \geq \frac{\varepsilon}{2L}\right) + \mathbb{P}\left(D_b \geq \frac{\varepsilon}{2L}\right)$$

Observe that the probability on the LHS is zero for $\varepsilon > \Lambda := \lambda_M - \lambda_m$ because $\hat{\lambda}_{K^{\pi}_{pB},N_p}$ and λ_0 both lie in $[\lambda_m, \lambda_M]$. Hence, it suffices to establish the inequality for $\varepsilon \in (0, \Lambda]$. To do so, I evaluate the bounds above on $u = \varepsilon/2L \in (0, u_M) := (0, \Lambda/2L)$. On a bounded range, the Bernstein exponent can be uniformly lower-bounded by a quadratic function of u, i.e.

$$\min\{au^2, bu\} \geq u^2 \cdot \min\left\{a, \frac{b}{u_M}\right\}$$

Therefore, the caps on probability can be upper bounded by simple expressions of the form $2\exp(-CK_{B-}\varepsilon^2)$, for positive finite constants (C_w, C_B, C_M). The constants only depend on the sub-exponential parameters, the window width C_0, Λ and L. Setting $C_1 := \min\{C_w, C_B, C_M\}$ delivers the statement in the Lemma with a prefactor of 6 for all $\varepsilon \in (\varepsilon_B, \Lambda] = (6Lr_B, \Lambda]$. Small deviations of the form $\varepsilon \in (0, \varepsilon_B)$ are inconsequential for large B because $\varepsilon_B = O(1/K_{B-})$ and $K_{B-} \underset{B}{\to} \infty$, so B_0 can be selected such that $C_1 K_{B-}\varepsilon_B^2 \leq \ln 6$ for all $B \geq B_0$. Then, for every $\varepsilon < \varepsilon_B$,

$$6\exp\left(-C_1 K_{B-}\varepsilon^2\right) \geq 6\exp\left(-C_1 K_{B-}\varepsilon_B^2\right) \geq 6e^{-\ln 6} = 1$$

and the probability statement holds trivially. The expectation bound is obtained by integrating over the tail, i.e.

$$\mathbb{E}\left[\left(\hat{\lambda}_{K^{\pi}_{pB},N_p} - \lambda_0\right)^2\right] = \int_0^{\infty} \mathbb{P}\left(|\ \hat{\lambda}_{K^{\pi}_{pB},N_p} - \lambda_0\ | > \sqrt{t}\right) dt \leq \int_0^{\infty} 6\exp\left(-C_1 K_{B-} t\right) dt = \frac{6}{C_1 K_{B-}}$$

□

Remark C.6. *Role of Clipping. Clipping plays two distinct roles in Lemma C.5. First, it bounds the estimation error by Λ, so that only deviations $\varepsilon \leq \Lambda$ carry probability and all Bernstein regimes collapse into the quadratic one, i.e. the clipped estimator is sub-gaussian on the effective range. Second, it bounds the denominator $\hat{\tau}^2$ away from zero. Both roles require the parameter space to be bounded away from degeneracy, i.e. $\tau_m^2 > 0$, as per Assumption 3.2. By contrast, the unclipped ratio $\tilde{\lambda}$ need not admit sub-exponential tails as it features extreme deviations coming from small-denominator issues.*

Proof of Theorem 5.4. I first argue that both algorithms satisfy the assumptions entering Lemma C.5. Both algorithms set $N^{\pi}_{pB} = N_m$ fixed. Moreover, $K^{\texttt{EtC-CS}}_{pB} = K_M$ deterministically, hence the window is degenerate $K_{B-} = K_{B+}$ and $C_0 = 1$. Similarly, $K^{\texttt{ACS}}_{pB} = \max\{s-1, K_M\} \in [K_M, K_m]$ a.s., therefore, the assumption is satisfied for $C_0 = K_m/K_M \leq 2h(\lambda_m)/h(\lambda_M)$ for B large enough. Finally, both `EtC-CS` and `ACS` trivially sample $K^{\pi}_{pB} \geq K_M \geq Bh(\lambda_M) - 1 \underset{B}{\to} \infty$.

Both policies are implementable, hence they both implement an optimal-map design with $\lambda = \hat{\lambda}_{K^{\pi}_{pB},N_m}$. Using the representation of the excess variance in (8) and the upper bound in the proof of Lemma 5.2

$$\mathbb{E}_{\pi}\left[\Delta_{\pi}\right] \leq \frac{\tau_M^2\sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{1}{B} \cdot \mathbb{E}\left[\left(\hat{\lambda}_{K^{\pi}_{pB},N_p} - \lambda_0\right)^2\right] \leq \frac{\tau_M^2\sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{6}{C_1\ B\ K_M}$$

where the second inequality follows from Lemma C.5. Recall $K_M = \Theta(B)$, so the right hand side is $O(B^{-2})$. Absorbing constants, $\mathbb{E}_{\pi}[\Delta_{\pi}] \leq C_{\pi}/B^2$ with C_{π} depending on $(\tau_M^2, \lambda_m, \lambda_M, F, V, N_m, C_0, C_1) \perp\!\!\!\perp B$. Setting $C_{\pi} = C_{\texttt{EtC-CS}}$ for `EtC-CS` and $C_{\pi} = C_{\texttt{ACS}}$ for `ACS` completes the proof with constants differing only in the window width C_0. □

C.3 Lower Bounds – Proof of Theorem 5.5

Policy Class. I first define the relevant comparison class Π. The policy class is fairly general and contains most reasonable policies as special cases. In particular, I consider policies which, in every period $t \in \{1, \dots, T\}$, select a cluster $k_t = k$ and sample at least one unit $N_t^{k_t} \geq 1$ until they run out of budget. As a result, the game contains at most $\lfloor B/V \rfloor$ rounds. If no additional observation can be sampled without violating the remaining budget, the algorithms must terminate. The same cluster k can be revisited several times. Let $N_T^k = \sum_{t=1}^T \mathbb{1}(k_t = k) N_t^{k_t}$ be the terminal number of samples of cluster k. Define as well $K_T = \sum_k \mathbb{1}(N_T^k > 0)$ and $\mathcal{K}_T = \{k : N_T^k > 0\}$. All policies must satisfy the budget constraint almost surely, i.e.

$$\mathbb{P}_\pi \left(\sum_{k \in \mathcal{K}_T} (F + V N_T^k) \leq B \right) = 1$$

Data Generating Process. Fix an instance $(\tau_0^2, \lambda_0) \in \text{int}(\Omega)$ with $\sigma^2 = \lambda_0 \tau_0^2$. Consider two DGPs (y_-, y_+) indexed by $\omega_- = (\tau_-^2, \lambda_-)$ and $\omega_+ = (\tau_+^2, \lambda_+)$, where $\lambda_\pm = \lambda_0 \pm \delta_B$ for some small positive $\delta_B \underset{B}{\to} 0$. $\tau_\pm^2 = \sigma^2/\lambda_\pm$, so that the individual-level variance σ^2 is common to both environments and only the cluster-level variance differs. For sufficiently large B, $(\tau_\pm^2, \lambda_\pm) \in \Omega$. The additive errors model parallels Equation (1),[59]

$$y_{\pm ki} = \mu + u_{\pm k} + \varepsilon_{ki} \qquad u_{\pm k} \sim N(0, \tau_\pm^2), \quad \varepsilon_{ki} \sim N(0, \sigma^2)$$

Separating Events. As is common practice in lower bound derivations, I characterize a separating event E, which in this case is a function of the terminal harmonic mean N_T^H. By linearity of the cost structure $K_T F + \sum_{k \in \mathcal{K}_T} V N_T^k = K_T(F + V \bar{N}_T)$, where $\bar{N}_T = \frac{1}{K_T} \sum_{k \in \mathcal{K}_T} N_T^k$ is the arithmetic mean. Recall that $N_T^H \leq \bar{N}_T$, hence $\frac{1}{K_T} \geq \frac{F + V N_T^H}{B}$. It follows

$$W(\boldsymbol{N}_T) = \frac{\tau^2}{K_T(\boldsymbol{N}_T)} \left(1 + \frac{\lambda}{N_T^H(\boldsymbol{N}_T)}\right) \geq \frac{\tau^2 (F + V N_T^H(\boldsymbol{N}_T))}{B} \left(1 + \frac{\lambda}{N_T^H(\boldsymbol{N}_T)}\right) = W^H(N_T^H)$$

Next, let $N_\pm = N^*(\lambda_\pm) = \sqrt{\lambda_\pm F/V}$, so $N_- < N_0 < N_+$. Define the separating event $E = \{N_T^H \leq N_0\}$. In environment ω_+, $W^H(\cdot; \omega_+)$ is strictly convex with minimum at $N_+ > N_0$, hence decreasing on $(0, N_+]$. It follows that

$$W(\boldsymbol{N}_T; \omega_+) - W(N_+; \omega_+) \geq W^H(N_T^H; \omega_+) - W^H(N_+; \omega_+) \underset{\text{on } E}{\geq} W^H(N_0; \omega_+) - W^H(N_+; \omega_+)$$

where the first inequality uses the harmonic mean relaxation and the fact that $W(N_+; \omega_+) = W^H(N_+; \omega_+)$, as per the balancedness argument in Appendix A. On E^c, $W^H(\cdot; \omega_-)$ is increasing on $[N_-, \infty)$ and $N_T^H > N_0 > N_-$, therefore the same inequality runs for ω_-. The same computation as in Equation (7)—with truth $\lambda_\pm$ and input λ_0—yields

$$W^H(N_0; \omega_\pm) - W^H(N_\pm; \omega_\pm) = \frac{\tau_\pm^2 \sqrt{FV}}{B} \cdot \frac{\left(\sqrt{\lambda_0} - \sqrt{\lambda_\pm}\right)^2}{\sqrt{\lambda_0}}$$

[59] Gaussian errors are exactly subgaussian, so the lower bound is established over a strict subclass of the environment considered in Theorem 5.5. The lower bound then carries over to the general environment class given in Assumption 3.1.

Finally, both branches can be uniformly bounded in terms of δ_B and common parameters by noting: (i) $\left(\sqrt{\lambda_0} - \sqrt{\lambda_\pm}\right)^2 = \delta_B^2/(\sqrt{\lambda_0} + \sqrt{\lambda_\pm})^2 \geq \delta_B^2/(4\lambda_M)$, (ii) $\lambda_0^{-1/2} \geq \lambda_M^{-1/2}$, and (iii) $\sigma^2/\lambda_M \leq \min\{\tau_-^2, \tau_+^2\}$

$$W^H(N_T^H; \omega_\pm) - W^H(N_\pm; \omega_\pm) \geq \frac{\sigma^2\sqrt{FV}}{4\lambda_M^{5/2}B}\,\delta_B^2 \qquad \text{on } E \text{ (for } \omega_+) \text{ and } E^c \text{ (for } \omega_-)$$

The result is completely symmetric to the excess variance of static policies in Lemma 5.2 with $\tau_\pm^2 \geq \sigma^2/\lambda_M$.

Augmented Experiment and Divergence Decomposition. For some policy $\pi \in \Pi$, let $(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the environments y_- and y_+, respectively. Now, consider the augmented experiment where, every time the learner samples cluster k, she observes both $(y_{ki})_{i\leq N^k}$ *and* u_k. Let $(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the augmented environments. By data processing,

$$D_{\mathrm{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \leq D_{\mathrm{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$$

The augmented game simplifies the analysis of the KL-divergence significantly because outcomes remain *iid* conditional on the history and revisiting clusters does not further increase the divergence.[60] In addition, unit-level errors share the same distribution across environments, so the KL-divergence analysis is limited to the cluster-level, i.e.

$$D_{\mathrm{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) = \mathbb{E}_{\pi-}\left[K_T\right] D_{\mathrm{KL}}(N(0,\tau_-^2), N(0,\tau_+^2))$$

Solving for D_{KL} explicitly,

$$\begin{aligned} D_{\mathrm{KL}}(N(0,\tau_-^2), N(0,\tau_+^2)) &= \frac{1}{2}\left[\frac{\tau_-^2}{\tau_+^2} - 1 - \ln\left(\frac{\tau_-^2}{\tau_+^2}\right)\right] = \frac{1}{2}\left[\frac{\lambda_+}{\lambda_-} - 1 - \ln\left(\frac{\lambda_+}{\lambda_-}\right)\right] \\ &= \frac{1}{2}\phi\left(\frac{\lambda_+}{\lambda_-}\right) = \frac{1}{2}\phi\left(1 + \frac{2\delta_B}{\lambda_0 - \delta_B}\right) \end{aligned}$$

where the third equality serves as definition of $\phi(x) = x - 1 - \ln(x)$. $\phi(x)$ is monotonically increasing in $x \geq 1$ because $\phi'(x) = 1 - 1/x$, so for all sufficiently large B choose $\delta_B \leq \lambda_0/2$, and using the fact that $\phi(1+x) \leq x^2/2$

$$\frac{1}{2}\phi\left(\frac{\lambda_+}{\lambda_-}\right) \leq \frac{1}{2}\phi\left(1 + \frac{4\delta_B}{\lambda_0}\right) \leq \frac{4\delta_B^2}{\lambda_0^2}$$

Finally, observe that for all policies $\pi \in \Pi$, $FK_T + \sum_k VN_T^k \leq B$ a.s., thus $K_T \leq B/F$ a.s.. Bringing all pieces together

$$D_{\mathrm{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \leq D_{\mathrm{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) \leq \frac{B}{F}\frac{4\delta_B^2}{\lambda_0^2} = C_{\mathrm{KL}} B \delta_B^2$$

[60] For a complete derivation of the chain rule under history-conditional *iid* outcomes see Lemma 15.1 in Lattimore and Szepesvári (2020).

Collecting Terms and BH Inequality. The argument is now completed through a standard Bretagnolle-Huber argument[61]

$$\tilde{\mathbb{E}}_{\pi-}\left[W(\boldsymbol{N};\omega_-)\right] - W(K^*(\lambda_-), N^*(\lambda_-);\omega_-) + \tilde{\mathbb{E}}_{\pi+}\left[W(\boldsymbol{N};\omega_+)\right] - W(K^*(\lambda_+), N^*(\lambda_+);\omega_+) \geq$$
$$\tilde{\mathbb{E}}_{\pi-}\left[W^H(N_T^H;\omega_-)\right] - W^H(N_-;\omega_-) + \tilde{\mathbb{E}}_{\pi+}\left[W^H(N_T^H;\omega_+)\right] - W^H(N_+;\omega_+) \geq$$
$$\tilde{\mathbb{P}}_{\pi-}\left(E^c\right)\cdot\left(\tilde{\mathbb{E}}_{\pi-}\left[W^H(N_T^H;\omega_-)\mid E^c\right] - W^H(N_-;\omega_-)\right) + \tilde{\mathbb{P}}_{\pi+}\left(E\right)\cdot\left(\tilde{\mathbb{E}}_{\pi+}\left[W^H(N_T^H;\omega_+)\mid E\right] - W^H(N_+;\omega_+)\right) \geq$$
$$\frac{\sigma^2\sqrt{FV}}{4B\lambda_M^{5/2}}\delta_B^2\left(\tilde{\mathbb{P}}_{\pi-}\left(E^c\right) + \tilde{\mathbb{P}}_{\pi+}\left(E\right)\right) \geq \frac{\sigma^2\sqrt{FV}}{4B\lambda_M^{5/2}}\delta_B^2\frac{1}{2}\exp\left(-D_{\mathrm{KL}}(\tilde{\mathbb{P}}_{\pi-},\tilde{\mathbb{P}}_{\pi+})\right) \geq$$
$$\frac{\sigma^2\sqrt{FV}}{4B\lambda_M^{5/2}}\delta_B^2\frac{1}{2}\exp\left(-D_{\mathrm{KL}}(\mathbb{P}_{\pi-},\mathbb{P}_{\pi+})\right) \geq \frac{\sigma^2\sqrt{FV}}{8B\lambda_M^{5/2}}\delta_B^2\exp\left(-C_{\mathrm{KL}}\delta_B^2 B\right)$$

Where the first inequality follows from the variance relaxation under the harmonic mean, the second inequality is a simple event partition, the third inequality follows from the uniform lower bound on excess variance, the fourth inequality follows from Bretagnolle-Huber inequality, the fifth inequality follows from a data processing argument, and the final inequality follows from the derivation above. Finally, by selecting $\delta_B = \sqrt{1/(C_{\mathrm{KL}}B)} \underset{B}{\to} 0$,

$$\frac{\sigma^2\sqrt{FV}}{8B\lambda_M^{5/2}}\frac{1}{C_{\mathrm{KL}}B}\exp\left(-1\right) \geq C_2/B^2$$

for some finite and strictly positive constant C_2 not depending on (π, B). The statement of the theorem is completed by applying the inequality $\max\{a,b\} \geq (a+b)/2$ and absorbing it into a constant C. □

D Design-Robust Inference for Asymptotically Even Policies

This section is a self-contained technical note on inference for adaptive designs under symmetry. In a nutshell, the thesis of this document is that, under a symmetry assumption on the errors, inference on sample averages remains invariant after conditioning on designs induced by even or asymptotically even policies. This principle extends well beyond the adaptive algorithms and the clustered sampling framework presented in the main text. The notation follows the literature on multi-armed adaptive designs: N is the horizon (indexed by i) and K the number of arms (indexed by k). It should not be conflated with the K versus N notation of the main text.

	Main text (Section 6)	**This Appendix**
Asymptotic index	Budget B	Horizon N
Groups	One population	K arms
Summands	Clusters $k \leq K_B^\pi$	Pulls $s \leq N_N^k$
Depth (design choice)	N_B^π	–

D.1 Setup and Definitions

Definition D.1. *Sampling Protocol. In each period $i = 1, 2, \ldots, N$: (i) the learner selects an arm $k_i \in [K] = \{1, \ldots, K\}$ according to a history-adapted policy π; (ii) the pull counts update, $N^k(i) =$*

[61] See Theorem 14.2 in Lattimore and Szepesvári (2020).

$N^k(i-1) + \mathbb{1}(k_i = k)$ *with* $N^k(0) = 0$*; and (iii) the learner observes*

$$X_i = \mu_{k_i} + \varepsilon^{k_i}_{N^{k_i}(i)}$$

Write $N^k_N := N^k(N)$ *for the terminal count, also referred to as terminal design. It is convenient to index arm-level statistics by pulls rather than periods, so that* $X^k_s = \mu_k + \varepsilon^k_s$ *denotes the* s*-th observation from arm* k*. Moreover, for* $M \geq 1$ *set*

$$\hat{\mu}^k_M = \frac{1}{M}\sum_{s=1}^{M} X^k_s, \qquad \bar{\varepsilon}^k_M = \frac{1}{M}\sum_{s=1}^{M} \varepsilon^k_s, \qquad Z^k_M = \sqrt{M}(\hat{\mu}^k_M - \mu_k) = \frac{1}{\sqrt{M}}\sum_{s=1}^{M} \varepsilon^k_s$$

with $Z^k_0 = 0$*. The terminal average of arm* k *is* $\hat{\mu}^k_{N^k_N}$*, defined on* $\{N^k_N > 0\}$*.*

Definition D.2. *Random Sequences. For each arm* $k \in [K]$ *define an infinite iid error sequence along the count dimension* $(\varepsilon^k_s)_{s\geq 1}$*. Let* $(\zeta^k_s)_{k\geq 1, s\geq 1}$ *be an iid Rademacher array, independent from the errors, used to break ties in the sign decomposition of Definition D.9, and let* $(U_i)_{i\geq 1}$ *be an iid* Unif$[0,1]$ *sequence, also independent from the errors, used as an independent randomization device.*[62]

Assumption D.3. *Errors.*

i) ***Moments.*** $\mathbb{E}[\varepsilon^k_s] = 0$ *and* $\mathbb{E}[(\varepsilon^k_s)^2] = \sigma^2_k \in (0, \infty)$ *for all* k, s*.*

ii) ***Independence.*** *The array* $(\varepsilon^k_s)_{k,s}$ *has mutually independent entries, and the three collections* $(\varepsilon^k_s)_{k,s}$*,* $(\zeta^k_s)_{k,s}$*,* $(U_i)_i$ *are mutually independent.*

iii) ***Symmetry.*** $\varepsilon^k_s \overset{d}{=} -\varepsilon^k_s$ *for all* k, s*.*

Remark D.4. *Independence across Both Indices. Assumption D.3.ii) emphasizes independence between arms and pulls. Every draw therefore carries its own sign, which makes possible the coordinate-wise definition of symmetry in Assumption D.3.iii) and the array factorization of Lemma D.11. Do not conflate the double indexing in the current appendix (arms and counts) with the double indexing in the main text (clusters and units). In particular,* $M_{k,N}$ *is certainly not independent along the* N *dimension.*

Definition D.5. *Even* σ*-algebra. The even* σ*-algebra is defined via* $\mathcal{E}_\infty := \sigma\big(((\varepsilon^k_s)^2)_{k,s},\ (U_i)_i\big)$*.*

D.2 The Gaussian Benchmark

Before turning to symmetry, I recall what exact Gaussianity buys compared to symmetry. Under Gaussian errors, the sample average of each arm is independent of its residuals, so any sampling policy driven by residuals—sample variances included—can be conditioned on at no cost. The inference-invariant policy class can then be extended from measurable wrt the even σ-algebra (squared errors) to measurable wrt the residual σ-algebra.

Definition D.6. *Residual* σ*-algebra and Nested Designs. Fix some deterministic pilot lengths* $(m^k)_{k\in[K]}$ *with* $m^k \geq 2$*. The residual* σ*-algebra is defined via*

$$\mathcal{R} := \sigma\Big(\big(\varepsilon^k_s - \bar{\varepsilon}^k_{m^k}\big)_{s\leq m^k,\ k\in[K]},\ (U_i)_i\Big)$$

[62] $(U_i)_i$ is absent in every argument below. It is carried so that policies employing external randomization, like a randomized tie-breaking or random arm selection, remain inside the even class of Definition D.17, and so that an asymptotically even surrogate may share the randomization of the policy it couples to.

Moreover, a design $(N_N^k)_{k\in[K]}$ is nested at $(m^k)_k$ if $N_N^k \geq m^k$ almost surely for every k.

Remark D.7. *By considering deterministic pilot lengths $(m^k)_{k\in[K]}$, the exposition is kept tidy. Random commitment stages are accommodated by considering the stopped residual filtration, as it is the case of `ACS` in the cluster specialization of Appendix E.1.*

Proposition D.8. *Exact Conditional Normality under Gaussian Errors. Let $\varepsilon_s^k \sim N(0,\sigma_k^2)$, independent across (k,s), and let π induce a design $(N_N^k)_{k\in[K]}$ that is $\mathcal{R}$-measurable and nested at $(m^k)_k$. Then, for every horizon N and every event $A \in \mathcal{R}$ with $\mathbb{P}(A) > 0$,*

$$\mathcal{L}\left(\left(Z^k_{N_N^k}/\sigma_k\right)_{k\in[K]} \,\middle|\, A\right) \;=\; N(\mathbf{0}, I) \qquad \textit{exactly}$$

Proof of Proposition D.8. *Step 1. Orthogonality.* Fix an arm k and any $M^k \geq m^k$. For every $s \leq m^k$,

$$\mathrm{Cov}\big(\bar\varepsilon^k_{M^k},\; \varepsilon^k_s - \bar\varepsilon^k_{m^k}\big) \;=\; \frac{\sigma_k^2}{M^k} - \frac{1}{m^k}\sum_{j\leq m^k}\mathrm{Cov}\big(\bar\varepsilon^k_{M^k}, \varepsilon^k_j\big) \;=\; \frac{\sigma_k^2}{M^k} - \frac{1}{m^k}\cdot m^k \cdot \frac{\sigma_k^2}{M^k} \;=\; 0$$

where I used $m^k \leq M^k$. All the variables involved are jointly Gaussian, so zero covariance implies independence, $\bar\varepsilon^k_{M^k} \perp\!\!\!\perp \big(\varepsilon^k_s - \bar\varepsilon^k_{m^k}\big)_{s\leq m^k}$. Independence across arms holds by assumption, and $(U_i)_i$ is independent of the errors. Since $Z^k_{M^k} = \sqrt{M^k}\,\bar\varepsilon^k_{M^k}$, it follows that for every deterministic vector $(M^k)_k$ with $M^k \geq m^k$,

$$\big(Z^k_{M^k}\big)_{k\in[K]} \;\perp\!\!\!\perp\; \mathcal{R}, \qquad \big(Z^k_{M^k}/\sigma_k\big)_{k\in[K]} \sim N(\mathbf{0}, I)$$

Step 2. Cell-wise factorization. The terminal counts take values in the finite grid $\mathcal{M} = \{(M^k)_k \in \mathbb{N}^K : \sum_k M^k = N,\; M^k \geq m^k\}$, and $\{(N_N^k)_k = (M^k)_k\} \in \mathcal{R}$ by $\mathcal{R}$-measurability of the design. For any design event $A \in \mathcal{R}$, define writing $A_{(M^k)_k} = A \cap \{(N_N^k)_k = (M^k)_k\} \in \mathcal{R}$, then for any fixed $(t_k)_k \in \mathbb{R}^K$,

$$\begin{aligned}\mathbb{E}\left[\exp\left(i\sum_k t_k \frac{Z^k_{N_N^k}}{\sigma_k}\right)\mathbb{1}_A\right] &= \sum_{(M^k)_k\in\mathcal{M}} \mathbb{E}\left[\exp\left(i\sum_k t_k\frac{Z^k_{M^k}}{\sigma_k}\right)\mathbb{1}_{A_{(M^k)_k}}\right] \\ &= \sum_{(M^k)_k\in\mathcal{M}} \exp\big(-\|t\|^2/2\big)\,\mathbb{P}\big(A_{(M^k)_k}\big) \;=\; \exp\big(-\|t\|^2/2\big)\,\mathbb{P}(A)\end{aligned}$$

where the second equality applies Step 1 cell by cell, and the nesting condition guarantees $M^k \geq m^k$ throughout. The proof concludes by dividing by $\mathbb{P}(A) > 0$ and applying Lévy's continuity theorem. □

Four Observations on Gaussianity vs Symmetry. First, strictly speaking $\mathcal{R}$ and $\mathcal{E}_\infty$ are incomparable, but in practice the set of $\mathcal{R}$-measurable policies is much bigger than the set of $\mathcal{E}_\infty$-measurable policies.[63] In that way, the two frameworks trade distributional assumptions against policy generality. Under Gaussianity, (i) the design may depend arbitrarily on the residuals—including through odd functionals of the data—and (ii) normality is preserved exactly at every horizon (Proposition D.8). Under symmetry, a substantial weakening of the error distribution class, the conditioning σ-algebra *de facto* contracts from the residuals to their magnitudes; thus, the invariant policy class

[63] Residuals themselves change sign under $\varepsilon \mapsto -\varepsilon$, hence $\mathcal{R} \not\subseteq \mathcal{E}_\infty$. However, the σ-algebras remain incomparable because while $\mathcal{E}_\infty$ records $(\varepsilon_s^k)^2$ at every s, $\mathcal{R}$ only sees the first m^k draws of each arm, so neither contains the other.

contracts accordingly from any mapping of the residuals to (asymptotically) even policies.

Second, neither framework covers policies that condition on the *level* of the data. The sample average is precisely the direction demeaning removes, so a design that reads a running mean like futility rules, threshold rules, or standard bandit algorithms induce designs outside both $\mathcal{R}$ and $\mathcal{E}_\infty$. Conditional normality fails even under Gaussian errors and the adjustment tradition reviewed in the main text becomes necessary. Third, nesting is the general-theory analogue of *implementability*: the observations used to choose the design must be a subset of the observations entering the terminal average. Finally, exactness is confined to standardization under the true variance σ_k. Any variance estimate will be a function of residuals themselves and so the distribution of the standardized object will be affected after conditioning. Exact studentized statements require the specific two-stage weighting of Stein (1945), and alternative studentized results remain purely asymptotic.

D.3 Signs and Magnitudes

For the remainder of this appendix, I move away from Gaussianity and focus on symmetry.

Definition D.9. *Signs and Magnitudes. For each (k,s), write $\varepsilon_s^k = \chi_s^k R_s^k$ with*

$$\chi_s^k = \mathbb{1}(\varepsilon_s^k \neq 0)\,\mathrm{sign}(\varepsilon_s^k) + \mathbb{1}(\varepsilon_s^k = 0)\,\zeta_s^k, \qquad R_s^k = |\varepsilon_s^k|$$

Lemma D.10. *Signs are Independent of Magnitudes. Under Assumption D.3, for each (k,s), $\chi_s^k \sim$ Rad and $\chi_s^k \perp\!\!\!\perp R_s^k$.*

Proof of Lemma D.10. Fix (k,s) and drop indices. For any Borel $A \subseteq [0,\infty)$,

$$\mathbb{P}(\chi = 1, R \in C) = \mathbb{P}\big(\varepsilon > 0, |\varepsilon| \in C\big) + \tfrac{1}{2}\,\mathbb{P}(\varepsilon = 0)\,\mathbb{1}(0 \in C)$$

where the second term uses $\zeta \sim$ Rad, independent of ε. By symmetry, $\mathbb{P}(\varepsilon > 0, |\varepsilon| \in C) = \mathbb{P}(-\varepsilon > 0, |-\varepsilon| \in C) = \mathbb{P}(\varepsilon < 0, |\varepsilon| \in C)$. Since $\{\varepsilon < 0, |\varepsilon| \in C\}$ and $\{\varepsilon > 0, |\varepsilon| \in C\}$ partition $\{|\varepsilon| \in C \setminus \{0\}\}$, each has probability $\frac{1}{2}\mathbb{P}(|\varepsilon| \in C\setminus\{0\})$. Adding the tie-break term gives $\mathbb{P}(\chi = 1, R \in C) = \frac{1}{2}\mathbb{P}(R \in C)$. The same argument holds for $\chi = -1$. The joint law factorizes, delivering both claims. □

Lemma D.11. *Signs are Independent of the Even σ-algebra. Under Assumption D.3, $(\chi_s^k)_{k\geq 1, s\geq 1} \perp\!\!\!\perp \mathcal{E}_\infty$.*

Proof of Lemma D.11. The pairs (χ_s^k, R_s^k) are functions of $(\varepsilon_s^k, \zeta_s^k)$, which by Assumption D.3.ii) are independent of $(U_i)_i$ and independent across both k and s (Remark D.4). Each pair factorizes by Lemma D.10, so the joint law of the array also factorizes with $(\chi_s^k)_{k,s} \perp\!\!\!\perp (R_s^k)_{k,s}$. Since

$$\mathcal{E}_\infty = \sigma\big(((\varepsilon_s^k)^2)_{k,s}, (U_i)_i\big) = \sigma\big(((R_s^k)^2)_{k,s}, (U_i)_i\big) = \sigma\big((R_s^k)_{k,s}, (U_i)_i\big)$$

and $(\zeta_s^k)_{k,s} \perp\!\!\!\perp \mathcal{E}_\infty$, it follows that $(\chi_s^k)_{k,s} \perp\!\!\!\perp \mathcal{E}_\infty$. □

D.4 Conditional CLT for a Fixed Design

Throughout this subsection I fix a single arm k and drop it from subscripts and superscripts. I further replace the possibly random count N_N^k by a deterministic M. Assumption D.3 is imposed throughout.

Lemma D.12. *Frozen Conditions. Let $(R_s)_s$ be iid with $\mathbb{E}[R_s^2] = \sigma^2 < \infty$. Then, there is an almost sure event $\Omega_0 \in \mathcal{E}_\infty$ on which, for every $\delta > 0$,*

$$(i) \quad \frac{1}{M}\sum_{s=1}^{M} R_s^2 \overset{a.s.}{\underset{M}{\to}} \sigma^2, \qquad (ii) \quad \frac{1}{M}\sum_{s=1}^{M} R_s^2 \, \mathbb{1}\{R_s > \delta\sqrt{M}\} \overset{a.s.}{\underset{M}{\to}} 0$$

Remark D.13. *Why "Frozen"? The squared errors become frozen once the even σ-algebra is conditioned upon, i.e. $\mathbb{E}[\varepsilon_s^2 \mid \mathcal{E}_\infty] = R_s^2$. Lemma D.12 verifies that the two inputs entering a Lindeberg–Feller Theorem hold pathwise on the frozen array.*

Proof of Lemma D.12. Part (i) follows directly from the SLLN, applicable since $\mathbb{E}[R_s^2] < \infty$. For the Lindeberg condition in part (ii), fix $C > 0$. Then, for every $M \geq C^2/\delta^2 \implies \delta\sqrt{M} \geq C$, hence $\mathbb{1}\{R_s > \delta\sqrt{M}\} \leq \mathbb{1}\{R_s > C\}$ and

$$\limsup_{M\to\infty} \frac{1}{M}\sum_{s=1}^{M} R_s^2 \mathbb{1}\{R_s > \delta\sqrt{M}\} \;\leq\; \limsup_{M\to\infty} \frac{1}{M}\sum_{s=1}^{M} R_s^2 \mathbb{1}\{R_s > C\} \;=\; \mathbb{E}\big[R_1^2 \mathbb{1}\{R_1 > C\}\big]$$

where the equality follows from the SLLN applied to the integrable *iid* summands $R_s^2\mathbb{1}\{R_s > C\}$. Dominated convergence sends the bound to zero as $C \to \infty$. Finally, define the almost sure event Ω_0 by intersecting the almost sure event of part (i) with an intersection over rational $\delta > 0$ and integer C—which can be extended to all $\delta > 0$ by monotonicity. □

Theorem D.14. *L^1 Convergence of the Conditional Characteristic Function. Let $\phi_M(t) := \mathbb{E}\big[\exp(itZ_M) \mid \mathcal{E}_\infty\big]$ denote the* conditional *characteristic function and $\phi(t) = \exp(-\sigma^2 t^2/2)$ the characteristic function of $N(0, \sigma^2)$. Then*

$$\mathbb{E}\big|\phi_M(t) - \phi(t)\big| \to 0 \qquad \forall t \in \mathbb{R}$$

Proof of Theorem D.14. *Step 1. Pathwise Lindeberg–Feller.* Fix ω in the almost sure event Ω_0 and $t \in \mathbb{R}$. Write $Y_{s,M} = \varepsilon_s/\sqrt{M} = \chi_s R_s/\sqrt{M}$, so $Z_M = \sum_{s\leq M} Y_{s,M}$. By Lemma D.11, when conditioning on $\mathcal{E}_\infty$, the magnitudes R_s are fixed and the signs χ_s remain *iid* Rademacher; hence, conditional on $\mathcal{E}_\infty$, the $Y_{s,M}$ are independent, mean zero, and have variance R_s^2/M, i.e.

$$\mathrm{Var}\big(Z_M \mid \mathcal{E}_\infty\big) = \sum_{s\leq M} \mathrm{Var}\big(Y_{s,M} \mid \mathcal{E}_\infty\big) = \frac{1}{M}\sum_{s\leq M} R_s(\omega)^2 \overset{\text{a.s.}}{\to} \sigma^2$$

by Lemma D.12.i), with conditional Lindeberg condition given by by Lemma D.12.ii),

$$\sum_{s\leq M} \mathbb{E}\big[Y_{s,M}^2 \mathbb{1}\{|Y_{s,M}| > \delta\} \,\big|\, \mathcal{E}_\infty\big] = \frac{1}{M}\sum_{s\leq M} R_s(\omega)^2 \, \mathbb{1}\{R_s(\omega) > \delta\sqrt{M}\} \overset{\text{a.s.}}{\to} 0$$

The Lindeberg–Feller CLT for independent, non-identically distributed triangular arrays therefore applies along $M \to \infty$, giving $\phi_M(t)(\omega) \to \exp(-\sigma^2 t^2/2)$. As this holds for every $\omega \in \Omega_0$, convergence is almost sure. *Step 2. L^1 convergence.* Characteristic functions are bounded by one, so $|\phi_M(t) - \phi(t)| \leq 2$. a.s. convergence plus uniform boundedness gives $\mathbb{E}|\phi_M(t) - \phi(t)| \to 0$ through a dominated convergence argument. □

Corollary D.15. *Uniform Mixing. Under the conclusion of Theorem D.14,*

$$\sup_{A\in\mathcal{E}_\infty,\ t\in\mathbb{R}} \left|\mathbb{E}[\phi_M(t)\mathbb{1}_A] - \phi(t)\mathbb{P}(A)\right| \to 0, \quad \iff \quad Z_M \implies N(0,\sigma^2)\ \ \mathcal{E}_\infty\text{-uniformly mixing}$$

Proof of Corollary D.15. For any $A \in \mathcal{E}_\infty$ and $t \in \mathbb{R}$,

$$\begin{aligned}\left|\mathbb{E}[\exp(itZ_M)\mathbb{1}_A] - \phi(t)\mathbb{P}(A)\right| &= \left|\mathbb{E}\big[\mathbb{E}[\exp(itZ_M)\mathbb{1}_A \mid \mathcal{E}_\infty]\big] - \mathbb{E}[\phi(t)\mathbb{1}_A]\right| \\ &= \left|\mathbb{E}\big[\big(\phi_M(t) - \phi(t)\big)\mathbb{1}_A\big]\right| \ \le\ \mathbb{E}\left|\phi_M(t) - \phi(t)\right| \ \to\ 0,\end{aligned}$$

where the second equality uses the $\mathcal{E}_\infty$-measurability of $\mathbb{1}_A$. The convergence is uniform over $A \in \mathcal{E}_\infty$ because the final bound does not depend on A. □

Remark D.16. *Is Uniform Mixing Necessary? Uniformity allows the conditioning event to move with the index in a way that standard pointwise mixing cannot accommodate. This is essential below, where the conditional design events of interest will move along the horizon.*

D.5 Even Policies and Terminal Events

Definition D.17. *Policy Class. Let $\nu_N^{\mathcal{K}} = (N_N^k/N)_{k\in\mathcal{K}}$ for some subset $\mathcal{K} \subseteq [K]$. A policy π is*

(i) ***Even*** *if the terminal design $(N_N^k)_{k\in[K]}$ is $\mathcal{E}_\infty$-measurable;*

(ii) ***Non-$\mathcal{K}$-starving*** *if $\mathbb{P}\big(\nu_N^{\mathcal{K}} \ge c\,\mathbf{1}\big) \to 1$ for some $c > 0$.*

Remark D.18. *Reading Definition D.17. Of the two, evenness is the substantive restriction. Were the policy able to select the next arm k_i on the basis of the raw errors $(\varepsilon_s^k)_{k,s<i}$, the terminal design and its associated events would not be $\mathcal{E}_\infty$-measurable. Non-starvation is required because asymptotic statements can only concern arms played infinitely often, and nothing in evenness prevents a policy from stopping play of an arm. Weaker divergence conditions, not necessarily linear in N, would suffice for most results in this note, but linear non-starvation becomes indispensable when transitioning towards asymptotically even policies. Divergence is then stated in linear terms throughout for consistency.*

D.6 Conditional CLT for Random Even Designs

Theorem D.19. *Mixing CLT Conditional on the Design. Write $\phi^k(t) = \exp(-\sigma_k^2 t^2/2)$. For a subset $\mathcal{K} \subseteq [K]$, let $(N_N^k)_{k\in[K]}$ be a terminal design induced by an even, non-$\mathcal{K}$-starving policy, and let $\big(A_N^{\mathcal{K}}\big)_N$ with $A_N^{\mathcal{K}} = \{(N_N^k)_{k\in\mathcal{K}} \in C\} \in \mathcal{E}_\infty$ be a sequence of design events. Then*

$$\left|\mathbb{E}\left[\exp\left(i\sum_{k\in\mathcal{K}} t_k Z^k_{N_N^k}\right)\mathbb{1}_{A_N^{\mathcal{K}}}\right] - \mathbb{P}\big(A_N^{\mathcal{K}}\big)\prod_{k\in\mathcal{K}}\phi^k(t_k)\right| \to 0 \qquad \forall (t_k)_k \in \mathbb{R}^{|\mathcal{K}|}$$

Moreover, for any such sequence with $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$,

$$\mathcal{L}\left(\left(Z^k_{N_N^k}\right)_{k\in\mathcal{K}} \,\middle|\, A_N^{\mathcal{K}}\right) \implies N\big(\mathbf{0}, \mathrm{diag}(\sigma_k^2)_{k\in\mathcal{K}}\big)$$

Proof of Theorem D.19. *Step 1: Decomposition of the joint conditional characteristic*

function. For any $(t_k)_k \in \mathbb{R}^{|\mathcal{K}|}$,

$$\begin{aligned}
\mathbb{E}\left[\exp\left(i\sum_{k\in\mathcal{K}} t_k Z^k_{N^k_N}\right)\mathbb{1}_{A^{\mathcal{K}}_N}\right] &= \mathbb{E}\left[\mathbb{E}\left[\exp\left(i\sum_{k\in\mathcal{K}} t_k Z^k_{N^k_N}\right)\mathbb{1}_{A^{\mathcal{K}}_N}\middle|\mathcal{E}_\infty\right]\right] \\
&= \mathbb{E}\left[\mathbb{E}\left[\sum_{(M^k)_{k\in\mathcal{K}}} \mathbb{1}\{(N^k_N)_{k\in\mathcal{K}} = (M^k)_{k\in\mathcal{K}}\}\exp\left(i\sum_{k\in\mathcal{K}} t_k Z^k_{M^k}\right)\middle|\mathcal{E}_\infty\right]\mathbb{1}_{A^{\mathcal{K}}_N}\right] \\
&= \mathbb{E}\left[\sum_{(M^k)_{k\in\mathcal{K}}} \mathbb{1}\{(N^k_N)_{k\in\mathcal{K}} = (M^k)_{k\in\mathcal{K}}\}\,\mathbb{E}\left[\exp\left(i\sum_{k\in\mathcal{K}} t_k Z^k_{M^k}\right)\middle|\mathcal{E}_\infty\right]\mathbb{1}_{A^{\mathcal{K}}_N}\right] \\
&= \mathbb{E}\left[\sum_{(M^k)_{k\in\mathcal{K}}} \mathbb{1}\{(N^k_N)_{k\in\mathcal{K}} = (M^k)_{k\in\mathcal{K}}\}\prod_{k\in\mathcal{K}}\phi^k_{M^k}(t_k)\,\mathbb{1}_{A^{\mathcal{K}}_N}\right] \;=\; \mathbb{E}\left[\prod_{k\in\mathcal{K}}\phi^k_{N^k_N}(t_k)\,\mathbb{1}_{A^{\mathcal{K}}_N}\right]
\end{aligned}$$

The second equality follows by $\mathcal{E}_\infty$-measurability of $A^{\mathcal{K}}_N$, the third by $\mathcal{E}_\infty$-measurability of the design indicators together with the finiteness of the grid of count vectors. The fourth equality is the load-bearing step: after conditioning on $\mathcal{E}_\infty$ the magnitudes entering each $Z^k_{M^k}$ are frozen and the signs remain *iid* and independent across arms, so the conditional expectation factorizes over k.

Step 2: L^1 convergence along the diverging counts. Define $\phi^k_M(t) = \mathbb{E}[\exp(itZ^k_M) \mid \mathcal{E}_\infty]$. Step 1 of Theorem D.14 established $|\phi^k_M(t) - \phi^k(t)| \xrightarrow[M]{\text{a.s.}} 0$ on Ω_0. Non-starvation gives $N^k_N \xrightarrow{p} \infty$ for each $k \in \mathcal{K}$, so Lemma B.1 yields $|\phi^k_{N^k_N}(t) - \phi^k(t)| \xrightarrow{p} 0$. A union bound over the finite set $\mathcal{K}$ delivers the vector statement. Uniform boundedness upgrades convergence in probability to L^1.

Step 3: Uniform mixing. By Corollary D.15, L^1 convergence of the conditional characteristic functions gives uniform mixing over $\mathcal{E}_\infty$, and $(A^{\mathcal{K}}_N)_N$ is $\mathcal{E}_\infty$-measurable by evenness. Finally, the first claim is established by noting that $\mathbb{E}\big|\prod_k \phi^k_{N^k_N}(t_k) - \prod_k \phi^k(t_k)\big| \to 0$, given that each factor converges in L^1 and all factors are bounded by one. The second claim follows by dividing by $\mathbb{P}(A^{\mathcal{K}}_N)$, which is eventually bounded away from zero by the theorem's hypothesis, and applying Lévy's continuity theorem. □

Lemma D.20. *Variance Consistency. For a deterministic diverging index M, the within-arm variance estimator*

$$\hat\sigma^2_{k,M} := \frac{1}{M}\sum_{s=1}^{M}\left(X^k_s - \hat\mu^k_M\right)^2 \qquad \textit{satisfies} \qquad \hat\sigma^2_{k,M} \xrightarrow[M]{a.s.} \sigma^2_k$$

Proof of Lemma D.20. Since $X^k_s - \hat\mu^k_M = \varepsilon^k_s - \bar\varepsilon^k_M$,

$$\hat\sigma^2_{k,M} = \frac{1}{M}\sum_{s=1}^{M}\left(\varepsilon^k_s\right)^2 - \left(\bar\varepsilon^k_M\right)^2 \xrightarrow[M]{\text{a.s.}} \sigma^2_k$$

by the CMT and the SLLN applied to the *iid* sequences $((\varepsilon^k_s)^2, (\varepsilon^k_s))_s$ with finite second moments. □

Corollary D.21. *Studentized Conditional Inference. Let $\sigma^2_k > 0$ for all $k \in \mathcal{K}$, let $(N^k_N)_{k\in[K]}$ be a design induced by an even, non-$\mathcal{K}$-starving policy, and let $(A^{\mathcal{K}}_N)_N$ be a sequence of $\mathcal{E}_\infty$-measurable*

design events with $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$. *Then*

$$\mathcal{L}\left(\left(\frac{\sqrt{N_N^k}\,(\hat{\mu}^k_{N_N^k} - \mu_k)}{\hat{\sigma}_{k,N_N^k}}\right)_{k\in\mathcal{K}} \,\middle|\, A_N^{\mathcal{K}}\right) \implies N(\mathbf{0}, I)$$

Proof of Corollary D.21. Decompose

$$\left(\frac{Z^k_{N_N^k}}{\hat{\sigma}_{k,N_N^k}}\right)_{k\in\mathcal{K}} = \left(\frac{Z^k_{N_N^k}}{\sigma_k}\right)_{k\in\mathcal{K}} \odot \left(\frac{\sigma_k}{\hat{\sigma}_{k,N_N^k}}\right)_{k\in\mathcal{K}}$$

Lemma D.20 gives $\hat{\sigma}^2_{k,M} \to \sigma_k^2$ a.s. along a deterministic M and non-starvation gives $N_N^k \xrightarrow{p} \infty$, so Lemma B.1 recovers convergence in probability $\sigma^2_{k,N_N^k} \xrightarrow[N]{p} \sigma_k^2$. Consequently, for every $\delta > 0$,

$$\begin{aligned}\mathbb{P}\left(\left\|(\sigma_k/\hat{\sigma}_{k,N_N^k})_{k\in\mathcal{K}} - \mathbf{1}\right\| > \delta \mid A_N^{\mathcal{K}}\right) &= \frac{\mathbb{P}\left(\left\{\left\|(\sigma_k/\hat{\sigma}_{k,N_N^k})_k - \mathbf{1}\right\| > \delta\right\} \cap A_N^{\mathcal{K}}\right)}{\mathbb{P}(A_N^{\mathcal{K}})} \\ &\le \frac{\mathbb{P}\left(\left\|(\sigma_k/\hat{\sigma}_{k,N_N^k})_k - \mathbf{1}\right\| > \delta\right)}{\mathbb{P}(A_N^{\mathcal{K}})} \;\to\; 0\end{aligned}$$

The ratio is well defined because $\mathbb{P}(\min_{k\in\mathcal{K}} \hat{\sigma}_{k,N_N^k} > 0) \to 1$ and $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$. By Theorem D.19, $\mathcal{L}\big((Z^k_{N_N^k}/\sigma_k)_{k\in\mathcal{K}} \mid A_N^{\mathcal{K}}\big) \implies N(\mathbf{0}, I)$. Applying a vectorial Slutsky theorem to the conditional law completes the proof. □

Remark D.22. *Studentization Does Not Require Evenness. The variance estimators* $(\hat{\sigma}^2_{k,N_N^k})_{k\in\mathcal{K}}$ *are* not $\mathcal{E}_\infty$*-measurable because they demean by the arm sample means* $\hat{\mu}^k$ *rather than by the true means* μ_k*, so the resulting cross terms depend on relative signs. Nonetheless, the corollary goes through because* $\hat{\sigma}^2$ *converges to a constant in the limit, which is absorbed by Slutsky regardless of measurability.*

D.7 Asymptotically Even Designs

Even policies are generally infeasible as the learner only observes the residuals, not the true errors. So the learner could, at best, emulate even policies by acting on functions of squared residuals, such as the within-arm sample variance $\hat{\sigma}^2_{k,M}$. However, policies depending on (sequences of) sample variances could in principle depend on the relative signs, therefore inducing designs outside the even σ-algebra. The example below describes one such policy.

Example D.23. *Sample Variances Load on Relative Signs. Fix a single arm and drop its index. The designer observes* $X_s = \mu + \varepsilon_s$ *with* μ *unknown, so neither* ε_s *nor* ε_s^2 *is observed. The sample variance is location invariant, hence computable from outcomes for* $m \ge 2$*. In particular, it can be written as*

$$\hat{\sigma}^2_m = \frac{m-1}{m^2}\sum_{s\le m}\varepsilon_s^2 \;-\; \frac{2}{m^2}\,U_m, \qquad U_m := \sum_{1\le i<j\le m}\varepsilon_i\varepsilon_j$$

The first term entering $\hat{\sigma}^2_m$ *is* $\mathcal{E}_\infty$*-measurable, thus* $\hat{\sigma}^2_m$ *is* $\mathcal{E}_\infty$*-measurable if and only if* $U_m \in \mathcal{E}_\infty$*. For* $m = 2$*,* $U_2 = |\varepsilon_1\varepsilon_2|\,\mathrm{sign}(\varepsilon_1)\mathrm{sign}(\varepsilon_2)$*, which is not* $\mathcal{E}_\infty$*-measurable by Lemma D.11. A policy that depends on* $\hat{\sigma}_2^2$ *may therefore induce a design that is not* $\mathcal{E}_\infty$*-measurable, and the hypothesis of Theorem D.19 fails.*

Remark D.24. *Enlarging $\mathcal{E}_\infty$ to contain the relative signs does not help. Conditioning on all relative signs (and all magnitudes) pins down the error path up to a single global flip, so Z_M becomes two-valued rather than Gaussian. In what follows, asymptotic evenness is not defined through leakage, but the intuition remains helpful. The odd component of $\hat{\sigma}_m^2$ is $-2U_m/m^2 = O_p(m^{-1})$, while the even component is $O_p(1)$. Thus, provided ε^2 is non-degenerate and $m \to \infty$, designs depending on terminal true variances can only differ from designs induced by terminal sample variances on events of vanishing probability.*

Definition D.25. *Asymptotically Even Policies. Let $\Xi_N^k = (N_N^k - N_{0N}^k)/\sqrt{N}$ for a deterministic centering sequence $(N_{0N}^k)_N$, and write $\Xi_N^{\mathcal{K}} = (\Xi_N^k)_{k\in\mathcal{K}}$. A policy π with terminal design $(N_N^k)_{k\in[K]}$ is asymptotically even if there exists an $\mathcal{E}_\infty$-measurable design $(\tilde{N}_N^k)_{k\in[K]}$, with $\tilde{\Xi}_N^{\mathcal{K}} = \big((\tilde{N}_N^k - N_{0N}^k)/\sqrt{N}\big)_{k\in\mathcal{K}}$, such that*

$$\big\|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\big\| \xrightarrow[N]{p} 0$$

Remark D.26. *The definition places little restrictions on the even and asymptotically even policies. The surrogate even policy need be neither feasible nor related to π in any substantive way. Moreover, π could a priori condition on events well outside $\mathcal{E}_\infty$, raw residuals included. However, in practice, the asymptotically even policy will be some functional of squared residuals, and the even surrogate, a functionally equivalent policy with the squared errors replacing the squared residuals.*

Remark D.27. *A Reduction in the Event Conditioning Class. For exactly even policies the event conditioning class is unrestricted. Theorem D.19 guarantees uniform mixing over $\mathcal{E}_\infty$, so any $\mathcal{E}_\infty$-measurable event of any scale, terminal or not, leaves the limiting law intact. The design events induced by asymptotically even policies are no longer $\mathcal{E}_\infty$-measurable, thus asymptotic normality is preserved by pairing the asymptotically even policy with an even surrogate. The class of conditioning events is reduced accordingly to the subset of events that agree asymptotically under both policies, in the sense $\mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \to 0$. The scale of the events that can survive this transition is at most of the same order as the scale at which the two designs are coupled.*

In the applications motivating this note, terminal counts fluctuate on the $\sqrt{N}$-scale around the deterministic optimal allocations. Therefore, in this context, the $\sqrt{N}$ normalization is the finest scale at which the feasible and surrogate designs can generally be coupled while retaining non-degeneracy of the conditioning events. Non-degeneracy is very important (i) because many events of interest live on this scale, such as the quantiles of N^k, and (ii) because it makes conditional and unconditional inference meaningfully distinct. At coarser scales like $\nu_N^{\mathcal{K}} = (N_N^k/N)_k$, coupling is usually trivial but the conditioning events are vacuous. In most cases, fractional designs converge in probability, and so every conditioning event has limiting probability in $\{0, 1\}$. In this regime, the conditional law degenerates to the unconditional law, such that a random-index central limit theorem in the spirit of Anscombe (1952) will generally suffice without ever invoking symmetry. By contrast, at the unit-scale coupling is usually not possible and requires stronger convergence than granted by Definition D.25. This trichotomy is discussed in greater detail in the main text.

Lemma D.28. *Asymptotically Even Events. Let $A_N^{\mathcal{K}} = \{\Xi_N^{\mathcal{K}} \in C\}$ for a Borel set $C \subseteq \mathbb{R}^{|\mathcal{K}|}$, generated by an asymptotically even, non-$\mathcal{K}$-starving policy π, and suppose C satisfies the margin condition*

$$\lim_{\eta\downarrow 0} \limsup_N \mathbb{P}\big(\operatorname{dist}(\Xi_N^{\mathcal{K}}, \partial C) \le \eta\big) = 0 \tag{17}$$

Then the event $\tilde{A}_N^{\mathcal{K}} = \{\tilde{\Xi}_N^{\mathcal{K}} \in C\}$ *belongs to* $\mathcal{E}_\infty$ *and satisfies* $\mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \to 0$.

Proof of Lemma D.28. Measurability of $\tilde{A}_N^{\mathcal{K}}$ is immediate, since $(\tilde{N}_N^k)_k$ is $\mathcal{E}_\infty$-measurable and the centering and scaling are deterministic. If the two indicators disagree at a point, that means that one of $(\Xi_N^{\mathcal{K}}, \tilde{\Xi}_N^{\mathcal{K}})$ lies in C and the other does not, so the segment joining them meets ∂C and

$$A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}} = \big\{\mathbb{1}_C(\Xi_N^{\mathcal{K}}) \neq \mathbb{1}_C(\tilde{\Xi}_N^{\mathcal{K}})\big\} \;\subseteq\; \big\{\operatorname{dist}(\Xi_N^{\mathcal{K}}, \partial C) \leq \|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\|\big\}$$

Introducing a separating $\eta > 0$,

$$\mathbb{P}\big(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}\big) \;\leq\; \mathbb{P}\big(\|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\| > \eta\big) + \mathbb{P}\big(\operatorname{dist}(\Xi_N^{\mathcal{K}}, \partial C) \leq \eta\big)$$

The first term vanishes for every fixed η by asymptotic evenness, so $\limsup_N \mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \leq \limsup_N \mathbb{P}(\operatorname{dist}(\Xi_N^{\mathcal{K}}, \partial C) \leq \eta)$ for every $\eta > 0$. The left-hand side does not depend on η, so letting $\eta \downarrow 0$ and invoking (17) sends this term to zero. □

Remark D.29. *Reading the Margin Condition. Condition* (17) *is an anti-concentration requirement on the scaled counts near* ∂C*: it forbids the design from piling up on the boundary, which is exactly where the feasible and surrogate designs can disagree. In applications it is usually verified by analyzing the limiting law of the scaled design. For example, assume that* $\Xi_N^{\mathcal{K}} \implies \Xi_\infty^{\mathcal{K}}$ *for a well-defined limit law* $\Xi_\infty^{\mathcal{K}}$ *with* $\mathbb{P}(\Xi_\infty^{\mathcal{K}} \in \partial C) = 0$*, then since the set* $\{x : \operatorname{dist}(x, \partial C) \leq \eta\}$ *is closed the Portmanteau Theorem gives*

$$\limsup_N \mathbb{P}\big(\operatorname{dist}(\Xi_N^{\mathcal{K}}, \partial C) \leq \eta\big) \;\leq\; \mathbb{P}\big(\operatorname{dist}(\Xi_\infty^{\mathcal{K}}, \partial C) \leq \eta\big)$$

where the right-hand side decreases to $\mathbb{P}(\Xi_\infty^{\mathcal{K}} \in \partial C) = 0$ *as* $\eta \downarrow 0$*. The same weak convergence argument implies that* $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$ *for any* C *whose interior receives positive limit mass.*

Remark D.30. *CLT and Fourth Moments in Asymptotically Even Policies. As hinted in Remark D.29, Condition* (17) *is usually verified by inspecting the limiting law of the scaled design. This procedure can be implemented by establishing a central limit theorem over the underlying statistic governing the design as, whenever the mass of the limiting law spreads smoothly, the anticoncentration requirement is immediately satisfied. Moreover, recall that the design is driven by second moments, thus such CLT must run over the pilot variances, whose own variance is* $\mu_{4,k} - \sigma_k^4$*. Existence of fourth moments is then a clean alternative to Condition* (17)*, under the following logic*

$$\textit{existence of fourth moments} \implies \textit{CLT over variances} \implies \textit{no piling at the boundary}$$

Finite fourth moments are a requirement of the limit theory of the design and not of the estimator, consequently, I impose such assumption only when discussing Lemma D.36 and Theorem 6.9, and nowhere else in this note.

Lemma D.31. *Increment Vanishes at Ratio-close Indices. Fix a single arm and drop its index. Let* $(M_N)_N$ *and* $(\tilde{M}_N)_N$ *be* $\{0, 1, \ldots, N\}$*-valued random indices with* $|M_N/N - \tilde{M}_N/N| \xrightarrow[N]{p} 0$ *and* $\mathbb{P}\big(M_N/N \wedge \tilde{M}_N/N \geq c\big) \to 1$ *for some* $c > 0$*. Then*

$$\big|Z_{M_N} - Z_{\tilde{M}_N}\big| \xrightarrow{p} 0$$

Proof of Lemma D.31. Work on the event $\{\tilde{M}_N \le M_N\}$, and write $\Sigma_M = \sum_{s\le M} \varepsilon_s$, so that $Z_M = \Sigma_M/\sqrt{M}$,

$$\left|Z_{M_N} - Z_{\tilde{M}_N}\right| \;\le\; \left|\sqrt{\tfrac{\tilde{M}_N}{M_N}} - 1\right| \cdot \left|Z_{\tilde{M}_N}\right| + \frac{\left|\Sigma_{M_N} - \Sigma_{\tilde{M}_N}\right|}{\sqrt{M_N}}$$

On $\{\tilde{M}_N \ge cN\}$, $|Z_{\tilde{M}_N}| \le \max_{cN\le m\le N}|\Sigma_m|/\sqrt{cN} = O_p(1)$ by Kolmogorov's maximal inequality applied over the *iid*, mean-zero, finite-variance error sequence. Convergence in probability of the scaled indices together with linear non-starvation gives $|\sqrt{\tilde{M}_N/M_N} - 1| \xrightarrow{p} 0$, and so the first product is $o_p(1)$.

For the second term, fix $\delta \in (0,1)$ and intersect the working event with $\{|M_N - \tilde{M}_N| \le \delta N\} \cap \{\tilde{M}_N \ge cN\}$, whose probability tends to one under the hypotheses. Therefore,

$$\frac{\left|\Sigma_{M_N} - \Sigma_{\tilde{M}_N}\right|}{\sqrt{M_N}} \;\le\; \max_{\substack{cN\le \tilde{m}\le m\le N \\ |m-\tilde{m}|\le \delta N}} \frac{|\Sigma_m - \Sigma_{\tilde{m}}|}{\sqrt{m}} \;\le\; \frac{1}{\sqrt{c}} \max_{\substack{0\le \tilde{m}\le m\le N \\ |m-\tilde{m}|\le \delta N}} \frac{|\Sigma_m - \Sigma_{\tilde{m}}|}{\sqrt{N}} \;\le\; \frac{1}{\sqrt{c}} \sup_{\substack{0\le s\le t\le 1 \\ |t-s|\le\delta}} \left|\Psi_N(t) - \Psi_N(s)\right|$$

where $\Psi_N(t) = \Sigma_{\lfloor Nt\rfloor}/\sqrt{N}$ is the rescaled partial-sum process. Under Assumption D.3, $(\Psi_N)_N$ is asymptotically uniformly equicontinuous (tight), as per Donsker's functional central limit theorem and continuity of the Brownian limit. Equivalently, $(\Psi_N)_N$ holds the local modulus-of-continuity property

$$\lim_{\delta\downarrow 0} \limsup_{N\to\infty} \mathbb{P}\left(\sup_{\substack{0\le s\le t\le 1 \\ |t-s|\le\delta}} \left|\Psi_N(t) - \Psi_N(s)\right| \ge \eta\right) = 0 \qquad \forall \eta > 0$$

Fix $\eta, \epsilon > 0$ and choose δ so that

$$\limsup_{N\to\infty} \mathbb{P}\left(\sup_{\substack{0\le s\le t\le 1 \\ |t-s|\le\delta}} \left|\Psi_N(t) - \Psi_N(s)\right| \ge \eta\sqrt{c}\right) \le \varepsilon/2$$

Since $\mathbb{P}(|M_N - \tilde{M}_N| > \delta N) \to 0$, for all N large

$$\mathbb{P}\left(\frac{\left|\Sigma_{M_N} - \Sigma_{\tilde{M}_N}\right|}{\sqrt{M_N}} \ge \eta\right) \;\le\; \mathbb{P}\big(|M_N - \tilde{M}_N| > \delta N\big) + \mathbb{P}\left(\sup_{\substack{0\le s\le t\le 1 \\ |t-s|\le\delta}} \left|\Psi_N(t) - \Psi_N(s)\right| \ge \eta\sqrt{c}\right) \;\le\; \epsilon$$

As η and ϵ were arbitrary the second term is $o_p(1)$ and the claim follows. For the case where $\{\tilde{M}_N > M_N\}$ the argument is completely symmetrical and so the unconditional statement follows too. □

Remark D.32. *Lemma D.31 Is Measurability-Free. Lemma D.31 is a statement about the random walk alone. It does not condition on $\mathcal{E}_\infty$ and imposes no measurability on M_N or $\tilde{M}_N$. As a result, it will be useful to bound the distance between $Z^k_{N^k_N}$ (non-measurable) and $Z^k_{\tilde{N}^k_N}$ (measurable). Additionally, this is exactly where **linear** non-starvation enters the analysis. Tightness of the sum process only manifests at indices which are negligible at the scale of counts. Asymptotic evenness controls $|N^k_N - \tilde{N}^k_N|$ only to order $o_p(\sqrt{N})$, so the difference between indices is negligible at the scale of counts just because the counts are assumed to be of order N. Slower forms of divergence will compromise convergence in probability between the scaled sums of the exactly even and asymptotically even policies.*

Theorem D.33. *Inference for Asymptotically Even Designs. Let π be an asymptotically even, non-$\mathcal{K}$-starving policy, and let $(A_N^{\mathcal{K}})_N$, with $A_N^{\mathcal{K}} = \{\Xi_N^{\mathcal{K}} \in C\}$, be a sequence of scaled design events satisfying the margin condition (17). Then*

$$\left| \mathbb{E}\left[\exp\left(i \sum_{k \in \mathcal{K}} t_k Z^k_{N^k_N} \right) \mathbb{1}_{A_N^{\mathcal{K}}} \right] - \mathbb{P}(A_N^{\mathcal{K}}) \prod_{k \in \mathcal{K}} \phi^k(t_k) \right| \to 0 \qquad \forall (t_k)_k \in \mathbb{R}^{|\mathcal{K}|}$$

If in addition $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$, then

$$\mathcal{L}\left((Z^k_{N^k_N})_{k \in \mathcal{K}} \;\middle|\; A_N^{\mathcal{K}} \right) \implies N\big(\mathbf{0}, \mathrm{diag}(\sigma_k^2)_{k \in \mathcal{K}}\big), \qquad \mathcal{L}\left(\left(\frac{\sqrt{N_N^k}(\hat{\mu}^k_{N^k_N} - \mu_k)}{\hat{\sigma}_{k, N^k_N}} \right)_{k \in \mathcal{K}} \;\middle|\; A_N^{\mathcal{K}} \right) \implies N(\mathbf{0}, I)$$

Remark D.34. *Not Uniform Mixing. The first statement cannot be phrased as $\mathcal{E}_\infty$-uniform mixing, since $A_N^{\mathcal{K}}$ is not even $\mathcal{E}_\infty$-measurable. In fact, convergence only holds pointwise along the design event sequence. For the same reason, the conditional characteristic function of the feasible design cannot be written as $\phi^k_{N^k_N}$. Yet, the essence of the result is precisely akin that of Theorem D.19.*

Proof of Theorem D.33. *Step 0: the surrogate inherits non-starvation.* Since $\tilde{\nu}_N^k - \nu_N^k = (\tilde{\Xi}_N^k - \Xi_N^k)/\sqrt{N}$, for any $\eta > 0$

$$\mathbb{P}\big(\tilde{\nu}_N^{\mathcal{K}} < (c - \eta)\mathbf{1}\big) \;\le\; \mathbb{P}\big(\|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\| > \sqrt{N}\,\eta\big) + \mathbb{P}\big(\nu_N^{\mathcal{K}} < c\,\mathbf{1}\big) \;\to\; 0$$

so $\tilde{\pi}$ is non-$\mathcal{K}$-starving with constant $c - \eta$. *Step 1: decomposition.* Writing $\tilde{A}_N^{\mathcal{K}}$ for the surrogate event of Lemma D.28,

$$\begin{aligned}
&\left| \mathbb{E}\Big[e^{i \sum_k t_k Z^k_{N^k_N}} \mathbb{1}_{A_N^{\mathcal{K}}} \Big] - \mathbb{P}(A_N^{\mathcal{K}}) \prod_{k \in \mathcal{K}} \phi^k(t_k) \right| \\
&\quad \le \underbrace{\left| \mathbb{E}\Big[e^{i \sum_k t_k Z^k_{N^k_N}} \mathbb{1}_{A_N^{\mathcal{K}}} \Big] - \mathbb{E}\Big[e^{i \sum_k t_k Z^k_{\tilde{N}^k_N}} \mathbb{1}_{\tilde{A}_N^{\mathcal{K}}} \Big] \right|}_{[1]} + \underbrace{\left| \mathbb{E}\Big[e^{i \sum_k t_k Z^k_{\tilde{N}^k_N}} \mathbb{1}_{\tilde{A}_N^{\mathcal{K}}} \Big] - \mathbb{P}(\tilde{A}_N^{\mathcal{K}}) \prod_{k \in \mathcal{K}} \phi^k(t_k) \right|}_{[2]} \\
&\qquad + \underbrace{\left| \prod_{k \in \mathcal{K}} \phi^k(t_k) \right| \cdot \left| \mathbb{P}(A_N^{\mathcal{K}}) - \mathbb{P}(\tilde{A}_N^{\mathcal{K}}) \right|}_{[3]}
\end{aligned}$$

Term [3] vanishes because $|\mathbb{P}(A_N^{\mathcal{K}}) - \mathbb{P}(\tilde{A}_N^{\mathcal{K}})| \le \mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \to 0$ by Lemma D.28, and the characteristic functions are bounded by one. Term [2] vanishes by Theorem D.19 applied to $\big((\tilde{N}_N^k)_{k \in \mathcal{K}}, (\tilde{A}_N^{\mathcal{K}})_N\big)$: the surrogate design and its events are $\mathcal{E}_\infty$-measurable by construction, and the surrogate counts diverge (linearly) with probability tending to one by Step 0, so the freezing identity $\mathbb{E}\big[e^{i \sum_k t_k Z^k_{\tilde{N}^k_N}} \mid \mathcal{E}_\infty\big] = \prod_k \phi^k_{\tilde{N}^k_N}(t_k)$ is available. *Step 2: the coupling term.* Split

$$[1] \;\le\; \left| \mathbb{E}\Big[\Big(e^{i \sum_k t_k Z^k_{N^k_N}} - e^{i \sum_k t_k Z^k_{\tilde{N}^k_N}} \Big) \mathbb{1}_{A_N^{\mathcal{K}}} \Big] \right| + \left| \mathbb{E}\Big[e^{i \sum_k t_k Z^k_{\tilde{N}^k_N}} \big(\mathbb{1}_{A_N^{\mathcal{K}}} - \mathbb{1}_{\tilde{A}_N^{\mathcal{K}}} \big) \Big] \right|$$

The second piece is bounded by $\mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \to 0$, the integrand having modulus one. For the first term, using the fact that $|e^{ia} - e^{ib}| \le \min\{2, |a - b|\}$ together with subadditivity of $x \mapsto \min\{2, x\}$

yields

$$\mathbb{E}\Big|e^{\,i\sum_k t_k Z^k_{N^k_N}} - e^{\,i\sum_k t_k Z^k_{\tilde N^k_N}}\Big| \;\le\; \sum_{k\in\mathcal{K}} \mathbb{E}\Big[\min\big\{2, |t_k|\cdot\big|Z^k_{N^k_N} - Z^k_{\tilde N^k_N}\big|\big\}\Big]$$

For each $k \in \mathcal{K}$, Step 0 gives $\mathbb{P}(N^k_N/N \wedge \tilde N^k_N/N \ge c') \to 1$ for some $c' > 0$, and asymptotic evenness gives $|N^k_N/N - \tilde N^k_N/N| = o_p(\sqrt{N})/N \xrightarrow{p} 0$, so the hypotheses of Lemma D.31 hold and each increment is $o_p(1)$. Uniform boundedness of the integrand by 2 upgrades convergence in probability to L^1 convergence. Moreover, the sum runs over finitely many arms, thus $[1] \to 0$, establishing the first statement.

Step 3. The conditional limits. Dividing by $\mathbb{P}(A^{\mathcal{K}}_N)$, eventually bounded away from zero by hypothesis, and applying Lévy's continuity theorem gives the first conditional statement. Studentization only requires $\hat\sigma^2_{k,N^k_N} \xrightarrow{p} \sigma^2_k$, which holds under non-starvation by Lemma D.20. The argument of Corollary D.21 therefore applies verbatim. Measurability of N^k_N and $A^{\mathcal{K}}_N$ were never invoked in the derivation. □

D.8 Policy Examples

Proposition D.35. *Two-Stage Neyman Allocation Is Asymptotically Even. Let $\sigma^2 = (\sigma^2_k)_k > \mathbf{0}$ and let $0 < d < \min_k f_k(\sigma)$ be known to the designer, with $dN \in \mathbb{N}$, $f_k(\sigma) = \sigma_k/\Sigma_\sigma$ and $\Sigma_\sigma = \sum_l \sigma_l$. Consider the policy π that samples uniformly for the first dKN periods—so every arm receives dN pulls—computes the pilot variances $\hat\sigma^2_{k,dN} = (dN)^{-1}\sum_{s\le dN}\big(X^k_s - \hat\mu^k_{dN}\big)^2$, and allocates the remaining periods so that the terminal counts equal $N^k_N = \lfloor N f_k(\hat\sigma)\rfloor$, with leftovers assigned by an arbitrary deterministic rule.*[64] *Then, π is non-$[K]$-starving and asymptotically even for $[K]$.*

Proof of Proposition D.35. *Surrogate evenness.* The pilot allocation is deterministic, hence so are the pilot counts. Define the surrogate counts $\tilde N^k_N = \lfloor N f_k(q)\rfloor$ under the same leftover rule, with $q^2_k = (dN)^{-1}\sum_{s\le dN}(\varepsilon^k_s)^2$. Each q^2_k is a function of $((\varepsilon^k_s)^2)_{s\le dN} \in \mathcal{E}_\infty$, and the rounding and leftover rules are deterministic, so $(\tilde N^k_N)_k$ is $\mathcal{E}_\infty$-measurable. *Non-starvation.* Every arm receives at least dN pulls in the pilot, which are later included within the terminal counts, so $N^k_N \ge dN$ almost surely. *Asymptotic evenness.* The pilot variance demeans by the pilot mean, so $q^2_k - \hat\sigma^2_{k,dN} = (\bar\varepsilon^k_{dN})^2 = O_p(1/N)$, provided $\bar\varepsilon^k_{dN} = O_p(N^{-1/2})$. Since $\sigma_k > 0$, the square root operator is Lipschitz on a neighbourhood of σ^2_k with probability tending to one, thus $|\hat\sigma_k - q_k| = O_p(1/N)$. Similarly, since $\Sigma_\sigma > 0$, f is Lipschitz on a neighbourhood of σ, thus $|f_k(\hat\sigma) - f_k(q)| = O_p(1/N)$. Finally, note that $|\lfloor Nx\rfloor - \lfloor Ny\rfloor| \le N|x-y|+1$ and that at most K units can be allocated to arm k through the leftover rule. Combining these three elements,

$$\big|N^k_N - \tilde N^k_N\big| \;\le\; N\big|f_k(\hat\sigma) - f_k(q)\big| + 1 + K \;=\; O_p(1)$$

so $|N^k_N - \tilde N^k_N|/\sqrt{N} \xrightarrow{p} 0$ and π is asymptotically even with even surrogate $(\tilde N^k_N)_k$. □

Lemma D.36. *Central Limit Theorem for the Scaled Counts. Together with Assumption D.3, assume that $\mu_{4,k} := \mathbb{E}[(\varepsilon^k_s)^4] < \infty$ and $\mu_{4,k} > \sigma^4_k$ for every k. Consider the usual definitions for $(\Xi^{[K]}_N, \tilde\Xi^{[K]}_N)$ centered at the oracle allocation $N^k_{\underline 0 N} = \lfloor N f_k(\sigma)\rfloor$. Then*

$$\Xi^{[K]}_N \implies G, \qquad \tilde\Xi^{[K]}_N \implies G, \qquad G = Jz$$

[64] Since $\min_k f_k(\sigma) > d$ strictly, $\mathbb{P}(\lfloor N f_k(\hat\sigma)\rfloor \ge dN\ \forall k) \to 1$ and the target allocation is implementable with probability approaching one. The complementary event has vanishing probability, thus affects none of the statements below.

where $z \sim N(\mathbf{0}, \mathrm{diag}(v_k)_k)$ *with* $v_k = (\mu_{4,k} - \sigma_k^4)/(4d\sigma_k^2) > 0$*, and* J *is the Jacobian of* f *at* σ *with entries* $J_{kl} = (\mathbb{1}\{k = l\}\Sigma_\sigma - \sigma_k)/\Sigma_\sigma^2$*.*

Proof of Lemma D.36. Consider the surrogate first. The variables $((\varepsilon_s^k)^2)_s$ are *iid* with mean σ_k^2 and variance $\mu_{4,k} - \sigma_k^4 < \infty$. The pilot size dN is deterministic, so $\sqrt{N}(q_k^2 - \sigma_k^2) \implies N\big(0, (\mu_{4,k} - \sigma_k^4)/d\big)$, jointly across arms with independent arm coordinates. The delta method applied to $x \mapsto \sqrt{x}$, whose derivative at σ_k^2 is $1/(2\sigma_k)$, gives $\sqrt{N}(q_k - \sigma_k) \implies N(0, v_k)$; applied to f it gives $\sqrt{N}(f(q) - f(\sigma)) \implies Jz = G$. Rounding and the leftover rule perturb $\tilde{\Xi}_N^k$ by at most $(1+K)/\sqrt{N} \to 0$, so $\tilde{\Xi}_N^{[K]} \implies G$ by Slutsky. Finally $\|\Xi_N^{[K]} - \tilde{\Xi}_N^{[K]}\| = O_p(1)/\sqrt{N} \xrightarrow{p} 0$ by Proposition D.35, so $\Xi_N^{[K]} \implies G$ as well. □

Corollary D.37. *Design-Conditional Inference for Two-Stage Neyman. Let* π *be the two-stage Neyman policy of Proposition D.35, and let* $A_N^{[K]} = \{\Xi_N^{[K]} \in C\}$ *for a Borel set* $C \subseteq \mathbb{R}^K$*, with* $\Xi_N^{[K]}$ *centered at* $N_{0N}^k = \lfloor N f_k(\sigma) \rfloor$*. (a) Under Assumption D.3, if* C *satisfies (17) and* $\liminf_N \mathbb{P}(A_N^{[K]}) > 0$*, then*

$$\mathcal{L}\left(\left(\frac{\sqrt{N_N^k}(\hat{\mu}^k_{N_N^k} - \mu_k)}{\hat{\sigma}_{k,N_N^k}} \right)_{k\in[K]} \,\middle|\, A_N^{[K]} \right) \implies N(\mathbf{0}, I)$$

(b) If in addition $\mu_{4,k} < \infty$ *for all* k*, (a) holds whenever* $\mathbb{P}(G \in \partial C) = 0$ *and* $\mathbb{P}(G \in C) > 0$*, with* G *as in Lemma D.36.*

Proof. (a) By Proposition D.35, π is asymptotically even and non-$[K]$-starving. The margin condition (17) holds by assumption, and the result follows from Theorem D.33. (b) Under fourth moments, Lemma D.36 demonstrates that $\Xi_N^{[K]} \implies G$. Assuming $\mathbb{P}(G \in \partial C) = 0$, the Portmanteau Theorem in Remark D.29 then yields (17) and $\mathbb{P}(A_N^{[K]}) \to \mathbb{P}(G \in C) > 0$. □

E Proofs of Section 6

This section translates the general theory of Appendix D into the clustered sampling problem and, in doing so, proves the results stated in Section 6. Two features of the cluster problem make this translation not mechanical. First, the cluster problem carries *two* error sequences, $(u_k)_k$ and $(\varepsilon_{ki})_{k,i}$, and *two* degrees of freedom (K, N), only one of which diverges with the budget. I handle this issue by establishing N-consistency and extending the results depth by depth. Second, establishing asymptotic evenness for `EtC-CS` follows the general recipe—replacing residuals by errors—outlined in the two-stage Neyman allocation example. Nonetheless, this replacement is insufficient for `ACS`, as doing so would still leave the analyst with the difficult task of pairing two random stopping times. The problem is avoided entirely by considering an even surrogate with fixed stopping time K_{0B} and localizing the stopping time of `ACS` in a window of order $O_p(\sqrt{B})$ around K_{0B}. The definitions and lemmas below are held to a minimum to avoid duplicities. Connections to the general theory are made throughout, so I recommend reading this section in light of the results in the standalone Appendix D.

E.1 Exact Gaussianity

Proposition 6.1 is the cluster instance of Proposition D.8 with two adaptations. Orthogonality must be established at two layers, and the stopped residual σ-algebra must be characterized appropriately, given that the commitment stage `ACS` is random.

Definition E.1. *Residual σ-algebras. For a fixed pilot depth N_p, an iid auxiliary sequence $(U_k)_k \sim \mathrm{Unif}[0,1]$ independent of the errors, and $j \geq 2$, define the residual σ-algebra of the first j clusters,*

$$\mathcal{R}_j := \sigma\Big(\big(\varepsilon_{ki} - \bar{\varepsilon}_{k,N_p}\big)_{i \leq N_p,\, k \leq j},\ \big(M_{k,N_p} - \bar{M}_{j,N_p}\big)_{k \leq j},\ (U_k)_{k \geq 1} \Big)$$

Definition E.2. *Residual Filtration and Pilot σ-algebras. Given $\mathcal{R}_j$ with $j \geq 2$, then,*

(i) Filtration. *$\mathcal{R}_k \subseteq \mathcal{R}_j$ for all $2 \leq k \leq j$, since the k-demeaned cluster residuals can be recovered from the j-demeaned ones through*

$$M_{l,N_p} - \bar{M}_{k,N_p} = \big(M_{l,N_p} - \bar{M}_{j,N_p}\big) - \frac{1}{k} \sum_{l' \leq k} \big(M_{l',N_p} - \bar{M}_{j,N_p}\big), \qquad l \leq k \leq j$$

(ii) Measurability. *For every $2 \leq k \leq j$, $(\hat{w}^2_{k,N_p}, \hat{b}^2_{k,N_p}, \hat{\lambda}_{k,N_p})$ are $\mathcal{R}_j$-measurable, where*

$$\hat{w}^2_{k,N_p} = \frac{1}{k} \sum_{l \leq k} \frac{1}{N_p - 1} \sum_{i \leq N_p} \big(\varepsilon_{li} - \bar{\varepsilon}_{l,N_p}\big)^2, \qquad \hat{b}^2_{k,N_p} = \frac{1}{k-1} \sum_{l \leq k} \big(M_{l,N_p} - \bar{M}_{k,N_p}\big)^2$$

(iii) Stopping. *The commitment stages $T^{\texttt{EtC-CS}} = K_M$ and $T^{\texttt{ACS}} = s$ are stopping times for $(\mathcal{R}_j)_j$. The first is deterministic so inclusion is trivial. Define the pilot σ-algebra as $\mathcal{P}^{\texttt{EtC-CS}} := \mathcal{R}_{K_M}$. For `ACS` see that*

$$\{s = j\} = \big\{ j + 1 > K^*(\hat{\lambda}_{j,N_p}) \big\} \cap \bigcap_{l = K_M}^{j-1} \big\{ l + 1 \leq K^*(\hat{\lambda}_{l,N_p}) \big\} \in \mathcal{R}_j$$

and so the stopped pilot σ-algebra is $\mathcal{P}^{\texttt{ACS}} := \mathcal{R}_s = \big\{ A : A \cap \{s = j\} \in \mathcal{R}_j \ \text{ for every } j \geq 2 \big\}$. For $\pi \in \{\texttt{EtC-CS}, \texttt{ACS}\}$, the terminal design (K_B^π, N_B^π) is $\mathcal{P}^\pi$-measurable by construction.

Proof of Proposition 6.1. *Step 1. Two-layer orthogonality.* Fix $j \geq 2$ and let $K' \geq j$, $N \geq N_p$. Since $\hat{\mu}_{K',N} - \mu = \bar{u}_{K'} + (K')^{-1} \sum_{k \leq K'} \bar{\varepsilon}_{k,N}$, and $u \perp\!\!\!\perp \varepsilon$ with independence across clusters, it suffices to verify orthogonality layer by layer against the generators of $\mathcal{R}_j$. At the unit layer, for $i \leq N_p$ and $k \leq j$, only the kth term contributes to the covariance, i.e.

$$\mathrm{Cov}\big(\bar{\varepsilon}_{k,N},\ \varepsilon_{ki} - \bar{\varepsilon}_{k,N_p}\big) = \frac{\sigma^2}{N} - \frac{1}{N_p} \cdot N_p \cdot \frac{\sigma^2}{N} = 0$$

both evaluations using $N_p \leq N$. At the cluster layer, for $k \leq j$,

$$\mathrm{Cov}\big(\bar{u}_{K'},\ u_k - \bar{u}_j\big) = \frac{\tau^2}{K'} - \frac{1}{j} \cdot j \cdot \frac{\tau^2}{K'} = 0$$

both evaluations using $j \leq K'$. The remaining piece of $M_{k,N_p} - \bar{M}_{j,N_p}$, namely $\bar{\varepsilon}_{k,N_p} - j^{-1} \sum_{l \leq j} \bar{\varepsilon}_{l,N_p}$, is also orthogonal to $(K')^{-1} \sum_k \bar{\varepsilon}_{k,N}$ by the same argument as in the unit layer. All variables are jointly Gaussian, so zero covariance implies independence: $\hat{\mu}_{K',N} \perp\!\!\!\perp \mathcal{R}_j$ with $j \leq K'$ and $N_p \leq N$.

Step 2. Nesting. $T^\pi \leq K_B^\pi$ and $N_p \leq N_B^\pi$ almost surely. For `EtC-CS` the result follows because $T^{\texttt{EtC-CS}} = K_M \leq K_B^{\texttt{EtC-CS}}$, and for `ACS` because $s \leq K^*(\hat{\lambda}_{\max\{s-1, K_M\}, N_p}) = K_B^{\texttt{ACS}}$. The hypotheses of Step 1 then hold on every cell of the design grid. The attentive reader will note that nesting is the inference analogue of implementability.

Step 3. Cell-wise factorization. The triple $(T^\pi, K_B^\pi, N_B^\pi)$ takes finitely many values. Fix $A \in \mathscr{P}^\pi$ with $\mathbb{P}(A) > 0$ and write $A_{(j,K,N)} := A \cap \{(T^\pi, K_B^\pi, N_B^\pi) = (j, K, N)\}$. Since (K_B^π, N_B^π) is $\mathscr{P}^\pi$-measurable, $A \cap \{(K_B^\pi, N_B^\pi) = (K, N)\} \in \mathscr{P}^\pi$, and intersecting with $\{T^\pi = j\}$ places $A_{(j,K,N)}$ in $\mathscr{R}_j$ by the stopped σ-algebra construction of Lemma E.2.iii). By Steps 1 and 2, $\sqrt{K}(\hat{\mu}_{K,N} - \mu)/\theta_N \sim N(0,1)$ independently of $\mathscr{R}_j \ni A_{(j,K,N)}$ on every cell, so

$$\mathbb{E}\left[\exp\left(\frac{itZ_B^\pi}{\theta_{N_B^\pi}}\right)\mathbb{1}_A\right] = \sum_{(j,K,N)} e^{-t^2/2}\,\mathbb{P}\big(A_{(j,K,N)}\big) = e^{-t^2/2}\,\mathbb{P}(A)$$

The proof is concluded by dividing by $\mathbb{P}(A)$ and applying Lévy's continuity theorem. □

E.2 Random Sequences in the Cluster Problem

Definition E.3. *Cluster Error Sequences. For a fixed depth N, define $M_{k,N} = u_k + N^{-1}\sum_{i\le N}\varepsilon_{ki}$, indexed by the cluster k. Let $(\zeta_k)_{k\ge1}$ be an iid Rademacher sequence, used to break ties in Lemma 6.3, and $(U_k)_{k\ge1}$ an iid* Unif$[0,1]$ *sequence available to the policy as an independent randomization device, both independent of $(C_k)_k$.*

Remark E.4. *What Carries Over? For each fixed N, Assumptions 3.1 and 6.2 make $(M_{k,N})_{k\ge1}$ an iid, mean-zero, symmetric sequence with $\mathbb{E}[M_{k,N}^2] = \theta_N^2 < \infty$. Hence, $(M_{k,N})_{k\ge1}$ with fixed N plays the role of the error sequence in Assumption D.3. Then every result of Appendix D applies over the "single-arm" sequence $(M_{k,N})_{k\ge1}$, after replacing the pull index by the cluster index, and the horizon N by the diverging cluster count K_B.*

E.3 Signs and Magnitudes

Proof of Lemma 6.3. Fix N. By Remark E.4, $(M_{k,N})_k$ is *iid*, mean zero and symmetric, so Lemma D.10 applies coordinatewise with ε_s^k replaced by $M_{k,N}$ and the tie-breaker ζ_s^k by ζ_k. Notably, independence is established only across the width dimension and not the depth dimension. □

Proof of Corollary 6.4. Fix N and let $r = (r_k)_k \in \{\pm1\}^\infty$. Define the cluster-flip map $g_r : (C_k)_k \mapsto (r_k C_k)_k$. By Assumption 6.2 and cross-cluster independence, g_r is measure preserving. In particular, it leaves every generator of $\mathcal{G}_\infty$ invariant, since

$$S^2_{k,N_p}(r_kC_k) = S^2_{k,N_p}(C_k), \qquad M^2_{k,N'}(r_kC_k) = \big(r_k M_{k,N'}(C_k)\big)^2 = M^2_{k,N'}(C_k) \quad \forall N' \ge 1$$

while sending $\chi_{k,N} \mapsto r_k\chi_{k,N}$. Hence for bounded measurable f and bounded $\mathcal{G}_\infty$-measurable h, $\mathbb{E}[f((\chi_{k,N})_k)h] = \mathbb{E}[f((r_k\chi_{k,N})_k)h]$ at every fixed r. Averaging over r drawn *iid* Rademacher independently of everything, $(r_k\chi_{k,N})_k$ is *iid* Rademacher for any $(\chi_{k,N})_k$, so the inner expectation is the constant $\mathbb{E}[f(\text{Rad})]$ and $\mathbb{E}\big[f\big((\chi_{k,N})_k\big)\,h\big] = \mathbb{E}\big[f(\text{Rad})\big]\,\mathbb{E}[h]$, delivering both claims. □

E.4 Conditional CLT for a Fixed Design

This subsection establishes uniform mixing of Z_B^π over $\mathcal{G}_\infty$ at a *deterministic* design. Consider $(K_{0B}, N_{0B}) \in \mathbb{N}^2$ with $K_{0B}, N_{0B} \ge 2$, $K_{0B}/B \to c' \in (0,1]$ and $N_{0B} \to N_{\underline{0}}$. Since N_{0B} is integer valued, there is B_0 with $N_{0B} = N_{\underline{0}}$ for all $B \ge B_0$.

Corollary E.5. *Frozen Conditions in the Cluster Problem. There is an almost sure event $\Omega_0 \in \mathcal{G}_\infty$ on which, for every $\delta > 0$,*

$$(i)\quad \frac{1}{K_{0B}} \sum_{k=1}^{K_{0B}} R_{k,N_{0B}}^2 \overset{a.s.}{\underset{B}{\to}} \theta_{N_{\underline{0}}}^2 = \tau^2 + \frac{\sigma^2}{N_{\underline{0}}}, \qquad (ii)\quad \frac{1}{K_{0B}} \sum_{k=1}^{K_{0B}} R_{k,N_{0B}}^2 \, \mathbb{1}\{R_{k,N_{0B}} > \delta\sqrt{K_{0B}}\} \overset{a.s.}{\underset{B}{\to}} 0$$

Proof of Corollary E.5. For $B \geq B_0$ the depth is fixed at $N_{\underline{0}}$, so both statements reduce to Lemma D.12 applied to the *iid* sequence $(R_{k,N_{\underline{0}}})_k$, which has $\mathbb{E}[R_{k,N_{\underline{0}}}^2] = \theta_{N_{\underline{0}}}^2 < \infty$ by Assumption 3.1. Take Ω_0 to be the corresponding almost sure event. □

Corollary E.6. *L^1 Convergence of the Conditional Characteristic Function. Let $Z_{0B} := K_{0B}^{-1/2} \sum_{k \leq K_{0B}} M_{k,N_{0B}}$, let $\phi_{0B}(t) := \mathbb{E}[\exp(itZ_{0B}) \mid \mathcal{G}_\infty]$, and let $\phi(t) = \exp(-\theta_{N_{\underline{0}}}^2 t^2/2)$. Then*

$$\big|\phi_{0B}(t) - \phi(t)\big| \overset{a.s.}{\underset{B}{\to}} 0 \qquad \text{and} \qquad \mathbb{E}\big|\phi_{0B}(t) - \phi(t)\big| \to 0 \qquad \forall t \in \mathbb{R}$$

Proof of Corollary E.6. Immediate from Theorem D.14 in light of Corollary E.5, the signs $(\chi_{k,N_{\underline{0}}})_k$ being *iid* Rademacher and independent of $\mathcal{G}_\infty$ by Corollary 6.4 at the fixed depth $N_{\underline{0}}$. □

E.5 Conditional CLT for Random Even Designs

Proof of Proposition 6.6. The argument mirrors Theorem D.19 and Corollary D.21. The only difference is that the depth margin must be frozen before the width asymptotics can run. *Step 1.* For fixed B, the decomposition of the conditional characteristic function in Step 1 of Theorem D.19 goes through by summing over the finite grid of budget-feasible designs $\{K_B^\pi = K_{0B}, N_B^\pi = N_{0B}\}$, which is $\mathcal{G}_\infty$-measurable by evenness. On each cell the depth is fixed, so Corollary 6.4 applies and the conditional expectation freezes.

Step 2. Let $\phi_{K_{0B},N_{0B}}(t)$ be conditional characteristic function at a deterministic design (K_{0B}, N_{0B}), and define $T_B := \sup_{K_0 \geq cB} |\phi_{K_{0B},N_{\underline{0}}}(t) - \phi(t)|$. On the probability-one frozen event $\{N_{0B} = N_{\underline{0}}\}$, $\phi_{K_{0B},N_{\underline{0}}}(t) \underset{B}{\to} \phi(t)$. Hence, $T_B \to 0$ a.s. for any $c \geq c'$. Then

$$\mathbb{P}\big(|\phi_B(t) - \phi(t)| > \eta\big) \;\leq\; \mathbb{P}\big(N_B^\pi \neq N_{\underline{0}}\big) + \mathbb{P}\big(K_B^\pi < cB\big) + \mathbb{P}(T_B > \eta) \;\to\; 0$$

The first term vanishes by N-consistency and the second by non-starvation. Uniform boundedness upgrades convergence in probability to L^1 convergence. *Step 3.* The events $(A_B)_B$ are $\mathcal{G}_\infty$-measurable by evenness, thus Corollary D.15 delivers the uniform mixing statement over $\mathcal{G}_\infty$. The first conditional statement follows by dividing by $\mathbb{P}(A_B)$, eventually bounded away from zero, and applying Lévy's continuity theorem. For the studentized statement, Lemma 4.1.ii) gives $\hat{b}_{K,N_{\underline{0}}}^2 \to \theta_{N_{\underline{0}}}^2$ almost surely along any diverging K, so the same composition argument as in Step 2 guarantees convergence in probability over the random index. Slutsky concludes exactly as in Corollary D.21. □

E.6 Asymptotically Even Designs

Corollary E.7. *Asymptotically Even Events in the Cluster Problem. Let π be an asymptotically even, N-consistent policy with surrogate $(\tilde{K}_B, \tilde{N}_B)$, and let $A_B = \{(\Xi_B^\pi, N_B^\pi) \in C\}$ for a Borel $C \subseteq \mathbb{R} \times \mathbb{N}$ satisfying the margin condition (10). Set $\tilde{A}_B = \{(\tilde{\Xi}_B, \tilde{N}_B) \in C\} \in \mathcal{G}_\infty$. Then $\mathbb{P}(A_B \triangle \tilde{A}_B) \to 0$.*

Proof of Corollary E.7. Because N_B^π and $\tilde{N}_B$ are integer valued, $|N_B^\pi - \tilde{N}_B| \xrightarrow{p} 0$ is equivalent to $\mathbb{P}(N_B^\pi = \tilde{N}_B) \to 1$. The depth coordinate is asymptotically fixed and can contribute to the symmetric difference only on an event of vanishing probability. Hence

$$\mathbb{P}\big(A_B \triangle \tilde{A}_B\big) \;\le\; \mathbb{P}\big(N_B^\pi \neq \tilde{N}_B\big) + \mathbb{P}\Big(\big\{\mathbb{1}_C(\Xi_B^\pi, N_B^\pi) \neq \mathbb{1}_C(\tilde{\Xi}_B, N_B^\pi)\big\}\Big)$$

where the second term is of the form of object bounded in Lemma D.28. In particular, the margin condition and $|\Xi_B^\pi - \tilde{\Xi}_B| \xrightarrow{p} 0$ send it to zero. The depth is eventually frozen and enters only through the section $C_{N_0} = \{x : (x, N_0) \in C\}$. □

Lemma E.8. *Increment Vanishes at Ratio-close Designs. Let (K_B^π, N_B^π) and $(\tilde{K}_B, \tilde{N}_B)$ be designs with bounded depths $2 \le N_B^\pi, \tilde{N}_B \le N_M < \infty$ a.s., such that $|\nu_B^\pi - \tilde{\nu}_B| \xrightarrow[B]{p} 0$, $\mathbb{P}(\nu_B^\pi \wedge \tilde{\nu}_B \ge c) \to 1$ for some $c > 0$, and $\mathbb{P}(N_B^\pi = \tilde{N}_B) \to 1$. Writing $\tilde{Z}_B = \tilde{K}_B^{-1/2} \sum_{k \le \tilde{K}_B} M_{k,\tilde{N}_B}$,*

$$\big|Z_B^\pi - \tilde{Z}_B\big| \xrightarrow[B]{p} 0$$

Proof of Lemma E.8. Split the difference at the common depth N_B^π,

$$\big|Z_B^\pi - \tilde{Z}_B\big| \;\le\; \underbrace{\left|\frac{1}{\sqrt{K_B^\pi}} \sum_{k \le K_B^\pi} M_{k,N_B^\pi} - \frac{1}{\sqrt{\tilde{K}_B}} \sum_{k \le \tilde{K}_B} M_{k,N_B^\pi}\right|}_{\text{width}} + \underbrace{\left|\frac{1}{\sqrt{\tilde{K}_B}} \sum_{k \le \tilde{K}_B} \big(M_{k,N_B^\pi} - M_{k,\tilde{N}_B}\big)\right|}_{\text{depth}}$$

The depth term cancels exactly at the probability-1 event $\{N_B^\pi = \tilde{N}_B\}$. For the width term, fix $\eta > 0$ and decompose over the finitely many admissible depths

$$\mathbb{P}\big(\text{width} > \eta\big) \;\le\; \sum_{N=2}^{N_M} \mathbb{P}\left(\left|\frac{1}{\sqrt{K_B^\pi}} \sum_{k \le K_B^\pi} M_{k,N} - \frac{1}{\sqrt{\tilde{K}_B}} \sum_{k \le \tilde{K}_B} M_{k,N}\right| > \eta\right)$$

For each fixed N, the sequence $(M_{k,N})_k$ is *iid*, mean zero, with finite variance θ_N^2, and the indices $K_B^\pi, \tilde{K}_B$ satisfy the hypotheses of Lemma D.31 with asymptotics running over B (not N). The sum runs over finitely many terms, and each of them vanishes completing the result. □

Theorem E.9. *Inference for Asymptotically Even Designs in the Cluster Problem. Let π be an asymptotically even, non-starving, N-consistent policy, and let $(A_B)_B$ be a sequence of scaled design events as in (9) satisfying the margin condition (10). Then*

$$\big|\mathbb{E}\big[\exp(itZ_B^\pi)\,\mathbb{1}_{A_B}\big] - \phi(t)\,\mathbb{P}(A_B)\big| \to 0 \qquad \forall t \in \mathbb{R}$$

with $\phi(t) = \exp(-\theta_{N_0}^2 t^2/2)$. If in addition $\liminf_B \mathbb{P}(A_B) > 0$, then

$$\mathcal{L}\Big(Z_B^\pi \;\Big|\; A_B\Big) \implies N\big(0, \theta_{N_0}^2\big), \qquad \mathcal{L}\left(\frac{\sqrt{K_B^\pi}\big(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu\big)}{\hat{b}_{K_B^\pi, N_B^\pi}} \;\middle|\; A_B\right) \implies N(0,1)$$

Proof of Theorem E.9. The argument is that of Theorem D.33 with minor changes. *Step 0: the surrogate also inherits N-consistency.* Non-starvation transfers exactly as in Step 0 of Theorem D.33. For N-consistency, $\mathbb{P}\big(\tilde{N}_B \neq N_0\big) \;\le\; \mathbb{P}\big(\tilde{N}_B \neq N_B^\pi\big) + \mathbb{P}\big(N_B^\pi \neq N_0\big) \;\to\; 0$, by asymptotic evenness

and N-consistency of π respectively. The surrogate is therefore even, non-starving and N-consistent, and so Proposition 6.6 applies.

Step 1: decomposition. Bound the target by the three terms of Step 1 in Theorem D.33, with $(\tilde{K}_B, \tilde{N}_B)$ and $\tilde{A}_B$ in place of the surrogate design and event. The third term is controlled by $|\mathbb{P}(A_B) - \mathbb{P}(\tilde{A}_B)| \leq \mathbb{P}(A_B \triangle \tilde{A}_B) \to 0$ from Corollary E.7, and the second term by Proposition 6.6 applied to the surrogate. *Step 2: the coupling term.* Decompose the first term as in Step 2 of Theorem D.33. The event-mismatch piece is bounded by $\mathbb{P}(A_B \triangle \tilde{A}_B) \to 0$. The increment piece is bounded using the same ingredients: $|e^{ia} - e^{ib}| \leq \min\{2, |a-b|\}$ and $\mathbb{E}[\min\{2, |t||Z_B^\pi - \tilde{Z}_B|\}] \to 0$, by Lemma E.8 and uniform boundedness. Lemma E.8 replaces here Lemma D.31 precisely because the cluster design carries a second, non-divergent coordinate. *Step 3: conditional limits.* Dividing by $\mathbb{P}(A_B)$ and applying Lévy's continuity theorem gives the first conditional statement. Studentization only requires $\hat{b}^2_{K_B^\pi, N_B^\pi} \xrightarrow{p} \theta^2_{N_0}$, which has been established in Step 3 of Proposition 6.6, together with $\liminf_B \mathbb{P}(A_B) > 0$. Measurability of (K_B^π, N_B^π) or A_B was never invoked. □

E.7 Application to `EtC-CS` and `ACS`

It remains to verify that both algorithms are asymptotically even, non-starving and N-consistent, and that the relevant terminal design events satisfy the margin condition. As anticipated in the main text and Appendix D, the margin condition is concretized through a CLT on the design statistic, which requires finite fourth moments. The assumption is strictly weaker than the subgaussian Assumption 5.3 needed for performance.

Assumption E.10. *Fourth Moments.* $\mathbb{E}[u_k^4] < \infty$ *and* $\mathbb{E}[\varepsilon_{ki}^4] < \infty$.

Throughout, recall $\alpha_{k,N_p} = S^2_{k,N_p} - \sigma^2$ and $\beta_{k,N_p} = M^2_{k,N_p} - \theta^2_{N_p}$, both *iid* across clusters, mean zero, and with finite variance under Assumption E.10.

Lemma E.11. *Uniform Rates for the Even Surrogate. Fix* N_p *and define*

$$\tilde{b}^2_{K,N_p} = \frac{1}{K}\sum_{k\leq K} M^2_{k,N_p}, \qquad \tilde{\lambda}_{K,N_p} = \Pi_{[\lambda_m,\lambda_M]}\left(\frac{\hat{w}^2_{K,N_p}}{\Pi_{[\tau_m^2,\tau_M^2]}\big(\tilde{b}^2_{K,N_p} - \hat{w}^2_{K,N_p}/N_p\big)}\right)$$

Let $K \in [K_M, K_m]$ *a.s., then under Assumptions 3.1, 3.2, 3.3 and E.10,*

i) $\hat{w}^2_{K,N_p}$, $\tilde{b}^2_{K,N_p}$ *and* $\tilde{\lambda}_{K,N_p}$ *are* $\mathcal{G}_\infty$*-measurable for every* K,

ii) $\sup_K|\hat{b}^2_{K,N_p} - \tilde{b}^2_{K,N_p}| = O_p(1/B)$, *and consequently* $\sup_K|\hat{\lambda}_{K,N_p} - \tilde{\lambda}_{K,N_p}| = O_p(1/B)$,

iii) $\sup_K|\hat{\lambda}_{K,N_p} - \lambda_0| = O_p(B^{-1/2})$ *and* $\sup_K|\tilde{\lambda}_{K,N_p} - \lambda_0| = O_p(B^{-1/2})$,

iv) for some fixed K_{0B}, $A < \infty$, *and* $J = \lceil A\sqrt{B}\rceil$, $\max_{|j|\leq J}\big|\tilde{\lambda}_{K_{0B}+j,N_p} - \tilde{\lambda}_{K_{0B},N_p}\big| = O_p\big(B^{-3/4}\big)$

Proof of Lemma E.11. *(i)* $\hat{w}^2_{K,N_p}$ and $\tilde{b}^2_{K,N_p}$ are averages of the $\mathcal{G}_\infty$-generators $(S^2_{k,N_p})_k$ and $(M^2_{k,N_p})_k$, and $\tilde{\lambda}_{K,N_p}$ is a measurable map of the two. Result follows.

(ii) Since $\hat{b}^2_{K,N_p} = (K-1)^{-1}\big[\sum_{k\leq K} M^2_{k,N_p} - K\bar{M}^2_{K,N_p}\big]$,

$$\hat{b}^2_{K,N_p} - \tilde{b}^2_{K,N_p} = \frac{1}{K(K-1)}\sum_{k\leq K} M^2_{k,N_p} \; - \; \frac{K}{K-1}\,\bar{M}^2_{K,N_p}$$

The first term is $(K-1)^{-1}$ times an average converging to $\theta^2_{N_p}$, hence $O_p(1/B)$ uniformly since $\Theta(B) = K_M \le K \le K_m = \Theta(B)$. For the second, Kolmogorov's maximal inequality gives $\max_{K\le K_m}|\sum_{k\le K} M_{k,N_p}| = O_p(\sqrt{K_m})$, so $\sup_{K\le K_m} \bar{M}^2_{K,N_p} \le O_p(K_m)/K_M^2 = O_p(1/B)$, using $K_m/K_M = O(1)$. Therefore, $\sup_{K\le K_m}|\hat{b}^2_K - \tilde{b}^2_K| = O_p(1/B)$. For the transfer to λ, note that $\hat{\tau}^2_{K,N_p}$ and $\tilde{\tau}^2_{K,N_p} := \Pi_{[\tau_m^2,\tau_M^2]}(\tilde{b}^2_{K,N_p} - \hat{w}^2_{K,N_p}/N_p)$ both share the term $\hat{w}^2_{K,N_p}/N_p$, which cancels, and so $|\tilde{\tau}^2 - \hat{\tau}^2| \le |\tilde{b}^2 - \hat{b}^2|$ by 1-Lipschitzness of the clip. The two clipped denominators exceed τ_m^2, so

$$\sup_K \left|\hat{\lambda}_{K,N_p} - \tilde{\lambda}_{K,N_p}\right| \;\le\; \sup_K \hat{w}^2_{K,N_p} \frac{|\tilde{\tau}^2_{K,N_p} - \hat{\tau}^2_{K,N_p}|}{\hat{\tau}^2_{K,N_p}\,\tilde{\tau}^2_{K,N_p}} \;\le\; \frac{\sup_K \hat{w}^2_{K,N_p}}{(\tau_m^2)^2} \sup_K \left|\hat{b}^2_{K,N_p} - \tilde{b}^2_{K,N_p}\right| = O_p(1/B)$$

using $\sup_K \hat{w}^2_{K,N_p} \le K_M^{-1}\sum_{k\le K_m} S^2_{k,N_p} = O_p(1)$ and the 1-Lipschitzness of the outer clip.

(iii) By Kolmogorov's maximal inequality applied to $(\alpha_{k,N_p})_k$ and $(\beta_{k,N_p})_k$, both with finite variance under Assumption E.10,

$$\sup_K \left|\hat{w}^2_{K,N_p} - \sigma^2\right| \le \frac{1}{K_M} \max_{K\le K_m} \left|\sum_{k\le K} \alpha_{k,N_p}\right| = O_p\big(B^{-1/2}\big), \qquad \sup_K \left|\tilde{b}^2_{K,N_p} - \theta^2_{N_p}\right| = O_p\big(B^{-1/2}\big)$$

Combining this result with part (ii), $\sup_K|\hat{b}^2 - \theta^2_{N_p}| = O_p(B^{-1/2})$. The clipped τ^2 estimator is Lipschitz as the clipping operator is 1-Lipschitz. Moreover, the clipped λ_0 estimator is also Lipschitz because the denominator is bounded below by τ_m^2 and Lipschitzness of the clipping map. Using $\lambda_0 \in [\lambda_m, \lambda_M]$, then $\sup_K|\hat{\lambda}_{K,N_p} - \lambda_0| = O_p(B^{-1/2})$. The statement for $\tilde{\lambda}$ follows from part (ii) and the triangle inequality.

(iv) This analysis is more subtle. Using the same Lipschitz argument for the clipping and $\lambda_0 \in [\lambda_m, \lambda_M]$, $|\tilde{\lambda}_{K_{0B}+j} - \tilde{\lambda}_{K_{0B}}| \le L_g\big(|\hat{w}^2_{K_{0B}+j} - \hat{w}^2_{K_{0B}}| + |\tilde{b}^2_{K_{0B}+j} - \tilde{b}^2_{K_{0B}}|\big)$. The two terms are handled identically, so I only discuss here $\hat{w}^2_{K,N_p}$. Decompose $\hat{w}^2_{K,N_p} = \sigma^2 + T_K/K$ with $T_K = \sum_{k\le K}\alpha_{k,N_p}$. Then, for every $j : |j| \le J$,

$$\hat{w}^2_{K_{0B}+j} - \hat{w}^2_{K_{0B}} = \frac{T_{K_{0B}+j} - T_{K_{0B}}}{K_{0B}+j} \;-\; \frac{j\,T_{K_{0B}}}{K_{0B}(K_{0B}+j)}$$

The first term may be interpreted as new noise, and the latter as a re-dilution term. For new noise, Kolmogorov's maximal inequality applied forwards on $(K_{0B}, K_{0B}+J]$ and backwards on $(K_{0B}-J, K_{0B}]$ around the deterministic centre K_{0B} gives $\max_{|j|\le J}|T_{K_{0B}+j} - T_{K_{0B}}| = O_p(\sqrt{J}) = O_p(B^{1/4})$. For the re-dilution term, $|T_{K_{0B}}| = O_p(\sqrt{B})$ by Chebyshev and the fact that $|j| \le J = O(\sqrt{B})$. Since, eventually, $K_{0B} - J = \Theta(B)$,

$$\max_{|j|\le J}\left|\hat{w}^2_{K_{0B}+j} - \hat{w}^2_{K_{0B}}\right| = O_p\left(\frac{B^{1/4}}{B}\right) + O_p\left(\frac{\sqrt{B}\cdot\sqrt{B}}{B^2}\right) = O_p\big(B^{-3/4}\big)$$

The proof is completed by carrying the same computation over $\tilde{b}^2_{K,N_p} = \theta^2_{N_p} + K^{-1}\sum_{k\le K}\beta_{k,N_p}$. □

Proposition E.12. *`EtC-CS` Is Asymptotically Even. Assume $N^*(\lambda_0) \notin \mathbb{N}$. `EtC-CS` is asymptotically even, non-starving and N-consistent, with even surrogate $(\tilde{K}_B, \tilde{N}_B) = \big(K^*(\tilde{\lambda}_{K_M,N_p}), N^*(\tilde{\lambda}_{K_M,N_p})\big)$ under the rounding convention of the feasible policy. Moreover $|K_B^{\texttt{EtC-CS}} - \tilde{K}_B| = O_p(1)$.*

Proof of Proposition E.12. Non-starvation and N-consistency are given by Corollary 4.7 and Lemma 4.4. The commitment stage K_M is deterministic, so $\tilde{\lambda}_{K_M,N_p}$ is $\mathcal{G}_\infty$-measurable by Lemma

E.11.i) and the surrogate design is even, provided the optimal map is trivially measurable. For the width coupling, observe that h is L_h-Lipschitz on $[\lambda_m, \lambda_M]$ and $K^*(\lambda) = Bh(\lambda)$, so by Lemma E.11.ii)

$$\left|K_B^{\texttt{EtC-CS}} - \tilde{K}_B\right| \;\le\; B\,L_h \left|\hat{\lambda}_{K_M,N_p} - \tilde{\lambda}_{K_M,N_p}\right| + 1 \;=\; O_p(1),$$

the 1 accounting for any rounding decisions. Therefore, $|K_B^{\texttt{EtC-CS}} - \tilde{K}_B|/\sqrt{B} \xrightarrow{p} 0$. For the depth, N^* is Lipschitz on $[\lambda_m, \lambda_M]$, so $|N^*(\hat{\lambda}_{K_M,N_p}) - N^*(\tilde{\lambda}_{K_M,N_p})| = O_p(1/B)$, since $N^*(\lambda_0) \notin \mathbb{N}$, the rounding (flooring) coincides with probability tending to one and $|N_B^{\texttt{EtC-CS}} - \tilde{N}_B| \xrightarrow{p} 0$. □

Proposition E.13. *`ACS` Is Asymptotically Even.* *`ACS` is asymptotically even, non-starving and N-consistent, with even surrogate $(\tilde{K}_B, \tilde{N}_B) = \big(K^*(\tilde{\lambda}_{K_{0B},N_p}), N^*(\tilde{\lambda}_{K_{0B},N_p})\big)$, $K_{0B} = \lfloor K^*(\lambda_0) \rfloor$. Moreover $|K_B^{\texttt{ACS}} - \tilde{K}_B| = O_p(B^{1/4})$.*

Proof of Proposition E.13. Non-starvation and N-consistency are given by Corollary 4.7 and Lemma 4.4. The surrogate is even by Lemma E.11.i) and determinism of K_{0B}. For the coupling I proceed in two stages.

Step 1: localizing the commitment stage. Let $D_B := \sup_{K \le K_m} |K^*(\hat{\lambda}_{K,N_p}) - K^*(\lambda_0)| \le BL_h \sup_K |\hat{\lambda}_{K,N_p} - \lambda_0| = O_p(\sqrt{B})$ by Lemma E.11.iii). Recall $s = \inf\{k \ge K_M : k+1 > K^*(\hat{\lambda}_{k,N_p})\}$ from (5). For any $k \ge K^*(\lambda_0) + D_B$ we have $k+1 > K^*(\lambda_0) + D_B \ge K^*(\hat{\lambda}_{k,N_p})$, so the algorithm stops; and, for any $k \le K^*(\lambda_0) - D_B - 1$ we have $k+1 \le K^*(\lambda_0) - D_B \le K^*(\hat{\lambda}_{k,N_p})$, so the stopping condition is not satisfied. Let the commitment stage be equal to $p_B = \max\{s-1, K_M\} \in [K_M, K_m]$. Then, $|p_B - K_{0B}| \le D_B + 2 = O_p(\sqrt{B})$, where the constant absorbs the rounding in K_{0B}.

Step 2: coupling. Fix some constant $A < \infty$ sufficiently large so that the event $\{|p_B - K_{0B}| \le J\}$ with $J = \lceil A\sqrt{B} \rceil$ has probability increasing to 1 (by Step 1). Then

$$\begin{aligned}
\left|K_B^{\texttt{ACS}} - \tilde{K}_B\right| &\le \left|K^*(\hat{\lambda}_{p_B,N_p}) - K^*(\tilde{\lambda}_{p_B,N_p})\right| + \left|K^*(\tilde{\lambda}_{p_B,N_p}) - K^*(\tilde{\lambda}_{K_{0B},N_p})\right| + 1 \\
&\le BL_h |\hat{\lambda}_{p_B,N_p} - \tilde{\lambda}_{p_B,N_p}| + BL_h |\tilde{\lambda}_{p_B,N_p} - \tilde{\lambda}_{K_{0B},N_p}| + 1 \\
&\le BL_h \sup_K \left|\hat{\lambda}_{K,N_p} - \tilde{\lambda}_{K,N_p}\right| + BL_h \max_{|j| \le J} \left|\tilde{\lambda}_{K_{0B}+j,N_p} - \tilde{\lambda}_{K_{0B},N_p}\right| + 1
\end{aligned}$$

which is $O_p(1) + O_p(B^{1/4})$ by Lemma E.11 parts (ii) and (iv). Hence $|K_B^{\texttt{ACS}} - \tilde{K}_B|/\sqrt{B} \xrightarrow{p} 0$. The depth statement follows as in Proposition E.12, since $|\hat{\lambda}_{p_B,N_p} - \tilde{\lambda}_{K_{0B},N_p}| \xrightarrow{p} 0$. □

Remark E.14. *On the Choice of `ACS` Surrogate. The surrogate for `ACS` exploits decisively the content in Remark D.26. The surrogate need be neither feasible nor the "even twin" of `ACS`, it only needs to be $\mathcal{G}_\infty$-measurable. The natural twin policy which runs the same stopping rule on the even sequence $(\tilde{\lambda}_{k,N_p})_k$ with its own stopping time $\tilde{s}$ would be admissible, but would further require coupling two random stopping times at the $\sqrt{B}$ scale. Instead, evaluating the even statistic at the deterministic stage K_{0B} reduces the problem to a single maximal inequality from a fixed center as in Lemma E.11.iv).*

Lemma E.15. *Central Limit Theorem for the Scaled Cluster Counts. Let*

$$V_\lambda = \frac{1}{\tau^8} \operatorname{Var}\Big(\theta_{N_p}^2 S_{k,N_p}^2 - \sigma^2 M_{k,N_p}^2\Big)$$

finite under Assumption E.10 and strictly positive by assumption. Then, for $\pi \in \{\texttt{EtC-CS}, \texttt{ACS}\}$,

$$\Xi_B^\pi \implies N\big(0, \Lambda_\pi^2\big), \qquad \tilde{\Xi}_B^\pi \implies N\big(0, \Lambda_\pi^2\big), \qquad \Lambda_\pi^2 := \frac{h'(\lambda_0)^2\, V_\lambda}{\nu^\pi}$$

with $\nu^{\texttt{EtC-CS}} = h(\lambda_M)$ *and* $\nu^{\texttt{ACS}} = h(\lambda_0)$.

Proof of Lemma E.15. By Lemma E.11.iii) and strict interiority, both clips are inactive with probability tending to one. On that event, $\hat{\lambda}_{K,N_p} = g(\hat{w}^2_{K,N_p}, \hat{b}^2_{K,N_p})$ (without clipping) and

$$\nabla g(\sigma^2, \theta^2_{N_p}) = \Big(\theta^2_{N_p}/\tau^4,\ -\sigma^2/\tau^4\Big)$$

The pairs $(S^2_{k,N_p}, M^2_{k,N_p})$ are *iid* across clusters with finite second moments under Assumption E.10, so the bivariate central limit theorem and the delta method give, along any deterministic $K \to \infty$, $\sqrt{K}(\hat{\lambda}_{K,N_p} - \lambda_0) \implies N(0, V_\lambda)$ with $V_\lambda = \nabla g' \Sigma \nabla g$ and Σ the covariance of the pair. By Lemma E.11.ii) the feasible and surrogate statistics differ by $O_p(1/B) = o_p(B^{-1/2})$, and so they share every $\sqrt{K}$-scale limit.

For `EtC-CS` the commitment stage K_M is deterministic with $K_M/B \to h(\lambda_M)$, giving $\sqrt{B}(\hat{\lambda}_{K_M,N_p} - \lambda_0) \implies N(0, V_\lambda/h(\lambda_M))$. For `ACS` the commitment stage p_B is random, but $p_B/K_{0B} \xrightarrow{p} 1$ by Step 1 of Proposition E.13 and $K_{0B}/B \to h(\lambda_0)$. Normalized *iid* partial sums are uniformly continuous in probability, so Anscombe's random-index central limit theorem (Anscombe, 1952) applies to the bivariate mean (jointly, via Cramér–Wold) over the random index p_B and the delta method gives $\sqrt{B}(\hat{\lambda}_{p_B,N_p} - \lambda_0) \implies N(0, V_\lambda/h(\lambda_0))$.

Finally, $\Xi_B^\pi = \sqrt{B}\big(h(\hat{\lambda}_{\cdot,N_p}) - h(\lambda_0)\big) + O(B^{-1/2})$, with the remainder absorbing the rounding in K_{0B}. One last application of the delta method step through h delivers the limits in the statement. The surrogate shares them because $|\Xi_B^\pi - \tilde{\Xi}_B^\pi| = |K_B^\pi - \tilde{K}_B^\pi|/\sqrt{B} = o_p(1)$, where the numerator uses $|K - \tilde{K}| = O_p(1)$ in `EtC-CS` and $O_p(B^{1/4})$ in `ACS` by Propositions E.12 and E.13. □

Proof of Theorem 6.9. By Propositions E.12 and E.13 both policies are asymptotically even, non-starving and N-consistent. Let $A_B = \{\Xi_B^\pi \in C\}$ for Borel $C \subseteq \mathbb{R}$ with Lebesgue-null boundary and $\mathbb{P}(N(0, \Lambda_\pi^2) \in C) > 0$. Since $\Lambda_\pi^2 > 0$ the limit law is absolutely continuous and assigns zero mass to ∂C, so Lemma E.15 and Remark D.29 guarantee the margin condition (10), and $\mathbb{P}(A_B) \to \mathbb{P}(N(0, \Lambda_\pi^2) \in C) > 0$. Theorem E.9 delivers both conditional statements. Intersecting A_B with $\{N_B^\pi = n\}$ is asymptotically innocuous because only $n = N_{\underline{0}}$ carries non-vanishing probability under N-consistency, as discussed extensively in the text. □

Remark E.16. *Theory Informed Remarks on the Simulation Evidence. Three comments. First, the quantile cells of* K_B^π *used in Section 7 belong to the relevant class of asymptotically even events: intervals in* Ξ_B^π *with Lebesgue-null boundary and limiting probability strictly inside* $(0, 1)$. *Second, the variances of the limiting designs are ranked. Since* h *is strictly decreasing and* $\lambda_0 < \lambda_M$ *under strict interiority,* $\nu^{\texttt{ACS}} = h(\lambda_0) > h(\lambda_M) = \nu^{\texttt{EtC-CS}}$ *and hence* $\Lambda^2_{\texttt{ACS}} < \Lambda^2_{\texttt{EtC-CS}}$. *By committing around* $K^*(\lambda_0)$ *rather than at the worst-case pilot* K_M, `ACS` *estimates the variance ratio on a strictly larger sample and induces a strictly less dispersed design in the limit. Third, Anscombe's theorem enters only for the pilot statistic* $\hat{\lambda}_{p_B,N_p}$, *where the commitment stage is random. It is silent about the outcome statistic of interest* $\hat{\mu}_{K_B^\pi, N_B^\pi}$, *whose design-conditional limit is not a random-index problem and is precisely what evenness and symmetry are needed for.*

F Additional Simulation Evidence

F.1 Performance

Figure 13 traces the median and mean terminal design against the the oracle map (3) evaluated across a grid of variance ratios. Both algorithms track the oracle closely, with a small tendency to oversample depth and undersample width. This reflects the right skew in $\hat{\lambda}_{pB}$ documented in Figure 3.Both algorithms display a similar performance with no clear domination. By definition, the static policies induce the same design invariantl across λ_0 values.

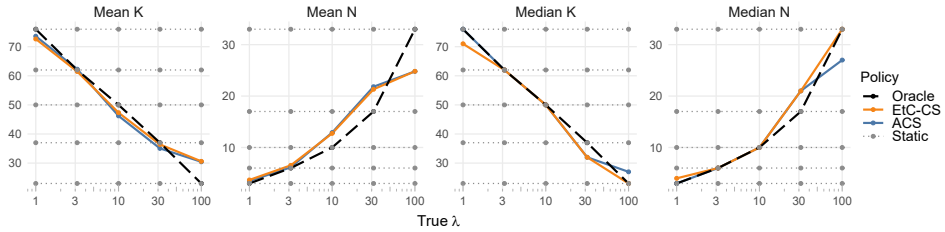


Figure 13: Mean and median terminal design (K_B^π, N_B^π) across λ_0

Notes: Static policies imputed at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $B = 2{,}000$, $F = 20$, $V = 2$.

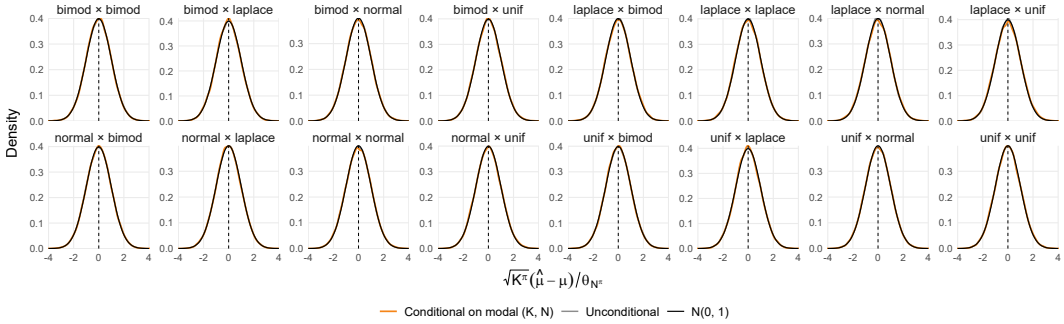


Figure 14: Empirical density of the design-conditional statistic $\sqrt{K_B^\pi}(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu)/\theta_{N_B^\pi}$ at the modal design across symmetric error distributions

Notes: Distribution of u_k (left) and ε_{ki} (right). 10^5 replications. $\mu = 0$, $\omega = (1, 5)$. $B = 2{,}000$, $F = 20$, $V = 2$.

G Appendix to Empirical Application

G.1 Data and Variable Construction

Sampling frame. All the details below have been obtained from the paper (Fink et al., 2020b), online appendices and pre-register (Fink et al., 2020a), repository data (Fink et al., 2024), and pre-analysis plan (Fink et al., 2014), as well as several communications with the authors and the relevant IPA division in charge of implementation. FJM began from a random list of 250 villages—five per administrative block, for 50 of the 53 blocks in Chipata District, Eastern Province, Zambia—with populations reported to comprise between 20 and 100 farming households. After inspection and logistical screening, 175 villages entered the experiment. The smallest of these villages contained 22 medium-sized farmers, and so the authors agreed to sample 22 households per village. Realized depth, however, ranged from 5 to 22, with an average of 17.9, leading to a baseline sample of 3,139 households, of which 3,028 appear in the year-one survey. Treatment was assigned at the village level after

the baseline survey using min-max T-randomization (Bruhn and McKenzie, 2009) resulting in three equally sized arms.

Outcome variables. The pre-registration lists harvest value as the primary outcome, while the published paper emphasizes reallocation of informal labor (*ganyu*). I therefore target log(1+total harvest value) and log(1 + ganyu mandays) as outcome variables. For the second, FJM report the binary variables `did_ganyu` and `plan_ganyu`. Binary outcomes cannot accommodated by the additive-error model (1), and so I consider continuous variable alternative. In particular, I use the continuous variable `did_ganyu_mandays_long`, which, despite being available at baseline and in both follow-up rounds, is not reported in the paper. Alternative continuous outcomes like `did_ganyu_hours_yest` and `did_ganyu_hours_short` were unavailable at baseline.

Trimming and residualization. The bottom 0.5% of baseline harvest values (16 observations) is trimmed for implausibly low values. The same absolute threshold (179.5 ZMW) is applied to the year-one data. Both outcomes are then residualized on the covariates entering the harvest value regression in Fink et al. (2020a), Table 4 Panel B, with three exclusions: the outcome's own baseline value, the assignment variable `control_gift`, which is not defined pre-assignment, and—for ganyu mandays only—the mechanically related outcomes `did_ganyu` and `plan_ganyu`. In both settings, harvest and ganyu mandays, I residualize by using the specification of harvest value because the outcome ganyu mandays does not enter any of the reported regressions. All specifications in the authors' analysis share nearly identical controls anyway, so this decision is probably without loss.

G.2 Homogeneity of the Variance Components

Section 8 identifies the design from baseline data alone. Therefore, there is an implicit running assumption requiring homogeneity of the variance components across survey rounds. I split this assumption in two testable hypotheses. Let y^d_{kit} be the potential outcome d of variable y in period $t \in \{0, 1\}$, then

$$\text{(A1) Time homogeneity} : y^d_{ki1} \overset{d}{=} y^d_{ki0} \text{ for all } d$$

(A1) is only testable on control units, since y^1_{kit} is unobserved at baseline. I run two versions of the test: one using control units in both rounds, which permits a paired bootstrap, i.e. $(\tau^2_{00}, \lambda_{00}) = (\tau^2_{10}, \lambda_{10})$ and one using all baseline units against year-one controls, which gains precision at the cost of pairing, i.e. $(\tau^2_{0,0+1}, \lambda_{0,0+1}) = (\tau^2_{10}, \lambda_{10})$. Secondly,

$$\text{(A2) Arm homogeneity at endline} : y^0_{ki1} \overset{d}{=} y^1_{ki1}$$

(A2) is stronger than required by the framework of Section 3, yet it supplies indirect evidence on the plausibility of (A1) for treated units, which is untestable: if time homogeneity holds for control units and the second moments agree across arms after treatment, then $(\tau^2_{01}, \lambda_{01}) = (\tau^2_{11}, \lambda_{11})$ is likely to hold in the data. Arm homogeneity at baseline holds by construction of the RCT, and so (A1-A2) allow us to drop subscripts entirely from the discussion, simplifying the exposition. Tests are conducted using a cluster bootstrap on log-ratios.

Figure 15 summarizes the test results. Homogeneity can only be rejected marginally in three out of the 20 tests. Two qualifications are worth mentioning. First, $\hat{\tau}^2$ is estimated from roughly 58 villages per arm, so the tests might be low-powered in this dimension. And second, a minority

of bootstrap draws produced numerically degenerate cluster variances ($\hat{\tau}^2 \leq 10^{-12}$). These draws (between 0 and 17% of the total) were discarded uniformly across $\hat{\tau}^2$ and $\hat{\lambda}$ estimation.[65]

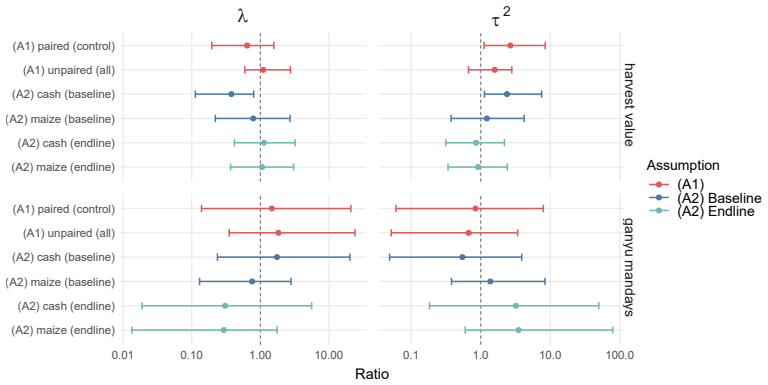


Figure 15: Variance-homogeneity tests

Notes: Cluster bootstrap on log-ratios. Dashed line at 1. 2,000 bootstrap draws.

G.3 Error Symmetry

The inference invariance theory relies on Assumption 6.2 which requires the entire error block $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ to satisfy sign-flip distributional invariance, i.e. $C_k \overset{d}{=} -C_k$. Unfortunately, when only cluster average residuals $M_{k,N} = u_k + \bar{\varepsilon}_{k,N}$ are available, this assumption is untestable. Testing $M_{k,N} \overset{d}{=} -M_{k,N}$ is necessary but not sufficient for testing $C_k \overset{d}{=} -C_k$. $M_{k,N}$ is an odd linear functional of C_k, so joint symmetry implies $M_{k,N} \overset{d}{=} -M_{k,N}$ at every depth, but the reverse is not true. Testing across a grid of depths is nonetheless informative. For small N, household noise ε_{ik} dominates, while for large N, $M_{k,N} \approx u_k$, and so the cluster-level noise dominates. Therefore, a joint test along the N dimension, i.e. $(M_{k,N})_N = -(M_{k,N})_N$, would examine the two error sources in different proportions. I run such test at $N \in \{4, 8, 12, 16, 20\}$, restricting the sample entering each test to villages with $N^k \geq N$ and randomly drawing N households from each village. Figure 16 plots nearly symmetric distributions for the cluster residuals $\hat{M}_{k,N} = \hat{\mu}^k_N - \hat{\mu}_{K,N}$ at three depths.

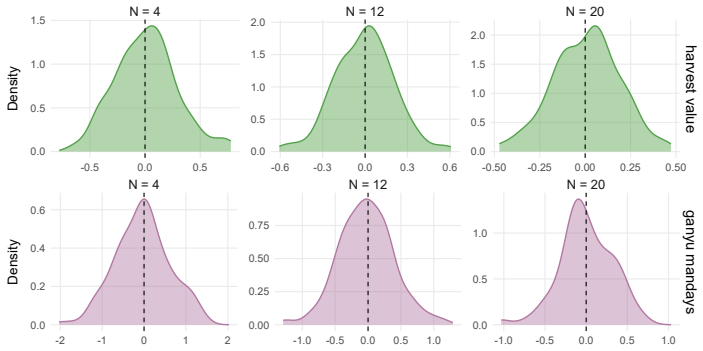


Figure 16: Density of the centered residuals $\hat{M}_{k,N}$

Notes: $N \in \{4, 12, 20\}$ with covariate-residualized data at baseline. Each panel uses a single random draw of N households per village. Similarly, the subsampling draws Table 7 aggregates over are not pooled. Only villages with $N^k \geq N$ are included. At $N = 20$, only $K = 66$ villages qualify, which accounts for the coarseness of the rightmost panel.

[65] All estimators $(\hat{\tau}^2, \hat{\lambda})$ use the unclipped counterparts to the estimators in Section 4.1 adapted to unequal village sizes N^k. Replacing the estimators by their standard uniform-depth form, i.e. $N = K^{-1} \sum_k N^k \approx 18$, leaves the input estimates nearly unchanged.

I conduct the symmetry test of Miao et al. (2006), calibrating the null via parametric bootstrap up to an $O(1/K)$ term.[66] The choice of N households per village is not unique, therefore I further bootstrap over sub-sampling draws and report the dependence-robust aggregate $\min\{1, 2 \times \text{median } p\text{-value}\}$ as in Meinshausen et al. (2009). Table 7 reports the results of the test. At standard confidence levels, symmetry cannot be rejected for either outcome at any depth. Although underpowered by construction, the results in this Appendix are conducive to block symmetry and support the implementation of standard inference.

Outcome	N	$k : N^k \geq N$	med(p)	Valid p-value	$Q_{0.10}(p)$	$Q_{0.90}(p)$
log(harvest value + 1)	4	175	0.501	1.000	0.101	0.904
	8	174	0.595	1.000	0.229	0.915
	12	166	0.756	1.000	0.361	0.948
	16	141	0.533	1.000	0.264	0.806
	20	66	0.615	1.000	0.487	0.780
log(ganyu mandays + 1)	4	175	0.461	0.922	0.072	0.864
	8	174	0.457	0.914	0.102	0.911
	12	166	0.496	0.992	0.097	0.894
	16	142	0.589	1.000	0.245	0.923
	20	67	0.500	1.000	0.342	0.854

Table 7: Symmetry test of Miao et al. (2006) on the centered residuals $\hat{M}_{k,N}$

Notes: For each depth, villages with $N^k \geq N$ are subsampled to depth N. The MGG statistic is calibrated by a Rademacher sign-flip null over 2,000 draws. Table reports the median, 10th and 90th percentile for the p-value recomputed over 200 independent sampling draws. Valid p-value $= \min\{1, 2\,\text{med}(p)\}$ is the dependence-robust aggregate of Meinshausen et al. (2009).

G.4 Semi-Synthetic Data Generation

Holding the budget fixed, any counterfactual design will require a population larger than the realized sample, as it will demand either more villages or more households per village than those actually surveyed. Thus, I construct and calibrate a semi-synthetic population for each outcome and each survey round (baseline and year 1).

Starting from the realized data, I residualize the outcome variables with baseline covariates according to the aforementioned procedure. Year-one data uses the specification exactly. Baseline data omits the outcome's own lag, the assignment variable `control_gift`, and, for ganyu mandays, `did_ganyu` and `plan_ganyu`. I then recover the design input estimates $(\hat{\mu}, \hat{\tau}^2, \hat{\sigma}^2)$ and decompose the residuals into an intercept, a village component and a household component. Residuals are re-scaled and re-centered such that sample moments are matched exactly. Synthetic data are generated under the additive model by sampling u_k and ε_{ki} independently and uniformly from the rescaled empirical residuals. This construction presents two limitations. First, it understates tail mass. Data are constructed by resampling realized residuals and so they cannot extrapolate beyond their support. Second, the model cannot reproduce point masses because it assumes the additive structure in Assumption 3.1, hence the mass at zero in ganyu mandays is smoothed. To gain perspective on the issue, the tenth percentile is exactly zero in the real data and -0.26 in the synthetic. Tables 8 and 9 compare the quantile distributions of the synthetic and realized data. Away from the kink at zero,

[66] The vector of residuals inherits the symmetry properties of the vector of errors, but the per-cluster Rademacher flips used to calibrate the null preserve the joint law only up to an $O(1/K)$ term arising from the common demeaning. At $K = 175$ this adjustment is immaterial.

both distributions track closely.

Outcome	Arm	Source	N	Mean	SD	$Q_{0.10}$	$Q_{0.25}$	$Q_{0.50}$	$Q_{0.75}$	$Q_{0.90}$
log(harvest value + 1)	NA	Simulated	36000	0.004	0.515	-0.638	-0.319	0.026	0.338	0.624
		Real	3123	-0.000	0.513	-0.631	-0.315	0.026	0.328	0.611
log(ganyu mandays + 1)	NA	Simulated	36000	0.000	1.193	-1.463	-0.819	-0.064	0.771	1.553
		Real	3136	-0.000	1.190	-1.405	-0.856	-0.080	0.793	1.571

Table 8: Validation of the calibrated DGP against the realized baseline sample

Notes: First and second moments are matched by construction.

Outcome	Arm	Source	N	Mean	SD	$Q_{0.10}$	$Q_{0.25}$	$Q_{0.50}$	$Q_{0.75}$	$Q_{0.90}$
log(harvest value + 1)	cash	Simulated	1044	0.123	0.652	-0.688	-0.259	0.156	0.551	0.913
		Real	1023	0.112	0.645	-0.737	-0.255	0.172	0.517	0.909
	control	Simulated	1044	-0.002	0.647	-0.830	-0.386	0.040	0.427	0.784
		Real	952	-0.000	0.660	-0.860	-0.414	0.044	0.442	0.791
	maize	Simulated	1062	0.023	0.659	-0.817	-0.368	0.062	0.462	0.816
		Real	1016	0.037	0.657	-0.800	-0.338	0.117	0.475	0.812
log(ganyu mandays + 1)	cash	Simulated	1044	-0.175	1.318	-1.877	-1.116	-0.202	0.768	1.559
		Real	600	-0.182	1.293	-1.921	-1.141	-0.214	0.768	1.499
	control	Simulated	1044	-0.004	1.314	-1.690	-0.953	-0.034	0.911	1.742
		Real	575	0.000	1.305	-1.730	-0.979	-0.060	0.990	1.799
	maize	Simulated	1062	-0.223	1.317	-1.910	-1.180	-0.240	0.712	1.518
		Real	636	-0.210	1.347	-1.983	-1.229	-0.285	0.811	1.561

Table 9: Validation of the calibrated DGP against the realized year-one sample.

Notes: First and second moments are matched by construction.

H Appendix to Section 9

Assumption H.1. *Assumptions for the Heterogeneous-Variances Extension. Consider the model with heterogeneous cluster variances in* (11).

(H1) ***Errors.*** u_k*:* $\mathbb{E}[u_k \mid X_k] = 0$, $\mathbb{E}[u_k^2 \mid X_k] = \tau_k^2 = \ell(X_k, \gamma)$ *with* $\gamma \in \Gamma$, $\dim(\gamma) < \infty$; ε_{ki}*:* $\mathbb{E}[\varepsilon_{ki}] = 0$, $\mathbb{E}[\varepsilon_{ki}^2] = \sigma^2$, *and* $\varepsilon_{ki} \perp\!\!\!\perp X_k$; $X \in \mathcal{X} = [x_m, x_M]$.

(H2) ***Compactness.*** $\mathcal{X}, \Gamma$ *known and compact;* ℓ *continuous and bounded away from zero on* $\mathcal{X} \times \Gamma$*; and* $\sigma^2 \in [\sigma_m^2, \sigma_M^2]$ *with* $\sigma_m^2 > 0$. *Consequently,* $\tau_k^2 \in [\tau_{k,m}^2, \tau_{k,M}^2]$ *with* $\tau_{k,m}^2 = \inf_{\gamma\in\Gamma} \ell(X_k, \gamma)$ *and* $\tau_{k,M}^2 = \sup_{\gamma\in\Gamma} \ell(X_k, \gamma)$. *Let* $\tau_m^2 = \min_k \tau_{k,m}^2$ *and* $\tau_M^2 = \max_k \tau_{k,M}^2$. *The implied* λ_0 *bounds are given by* $\lambda_m = \sigma_m^2/\tau_M^2$ *and* $\lambda_M = \sigma_M^2/\tau_m^2$. *Observe that both aggregate pairs* $(\bar{\tau}^2, \bar{\lambda}_0)$ *and* $(\tau_{\mathrm{h}}^2, \lambda_{0\mathrm{h}})$ *lie in the interior of* $[\tau_m^2, \tau_M^2] \times [\lambda_m, \lambda_M]$.

(H3) ***Interiority.*** $\dfrac{\tau_m^2}{\tau_M^2}\sqrt{\lambda_m F/V} \geq 2$ *and* $B \geq 2\big(F + V N_{\max}\big)$ *with* $N_{\max} = \dfrac{\tau_M^2}{\tau_m^2}\sqrt{\lambda_M F/V}$.

(H4) ***Conditional Subgaussianity.*** $u_k \mid X_k$ *is* κ_u*-subgaussian uniformly in* X_k, ε_{ki} *is* κ_ε*-subgaussian.*

Variance of Weighted Averages. Under (H1), the cluster average estimator is unbiased for μ: $\mathbb{E}[\hat{\mu}_{N^k}^k \mid X^K] = \mu$, and so is any weighted combination (conditional on the weights): $\mathbb{E}[\hat{\mu}_\omega \mid X^K] = \mu$. Therefore mean-squared error minimization reduces to minimizing the expected variance $\mathbb{E}[\mathrm{Var}(\hat{\mu}_\omega \mid X^K)]$. Conditional on X^K, cluster means are independent with $\mathrm{Var}(\hat{\mu}_{N^k}^k \mid X^K) = \theta_k^2 := \tau_k^2 + \sigma^2/N^k$,

so

$$\mathbb{E}\big[\mathrm{Var}(\hat{\mu}_\omega \mid X^K)\big] = \mathbb{E}\left[\frac{1}{\big(\sum_k \omega_k\big)^2}\sum_k \omega_k^2\,\theta_k^2\right] \tag{18}$$

For weights that depend on the outcomes—such as the feasible weights of Algorithm 3, which plug in estimated variance components—the display is understood conditionally on the design σ-algebra.

Proof of Lemma 9.1 (Unweighted Averages). With $\omega_k = 1/K$, (18) simplifies to $\mathbb{E}\big[K^{-2}\sum_k \tau_k^2 + \sigma^2 K^{-2}\sum_k 1/N^k\big]$. Conditional on every X^K, depth does not interact with τ_k^2, so the rest of the proof tracks closely the generalized problem in Appendix A: conditional on K, the balanced solution $N^k = N$ follows from convexity of the map $x \mapsto 1/x$. Moreover, by random recruitment $\mathbb{E}\big[K^{-1}\sum_k \tau_k^2\big] = \bar{\tau}^2$, and so the objective reduces to the baseline objective evaluated at $\bar{\tau}^2$. On every realization of X^K the optimal map takes the form

$$N^{k*}(\bar{\lambda}_0) = N^*(\bar{\lambda}_0) = \sqrt{\bar{\lambda} F/V}, \quad K^*(\bar{\lambda}_0) = \frac{B}{F + \sqrt{\bar{\lambda}_0 FV}}$$

Plugging the oracle allocations into the estimator's variance recovers $W_u^* = B^{-1}\,\bar{\tau}^2\left(\sqrt{\bar{\lambda}_0 V} + \sqrt{F}\right)^2$. □

Proof of Corollary 9.2. Under (H1), $(X_k, u_k, (\varepsilon_{ki})_i)$ is iid across k, so the cluster errors $M_{k,N}$ are unconditionally iid with variance $\bar{\tau}^2 + \sigma^2/N$, and the baseline estimators $(\hat{\tau}^2, \hat{\lambda})$ are consistent for $(\bar{\tau}^2, \bar{\lambda}_0)$. The concentration machinery of Appendix C applies verbatim under compactness of the map inputs $(\bar{\tau}^2, \bar{\lambda}_0)$ in (H2) and the (conditional) subexponential guarantees of $M_{k,N}$ in (H4). By Theorem 5.4, `EtC-CS` therefore attains $\mathbb{E}[\Delta] = O(1/B^2)$ against W_u^*. □

Proof of Proposition 9.3 (GLS Estimator). Throughout, depths are considered under the continuous relaxation, as in the main text. The proof operates in three steps:

Step 1: the clairvoyant problem. Condition on (K, X^K) and let clusters be indexed by the first time they are explored. Imagine a clairvoyant learner whose selected depths could depend on the full vector X^K. Because the learner may condition on a richer class of events, the clairvoyant's optimal allocation achieves lower variance than the optimal allocation available to a standard oracle learner, whose depth rules may only depend on $(X_j)_{j\le k}$. Clearly, the clairvoyant's variance is also a lower bound to the variance of any feasible policy. By construction of the GLS estimator, $\mathrm{Var}(\hat{\mu}_\omega \mid X^K) = 1/\sum_k \omega_k$. Conditional on (K, X^K) weights are deterministic, so variance minimization is equivalent to maximizing total precision $\sum_k \omega_k = \sum_k N^k/(\tau_k^2 N^k + \sigma^2)$. The objective is strictly increasing in each N^k, so the budget binds and total depth is given by $S = (B - KF)/V$, which is constant given K. Since $\partial\,\omega_k/\partial N^k = \sigma^2/(\tau_k^2 N^k + \sigma^2)^2 > 0$ and $\partial^2\omega_k/\partial(N^k)^2 < 0$, the first-order conditions equalize $\tau_k^2 N^k + \sigma^2$ across clusters with $N^k = \alpha/\tau_k^2$ for a constant α pinned down by the budget constraint, namely

$$N^k = \bar{N}\cdot\left(\tau_{\mathrm{h},K}^2/\tau_k^2\right), \qquad \bar{N} = S/K \tag{19}$$

Step 2: exact collapse. At (19), $\tau_k^2 N^k = \bar{N}\tau_{\mathrm{h},K}^2$ for every k, so

$$\sum_k \frac{N^k}{\tau_k^2 N^k + \sigma^2} = \frac{\sum_k N^k}{\bar{N}\tau_{\mathrm{h},K}^2 + \sigma^2} = \frac{\bar{N}K}{\bar{N}\tau_{\mathrm{h},K}^2 + \sigma^2} \qquad \Longrightarrow \qquad \mathrm{Var}\big(\hat{\mu}_\omega \mid K, X^K\big) = \frac{1}{K}\left(\tau_{\mathrm{h},K}^2 + \frac{\sigma^2}{\bar{N}}\right)$$

Thus the problem of the clairvoyant designer mimics the baseline objective with the realized harmonic

mean in place of τ^2 and $\bar{N}$ in place of N. Minimizing over $\bar{N}$ returns

$$\bar{N}^* = \sqrt{\lambda_{0\mathrm{h},K}\frac{F}{V}}, \qquad \lambda_{0\mathrm{h},K} = \sigma^2/\tau^2_{\mathrm{h},K}$$

Step 3: lower bound. The inequality $1/K \geq (F + V\bar{N})/B$ must hold pathwise, hence

$$\frac{1}{K}\left(\tau^2_{\mathrm{h},K} + \frac{\sigma^2}{\bar{N}}\right) \;\geq\; \frac{(F+V\bar{N})\big(\tau^2_{\mathrm{h},K} + \sigma^2/\bar{N}\big)}{B} \;\geq\; \frac{\big(\sqrt{F\tau^2_{\mathrm{h},K}} + \sqrt{V\sigma^2}\big)^2}{B}$$

where the last step minimizes over $\bar{N}$ (attained at $\bar{N}^*$). This chain holds realization by realization for any feasible policy, adapted or clairvoyant. Summands $1/\tau_k^2$ are iid and bounded under (H2), and the maps $y \mapsto 1/y$ and $y \mapsto 1/\sqrt{y}$ are convex, so Jensen gives $\mathbb{E}[\tau^2_{\mathrm{h},K}] \geq \tau^2_{\mathrm{h}}$ and $\mathbb{E}[\tau_{\mathrm{h},K}] \geq \tau_{\mathrm{h}}$, where expectations are taken at a deterministic K. It follows that

$$\mathbb{E}\left[\frac{1}{K}\left(\tau^2_{\mathrm{h},K} + \frac{\sigma^2}{\bar{N}}\right)\right] \geq W^*_\omega = \frac{\big(\sqrt{F\tau^2_{\mathrm{h}}} + \sqrt{V\sigma^2}\big)^2}{B}$$

Step 4: attainment. Under the pseudo-oracle map N^{k*}_ω in (14) with width adjusted to exhaust the budget along the realized covariate path, the realized objective is (13) evaluated within $O_p(\sqrt{B})$ of $(K^*_\omega, \bar{N}^*_\omega)$. Additionally, a second-order expansion under (H2) delivers $\mathbb{E}[\tau^2_{\mathrm{h},K}] = \tau^2_{\mathrm{h}} + O(1/K)$ and $\mathbb{E}[\tau_{\mathrm{h},K}] = \tau_{\mathrm{h}} + O(1/K)$. The realized width is linear in B, so the Jensen gaps are $O(1/B)$ and contribute to the variance $O(1/B^2)$, as they are pre-factored by the usual $1/B$ term. First-order stationarity of the collapsed objective at the oracle absorbs the $O_p(\sqrt{B})$ width deviations at the same order. □

Remark H.2. *Data-Dependent Widths and the Jensen Step. The application of Jensen in Step 3 hides the role of holding K fixed. In particular, $\mathbb{E}[K^{-1}\sum_{k\leq K} 1/\tau_k^2] = 1/\tau^2_{\mathrm{h}}$ only if K is fixed. By contrast, the equality may fail for data-dependent widths, as the stopping rule governing width could stop at a favorable realized mix—for instance, recruiting new clusters until the running harmonic mean dips below τ^2_{h} and stopping there. Such a policy can push $\mathbb{E}[\tau^2_{\mathrm{h},K}]$ below τ^2_{h}, moving the expected pathwise bound below W^*_ω. This improvement upon the oracle is not driven by more efficient sampling but instead by cherry-picking the composition of the sample. There is no analogue to this problem in the homogeneous setting because the relevant population parameter τ^2 is constant and design independent, unlike $\tau^2_{\mathrm{h},K}$. The damage of cherry-picking is bounded. Admissible widths lie in a $\Theta(B)$ window, so the running average of the summands $1/\tau_k^2$—bounded under (H2)—deviates from its mean by $O(B^{-1/2})$ uniformly, invoking the same maximal-inequality arguments of Appendix C. As a result, even an adversarial stop depresses $\tau^2_{\mathrm{h},K}$ by at most $O(B^{-1/2})$, worth $O(B^{-3/2})$ in the bound after accounting for the $1/B$ prefactor. The implications for an optimality theory are mixed. The rate sits below the oracle scale, so W^*_ω remains a correct benchmark up to first order and any upper-bound statement is unaffected. On the other hand, $B^{-3/2}$ lies above the excess-variance scale B^{-2}, so minimax lower bounds cannot be centered at W^*_ω: the cherry-picking stopping rule could itself swamp the $1/B^2$ excess variance statement. Lower bounds should therefore either restrict to deterministic-width policies or center on the random benchmark $\mathbb{E}\big[\big(\sqrt{F\tau^2_{\mathrm{h},K}} + \sqrt{V\sigma^2}\big)^2\big]/B$.*

Proof of Proposition 9.4. Expanding $W^*_u = \bar{\tau}^2(\sqrt{V\bar{\lambda}_0} + \sqrt{F})^2/B$ using $\bar{\tau}^2\bar{\lambda}_0 = \sigma^2$ and

$\bar{\tau}^2\sqrt{V\bar{\lambda}_0} = \bar{\tau}\sqrt{V\sigma^2}$,

$$B \cdot W_u^* = V\sigma^2 + 2\sqrt{VF\sigma^2}\,\bar{\tau} + F\bar{\tau}^2, \qquad B \cdot W_\omega^* = V\sigma^2 + 2\sqrt{VF\sigma^2}\,\tau_{\mathrm{h}} + F\tau_{\mathrm{h}}^2$$

and differencing gives (15). Nonnegativity of both terms follows from the arithmetic–harmonic mean inequality, $\bar{\tau}^2 \geq \tau_{\mathrm{h}}^2$, with equality if and only if τ_k^2 is degenerate. Both gaps are increasing in the dispersion of τ_k^2 in the usual convex-order sense. □

Estimation of the Variance Link. The unit-level variance is estimated as in the main text after accommodating for heterogeneous depths, i.e.

$$\hat{w}_K^2 = \frac{1}{K}\sum_k \frac{1}{N^k - 1}\sum_{i \leq N^k} \left(y_{ki} - \hat{\mu}_{N^k}^k\right)^2, \qquad \hat{w}_K^2 \overset{\text{a.s.}}{\underset{K}{\to}} \sigma^2$$

Each cluster contributes a single draw of u_k, so τ_k^2 is identified only by pooling through the link's parametric reduction. The estimation strategy resembles that of non-linear least squares with pseudo-outcomes $(q_{k,K})_k$ in place of u_k,

$$q_{k,K} = \frac{K(\hat{\mu}_k - \hat{\mu})^2 - \hat{b}_K^2}{K-2} - \frac{\hat{w}_K^2}{N^k} = \frac{K(M_k - \bar{M})^2 - \hat{b}_K^2}{K-2} - \frac{\hat{w}_K^2}{N^k} \tag{20}$$

where $\hat{\mu} = K^{-1}\sum_k \hat{\mu}_{N^k}^k$ is the unweighted grand-mean, $\hat{\mu}_k = \hat{\mu}_{N^k}^k$, $M_k = u_k + \bar{\varepsilon}_{k,N^k}$ and $\hat{b}_K^2 = (K-1)^{-1}\sum_k(\hat{\mu}_k - \hat{\mu})^2$. The pseudo-outcomes can be shown to be conditionally unbiased for $K \geq 3$,

$$\begin{aligned}
\mathbb{E}[q_{k,K} \mid X^K] &= \frac{1}{K-2}\left(K\mathbb{E}[(M_k - \bar{M})^2 \mid X^K] - \bar{\theta}^2\right) - \frac{\sigma^2}{N^k} \\
&= \frac{1}{K-2}\left(K\left(1 - \frac{1}{K}\right)^2\theta_k^2 + \frac{1}{K}\sum_{j \neq k}\theta_j^2 - \bar{\theta}^2\right) - \frac{\sigma^2}{N^k} \\
&= \theta_k^2 - \frac{\sigma^2}{N^k} = \tau_k^2
\end{aligned}$$

where the first equality uses $\mathbb{E}[\hat{b}_K^2 \mid X^K] = \bar{\theta}^2$ and $\mathbb{E}[\hat{w}_K^2 \mid X^K] = \sigma^2$, similarly to the results in Appendix B; and the second equality uses cross-cluster independence. Next, consider the nonlinear least squares estimator of γ,

$$\hat{\gamma} = \arg\min_{\gamma \in \Gamma} \frac{1}{K}\sum_k \left(\ell(X_k, \gamma) - q_{k,K}\right)^2$$

Under the usual assumptions, $\hat{\gamma}$ is $\sqrt{K}$-consistent by standard M-estimation arguments, namely (i) identification, the map $\gamma \mapsto \mathbb{E}[(\ell(X,\gamma) - \ell(X,\gamma_0))^2]$ is uniquely minimized at γ_0 with $\mathbb{E}[\nabla_\gamma \ell \nabla_\gamma \ell']$ nonsingular; (ii) compactness (H2); and (iii) finite conditional fourth moments of the outcomes. There is one non-standard argument in this process: the pseudo-outcomes are not independent across k, as they share $\hat{\mu}$, $\hat{w}_K^2$ and $\hat{b}_K^2$. However, the induced perturbation of the NLS score is $O_p(1/K)$ and does not affect rates. Notably, the dependence does affect the asymptotic variance and standard errors of $\hat{\gamma}$. Finally, define the clipped plug-ins in the usual way,

$$\hat{\tau}_k^2 = \Pi_{[\tau_{k,m}^2,\, \tau_{k,M}^2]}\left(\ell(X_k, \hat{\gamma})\right), \qquad \hat{\tau}_{\mathrm{h}}^2 = \Pi_{[\tau_m^2,\, \tau_M^2]}\left(\frac{K}{\sum_k 1/\hat{\tau}_k^2}\right), \qquad \hat{\lambda}_{\mathrm{h}} = \Pi_{[\lambda_m,\, \lambda_M]}\left(\frac{\hat{w}_K^2}{\hat{\tau}_{\mathrm{h}}^2}\right)$$

Algorithm 3. I now introduce the algorithm and discuss its properties alongside Conjecture 9.5.

Algorithm 3 `wEtC-CS`

Input $B, F, V, \Gamma, [x_m, x_M], \ell(\cdot,\cdot), [\sigma_m^2, \sigma_M^2]$
derive $\tau_m^2, \tau_M^2, \lambda_m, \lambda_M, \quad K_M^\omega = \lfloor K_\omega^*(\lambda_M) \rfloor, \quad N_{\max} = \left\lfloor \frac{\tau_M^2}{\tau_m^2}\sqrt{\lambda_M F/V} \right\rfloor$
for clusters $1 \le k \le K_M^\omega$
  visit cluster k, **observe** X_k, **sample** $N_p^k = \left\lfloor \frac{\tau_m^2}{\tau_{k,M}^2}\sqrt{\lambda_m F/V} \right\rfloor$ units, with $\tau_{k,M}^2 = \sup_{\gamma\in\Gamma} \ell(X_k,\gamma)$
estimate $\hat{w}^2$, $\hat{\gamma}$, $\hat{\tau}_k^2 = \ell(X_k, \hat{\gamma})$, $\hat{\tau}_{\mathrm{h}}^2$, $\hat{\lambda}_{\mathrm{h}}$
complete every pilot cluster $k \le K_M^\omega$ up to $N^k = \left\lfloor \frac{\hat{\tau}_{\mathrm{h}}^2}{\hat{\tau}_k^2}\sqrt{\hat{\lambda}_{\mathrm{h}} F/V} \right\rfloor$
while remaining budget $\ge F + V N_{\max}$
  visit a new cluster k, **observe** X_k, **sample** $N^k = \left\lfloor \frac{\hat{\tau}_{\mathrm{h}}^2}{\hat{\tau}_k^2}\sqrt{\hat{\lambda}_{\mathrm{h}} F/V} \right\rfloor$ units
Output the design $\{(X_k, N^k)\}_k$ and $\hat{\mu}_{\hat{\omega}}$ with weights $\hat{\omega}_k = \left(\hat{\tau}_k^2 + \hat{w}^2/N^k\right)^{-1}$

Pilot implementability. Under (H3), $N_p^k = \lfloor (\tau_m^2/\tau_{k,M}^2)\sqrt{\lambda_m F/V} \rfloor \ge 2$, hence for every admissible estimate the terminal depth satisfies $N^k \ge (\tau_m^2/\tau_{k,M}^2)\sqrt{\lambda_m F/V} \ge N_p^k$ by clipping. The pilot is trivially nested in the terminal design cluster by cluster.

Top-up feasibility. Recall $K_p^{\texttt{wEtC-CS}} = K_M^\omega = \lfloor K_\omega^*(\lambda_M) \rfloor$. Moreover, by definition of the empirical harmonic mean over the pilot clusters $\hat{\tau}_{\mathrm{h}}^2$, $\hat{\tau}_{\mathrm{h}}^2 \sum_{k\le K_p} 1/\hat{\tau}_k^2 = K_p$, hence

$$FK_p + V\sum_{k\le K_p} N^k \;\le\; FK_p + V\,\hat{\tau}_{\mathrm{h}}^2 \sum_{k\le K_p} \frac{1}{\hat{\tau}_k^2}\sqrt{\hat{\lambda}_{\mathrm{h}} F/V} \;=\; K_p\Big(F + \sqrt{\hat{\lambda}_{\mathrm{h}} FV}\Big) \;\le\; K_\omega^*(\hat{\lambda}_{\mathrm{h}})\Big(F + \sqrt{\hat{\lambda}_{\mathrm{h}} FV}\Big) \;=\; B$$

where the last inequality uses $K_p \le K_\omega^*(\hat{\lambda}_{\mathrm{h}})$ by clipping and monotonicity. Pilot completion is then implementable, as it never exceeds the budget, and the algorithm requires no safeguards on that line.

Width as an outcome. Unlike the homogeneous algorithms, `wEtC-CS` does not implement an optimal map of the form $K_B^{\texttt{wEtC-CS}} = K^*(\hat{\lambda})$ for a λ estimate. Per-cluster costs $F + VN^k(X_k)$ are random before recruitment, so terminal width cannot be fixed after the pilot. The algorithm instead commits to a depth profile of the form $N^k = \frac{\hat{\tau}_{\mathrm{h}}^2}{\hat{\tau}_k^2}\sqrt{\hat{\lambda}_{\mathrm{h}} F/V}$, and recruits until the remaining budget cannot cover a maximum cost cluster $F + VN_{\max}$. The conservative buffer wastes at most $F + VN_{\max} = O(1)$, so the fixed profile protocol implements a near-optimal design for an effective budget of $B - O(1)$, which is worth $O(1/B^2)$ in variance.

Remark H.3. *Possible Refinements. The algorithm admits several natural improvements. For instance, the learner may implement an* `ACS`*-style variant that re-estimates* $(\hat{\gamma}, \hat{\tau}_{\mathrm{h}}^2)$ *after every cluster rather than committing at the pilot. Similarly, the* $O(1)$ *leftover can be used to improve precision by deepening visited clusters or exploring a final truncated cluster. Finally, the NLS step could be replaced by a design-targeted variant that minimizes the plug-in design variance rather than the* L^2 *fit.*